



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



First Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamical Model

Robot Control



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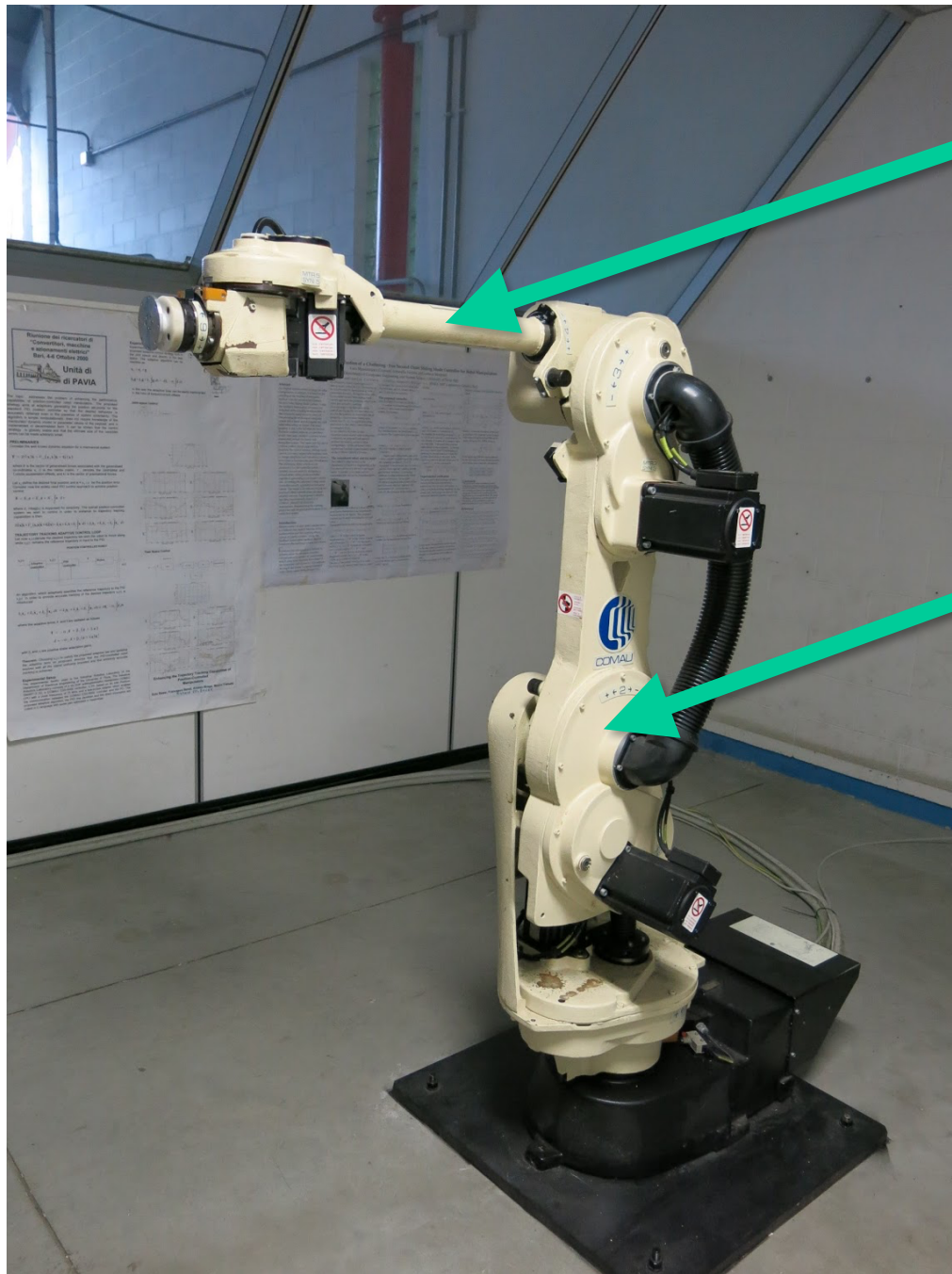
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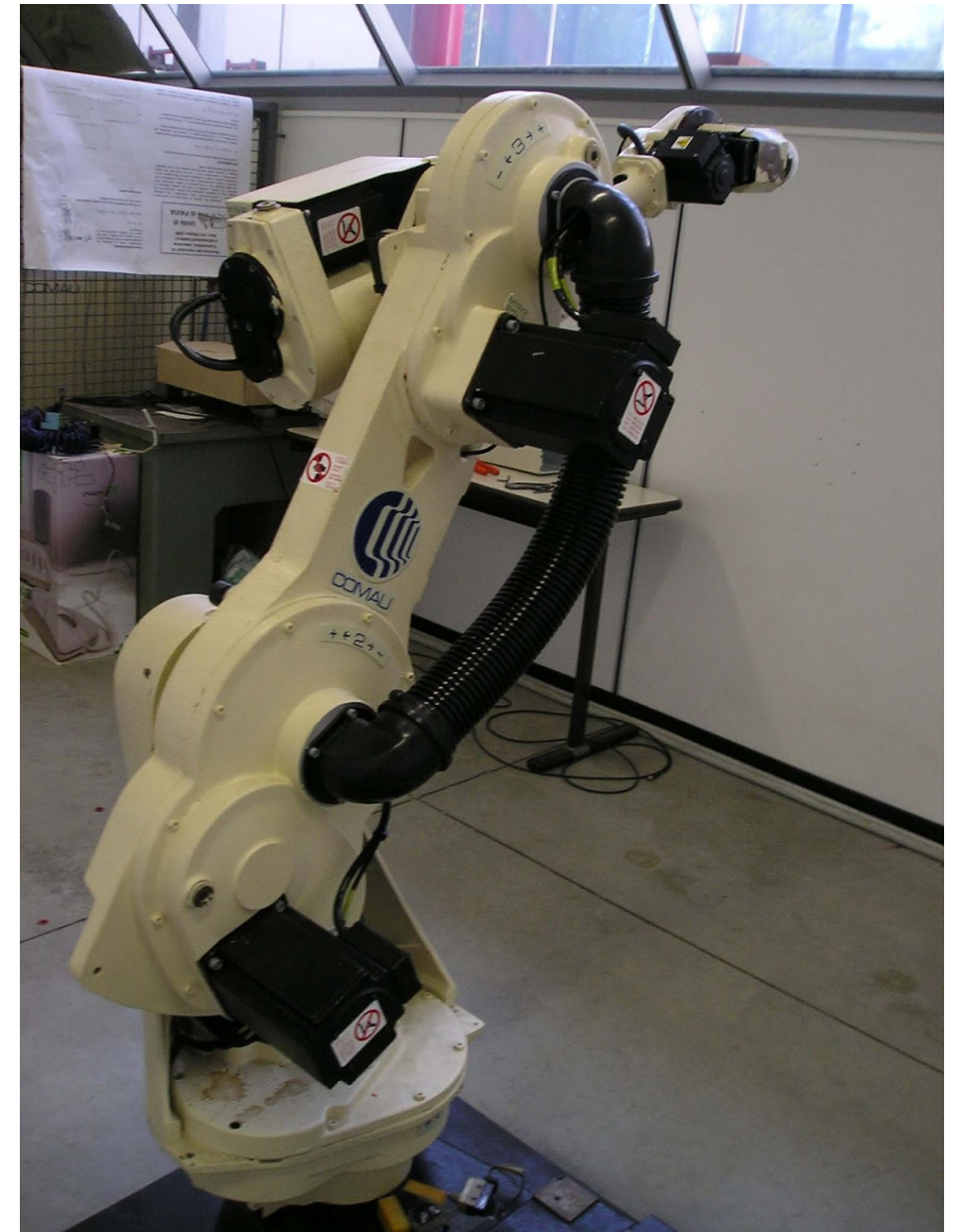
Basic Terminology

Basic Terminology and Concepts

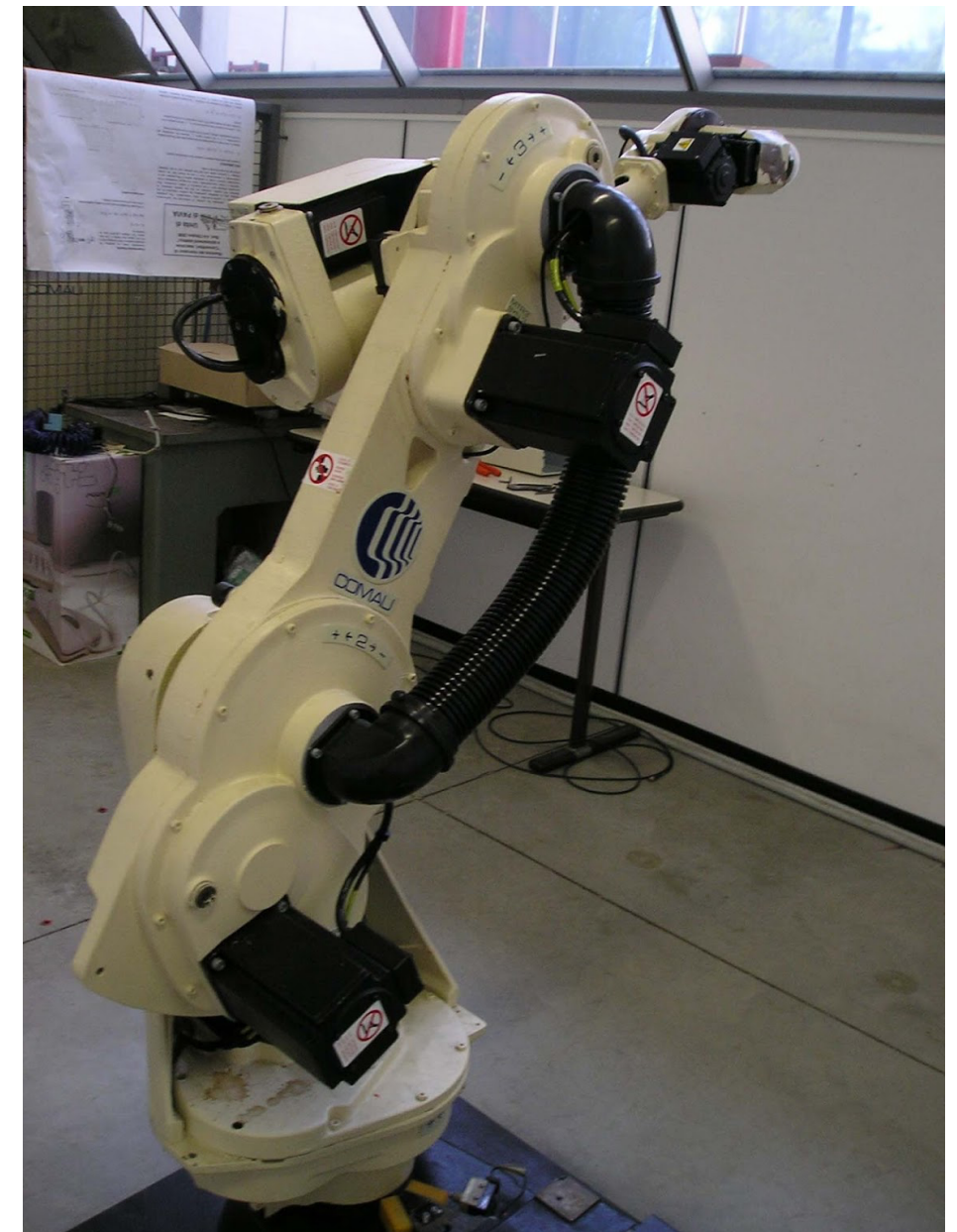
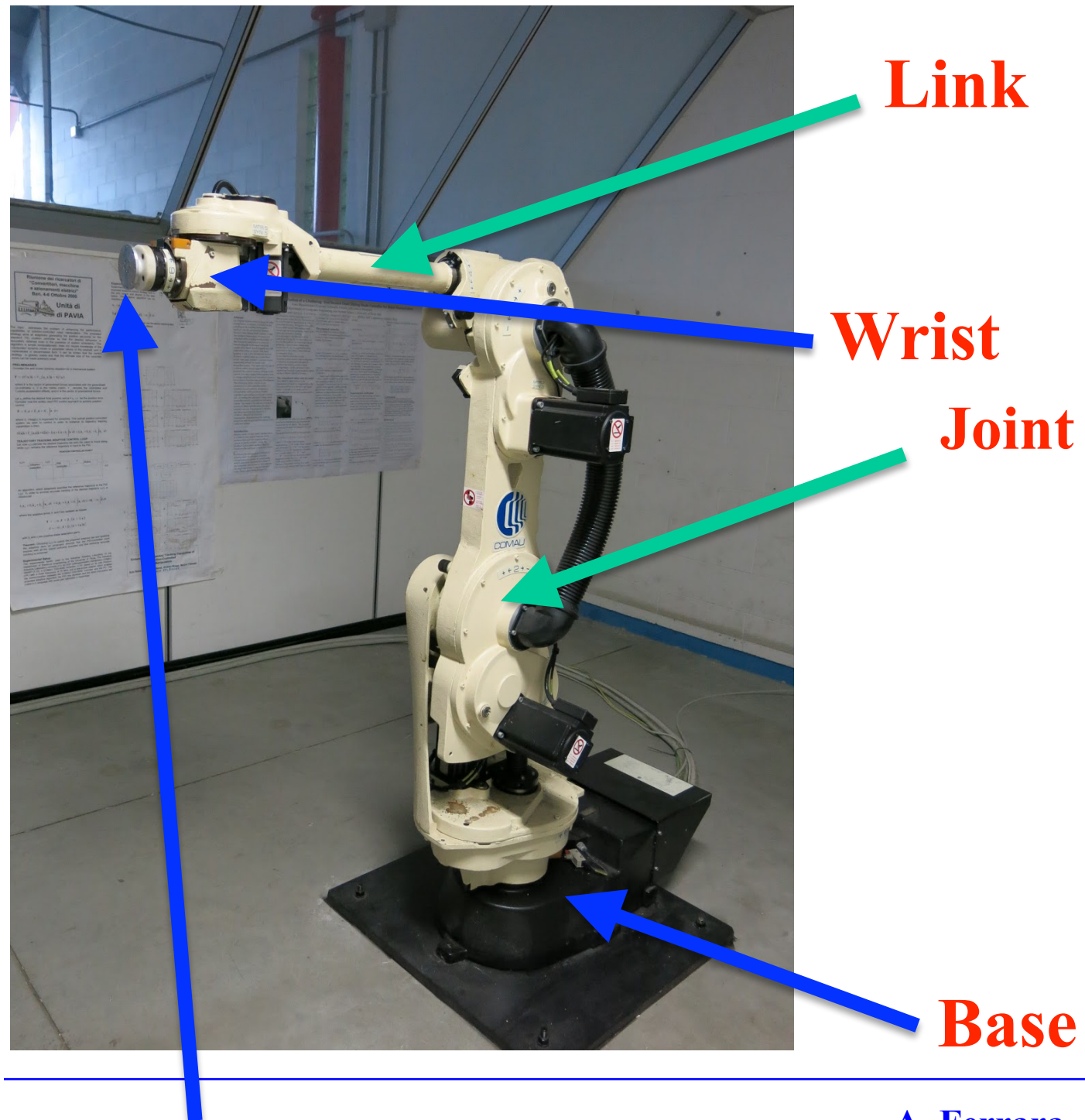


Link

Joint



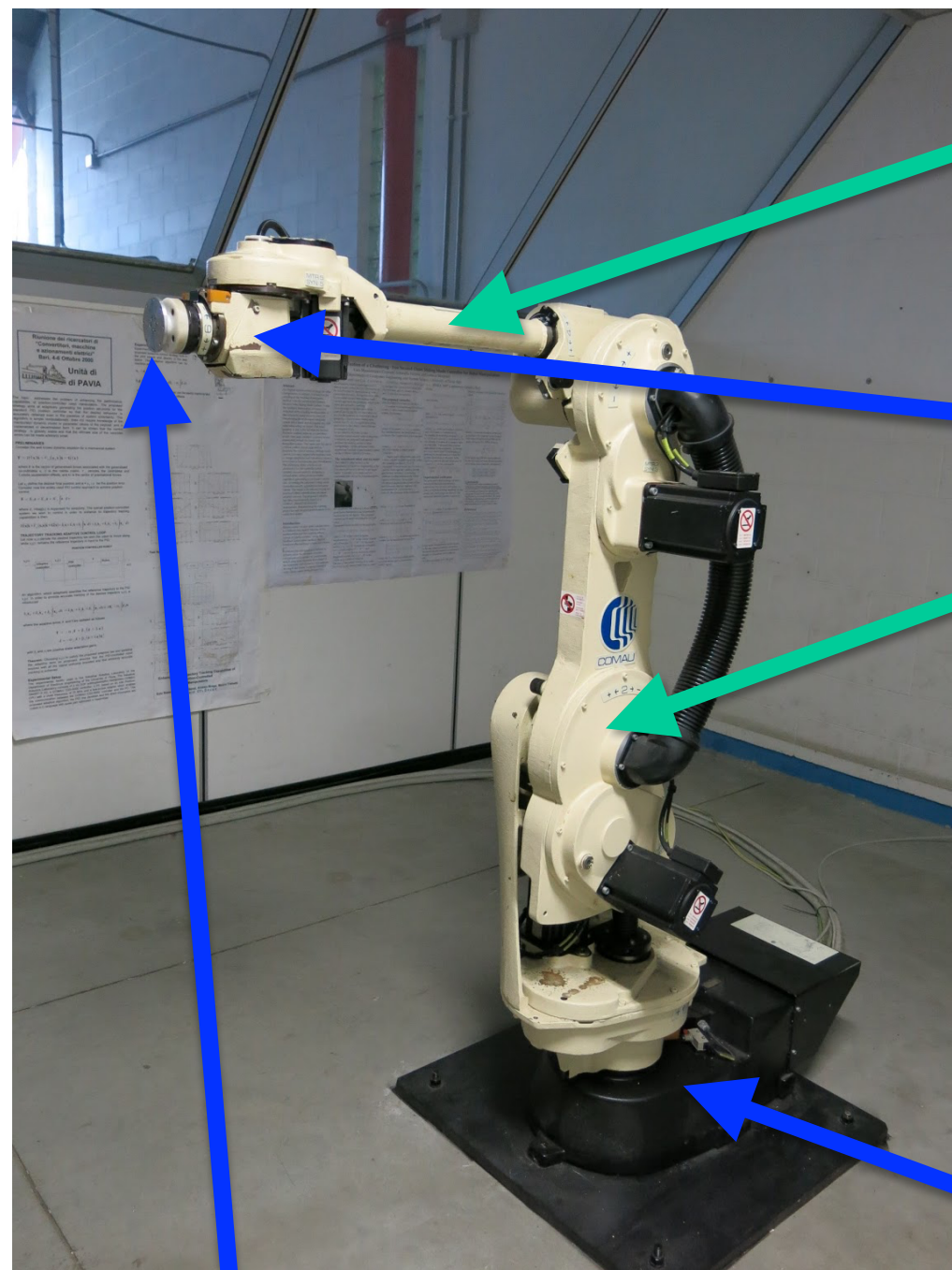
Basic Terminology and Concepts



End-Effector

A. Ferrara

Basic Terminology and Concepts



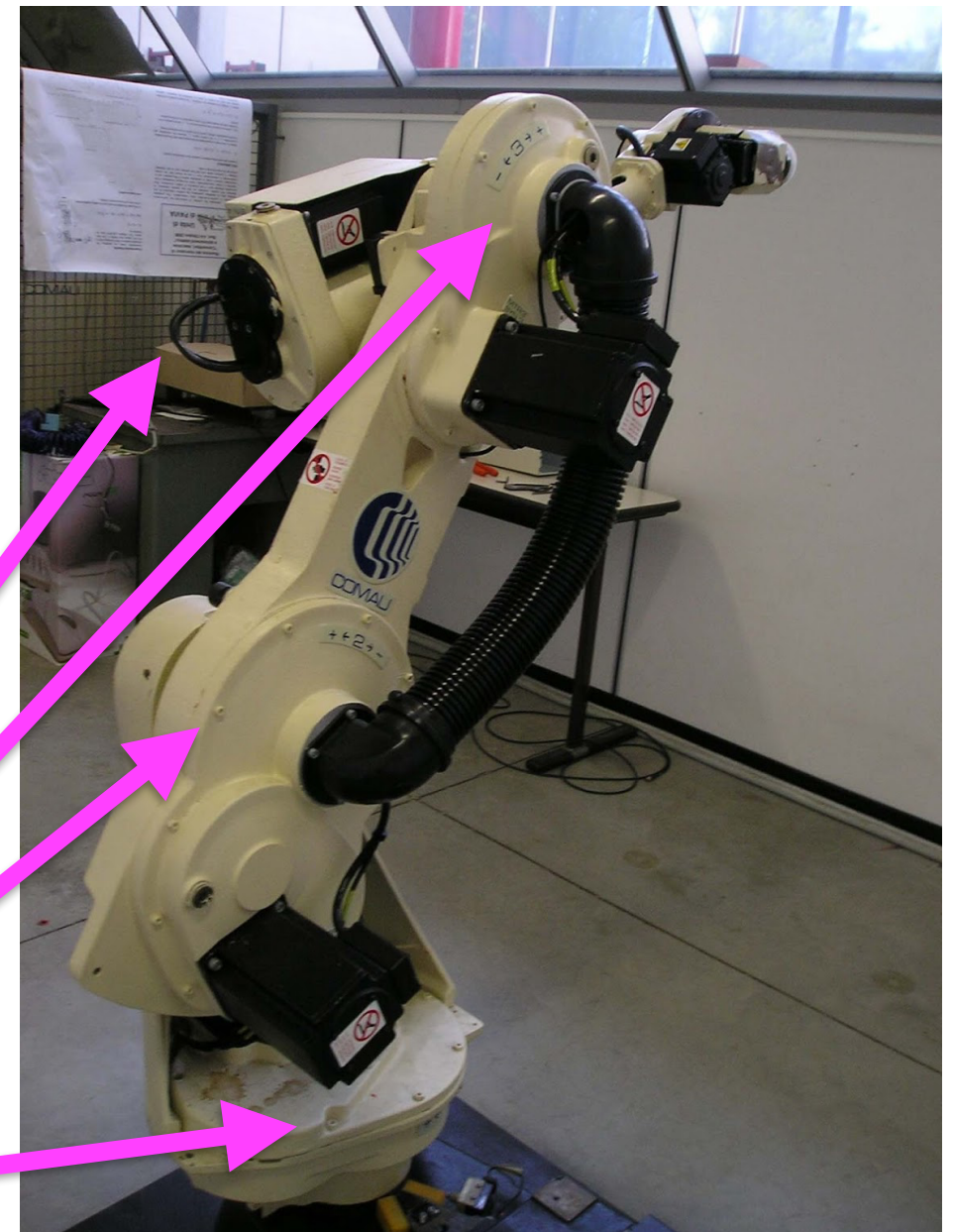
Link

Wrist

Joint

DoF or DoM

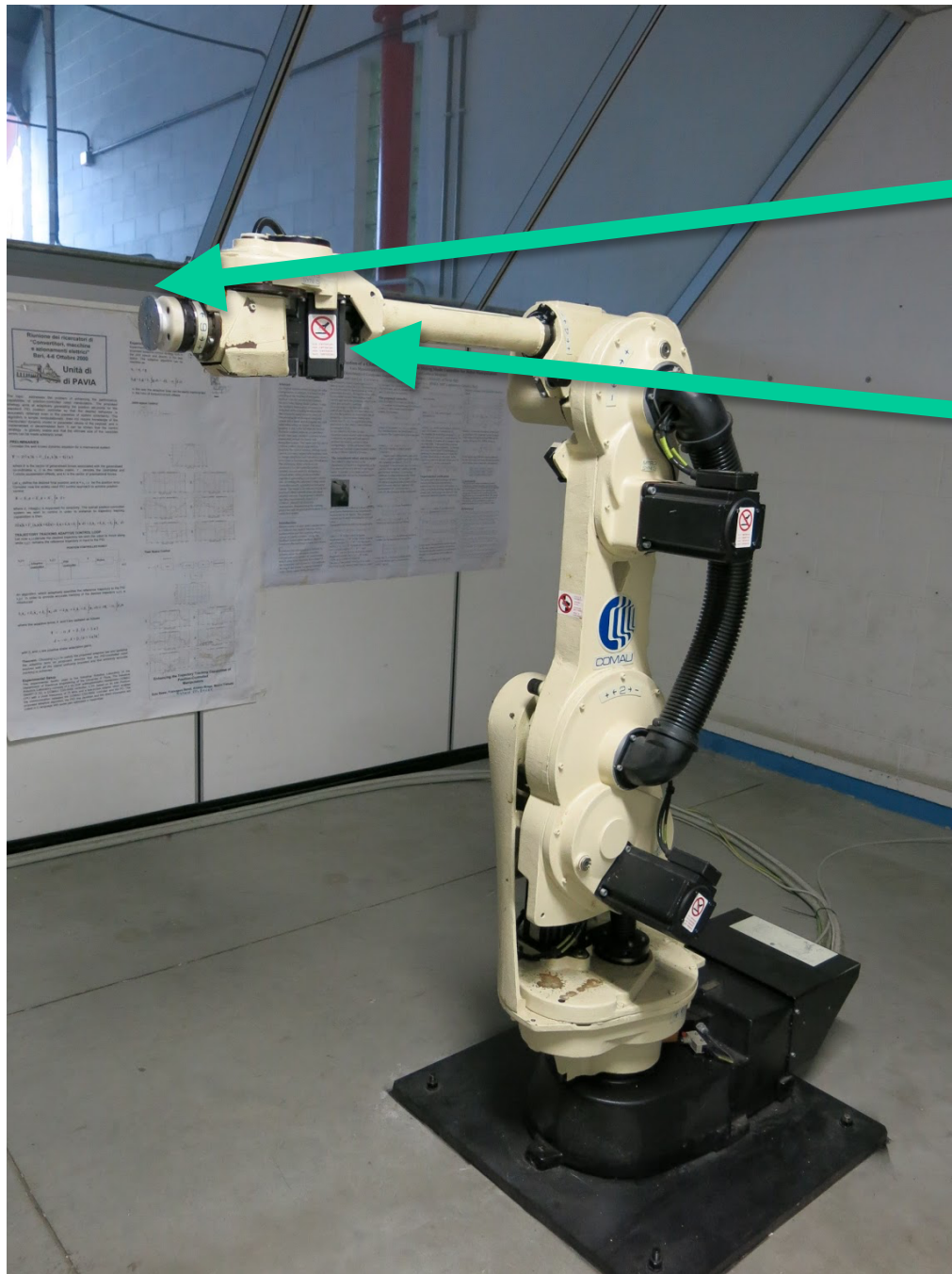
Base



End-Effector

A. Ferrara

Basic Terminology and Concepts

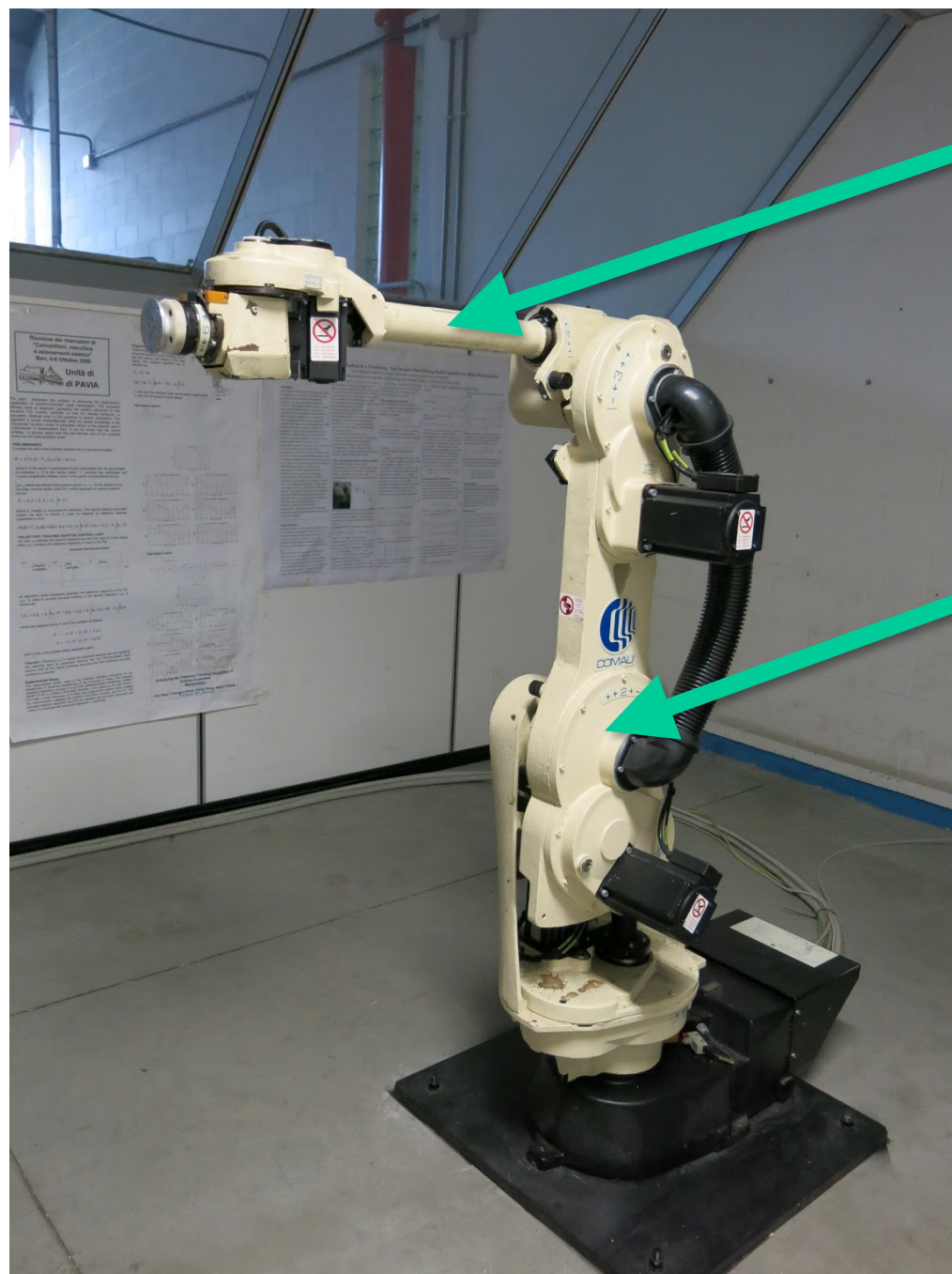


End-Effector: performs the required task

Wrist: provides **dexterity** (the ability to perform a difficult action quickly and skilfully with the hand, *Cambridge Dict.*)

ARM: ensures mobility

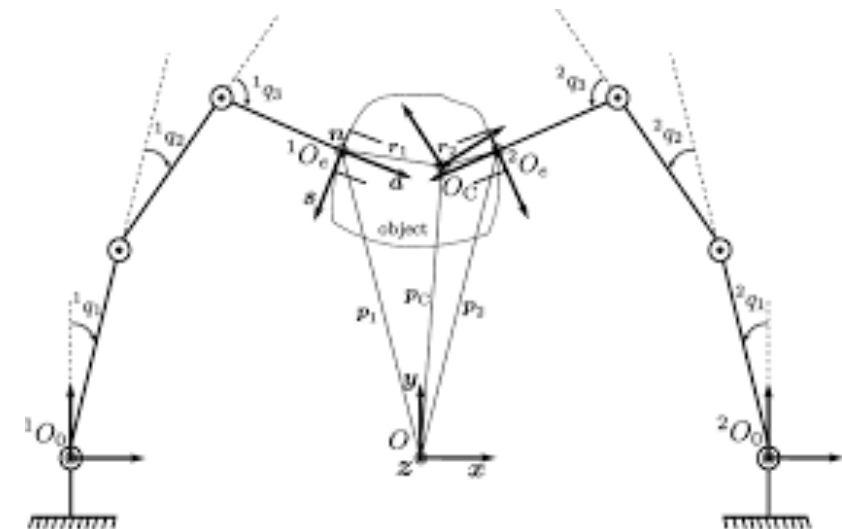
Robot Manipulators = Kinematic Chains



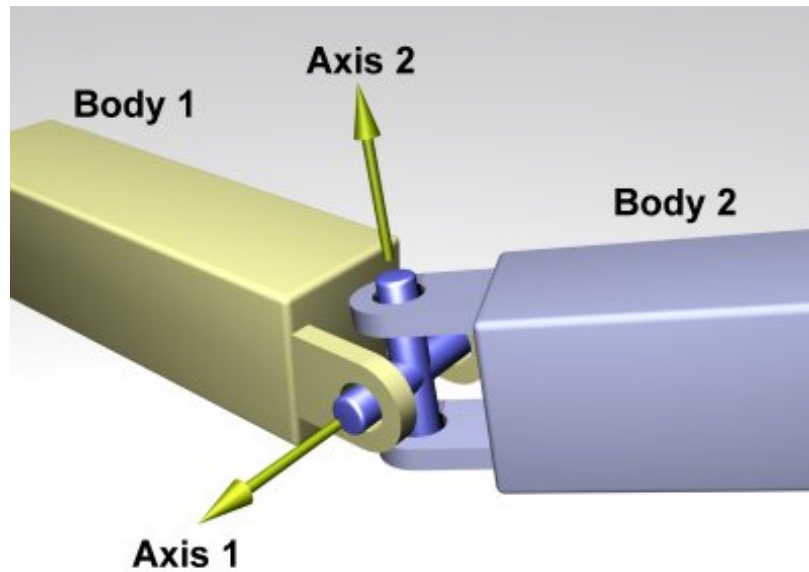
Link

Joint

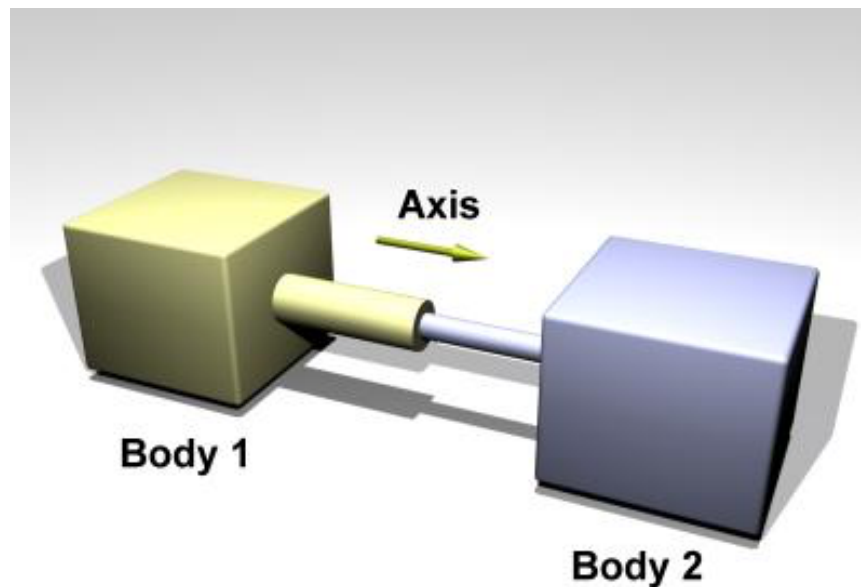
Kinematic Chain:
An assembly of
rigid bodies (links)
connected by **joints** so
that they are able to
move relatively to one
another



Types of Joints

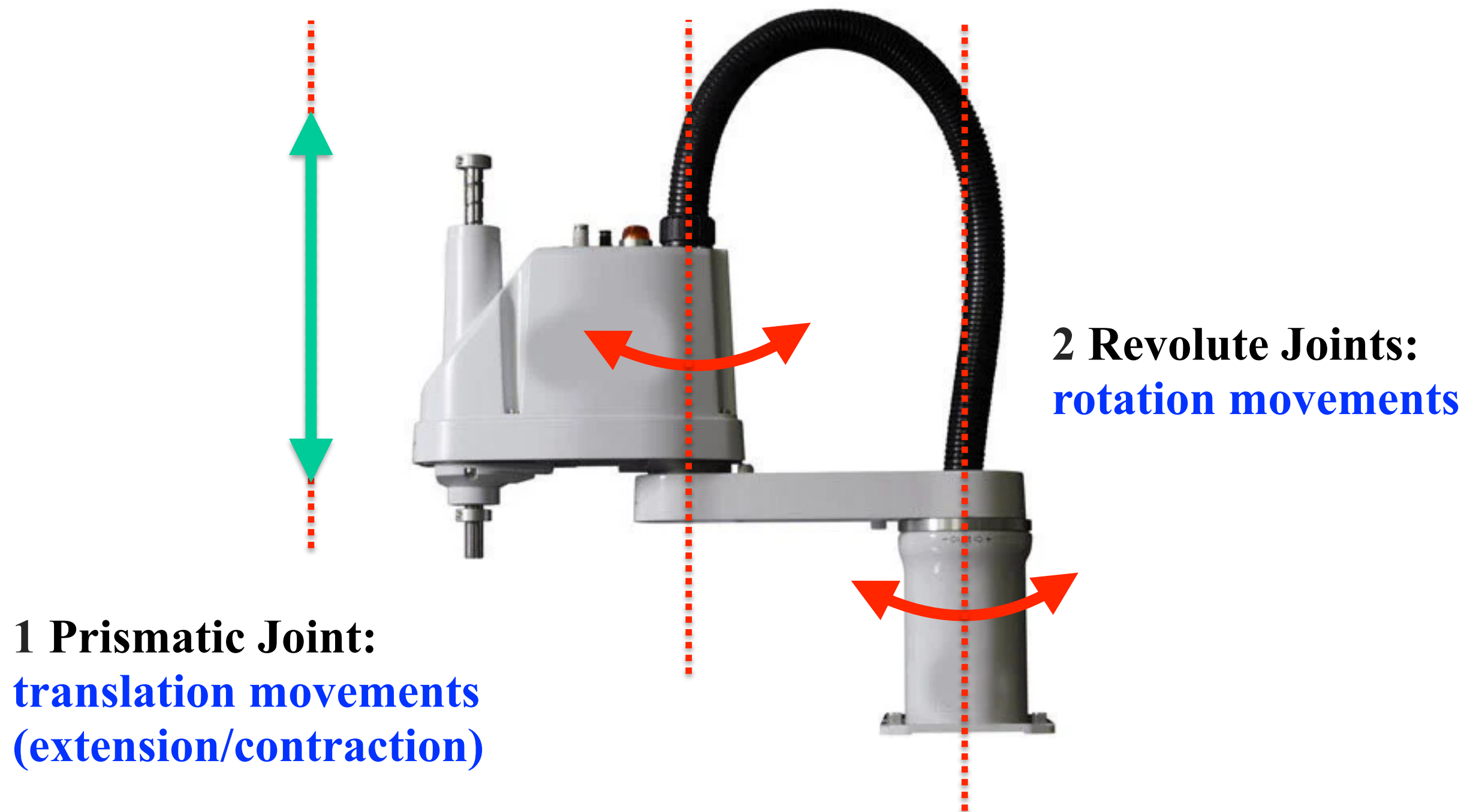


Revolute Joint:
rotation movements

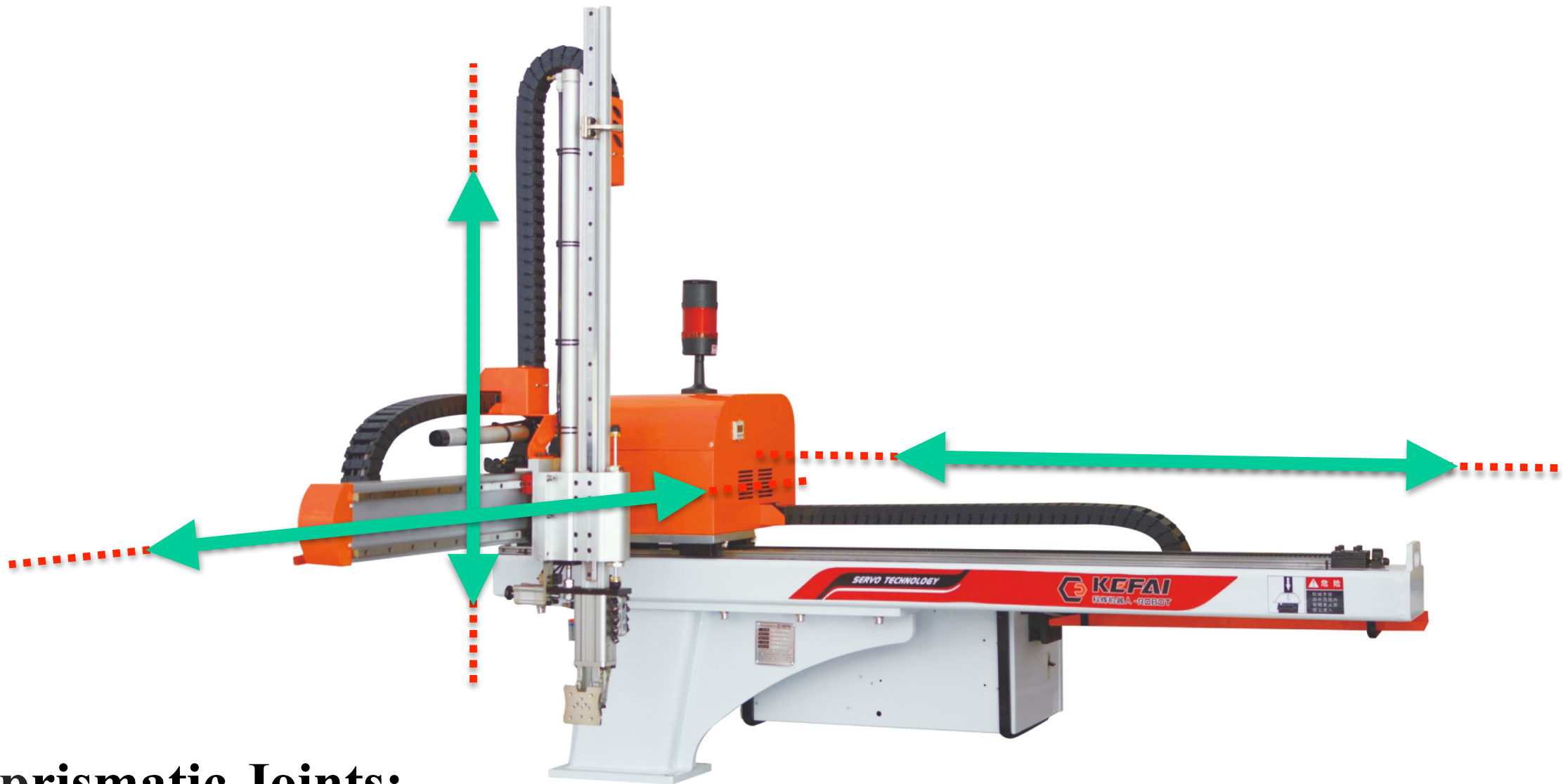


Prismatic Joint:
translation movements
(extension/contraction)

Types of Joints: Example 1



Types of Joints: Example 2



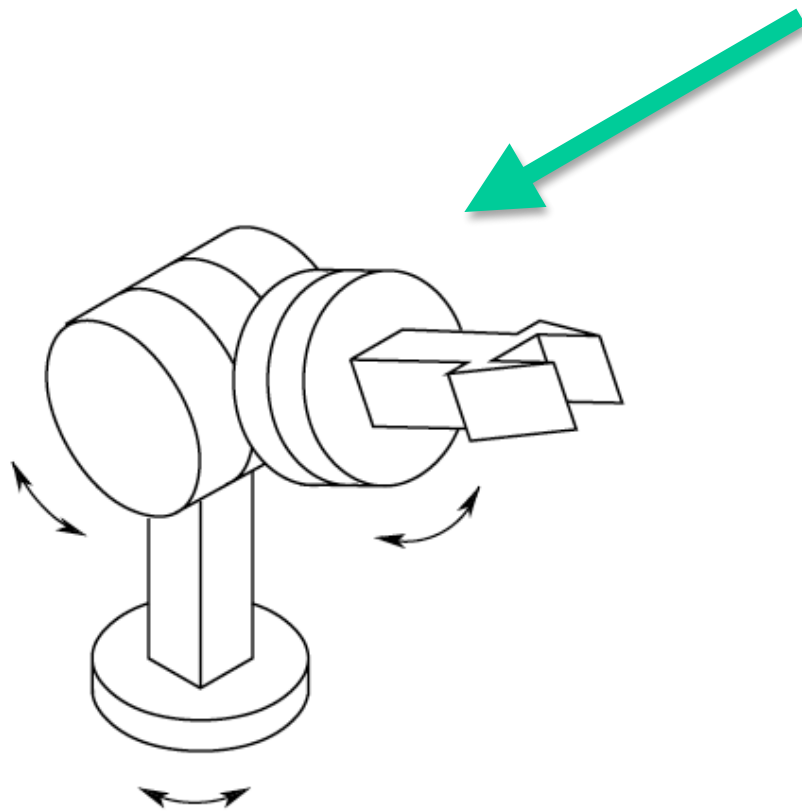
Only prismatic Joints:
translation movements
(extension/contraction)

Types of Joints

Spherical wrist (highest dexterity)

three revolute joint axes intersect at a single point

decoupling between position and orientation of the end-effector: the arm performs the task of positioning the point of intersection, while the wrist determines the end-effector orientation



Types of Joints



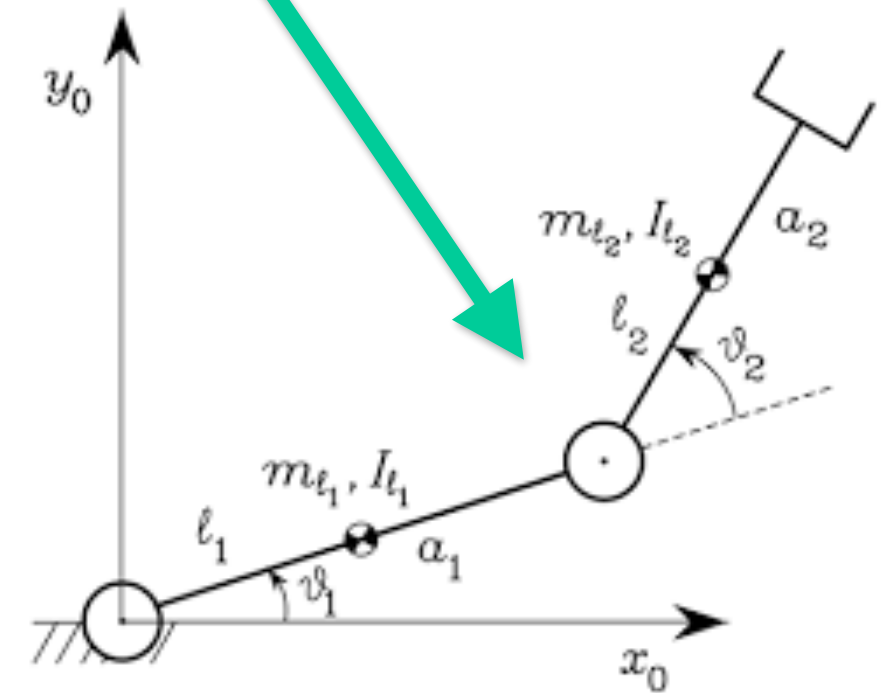
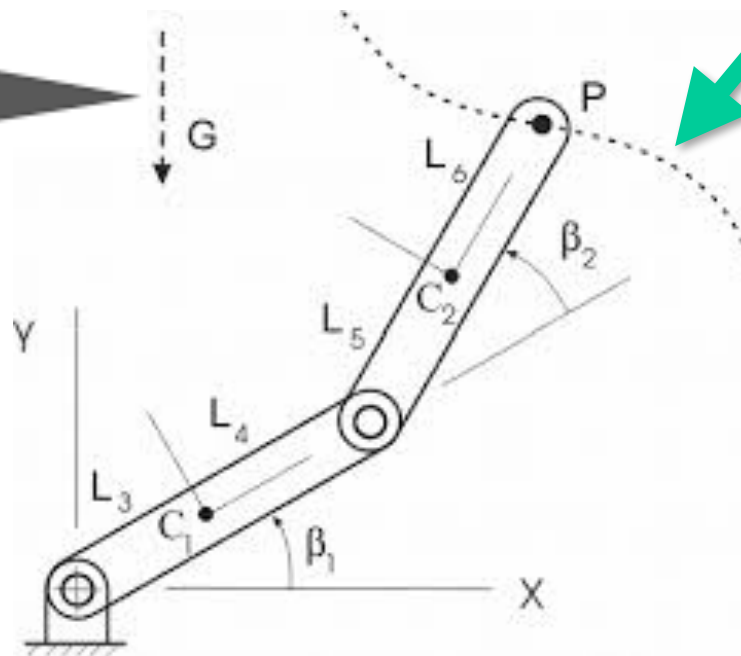
Source: <http://www.hephaist.co.jp/e/pro/ball.html>

Spherical rolling joint:
rotation movements
(greater rigidity and
compactness than
joints made from
integrating multiple
revolute joints)

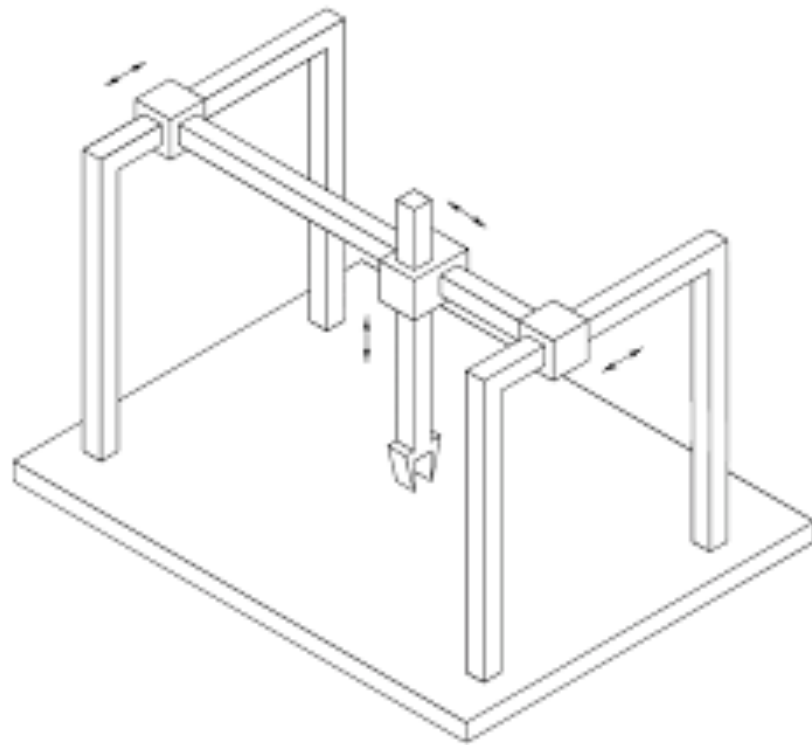
Types of Joints: Pictorial Representation



**Revolute (or Rotoidal)
Joint: 3D and 2D
representation**

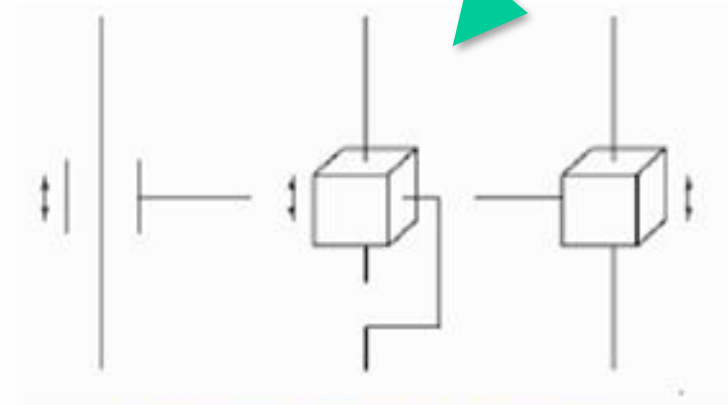


Types of Joints: Pictorial Representation

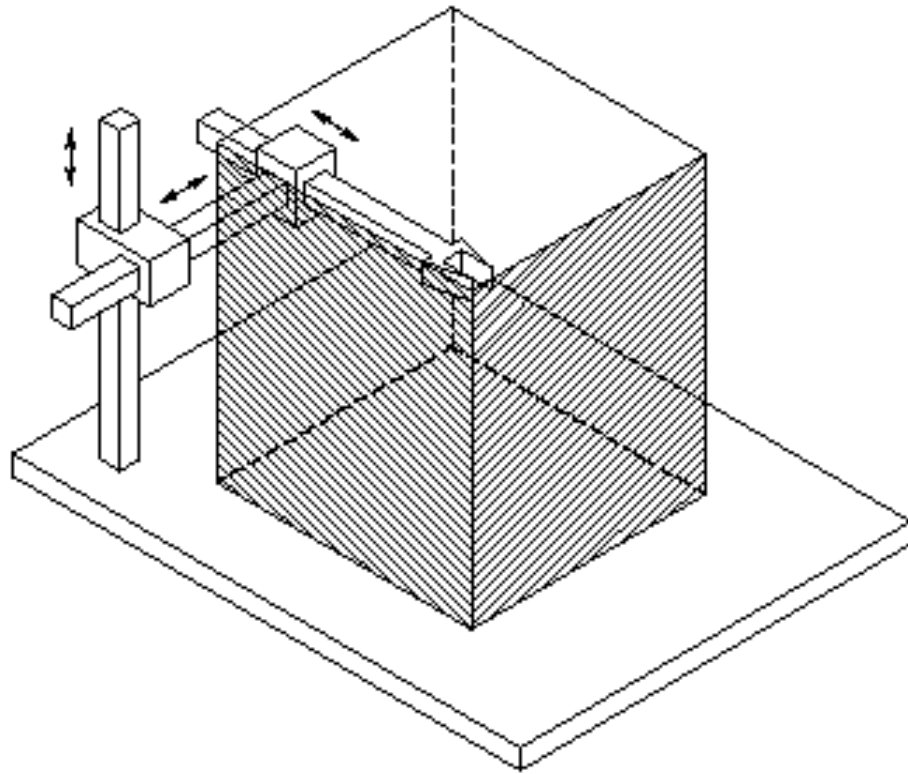


Gantry manipulator

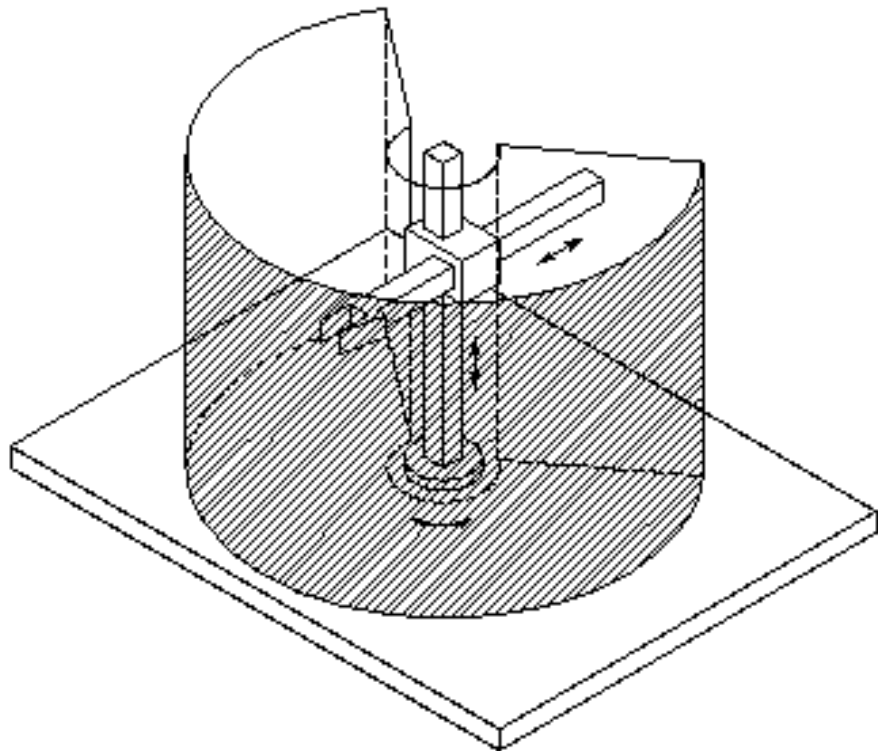
Prismatic Joint:
3D and 2D
representation



Types of Robot: Classification

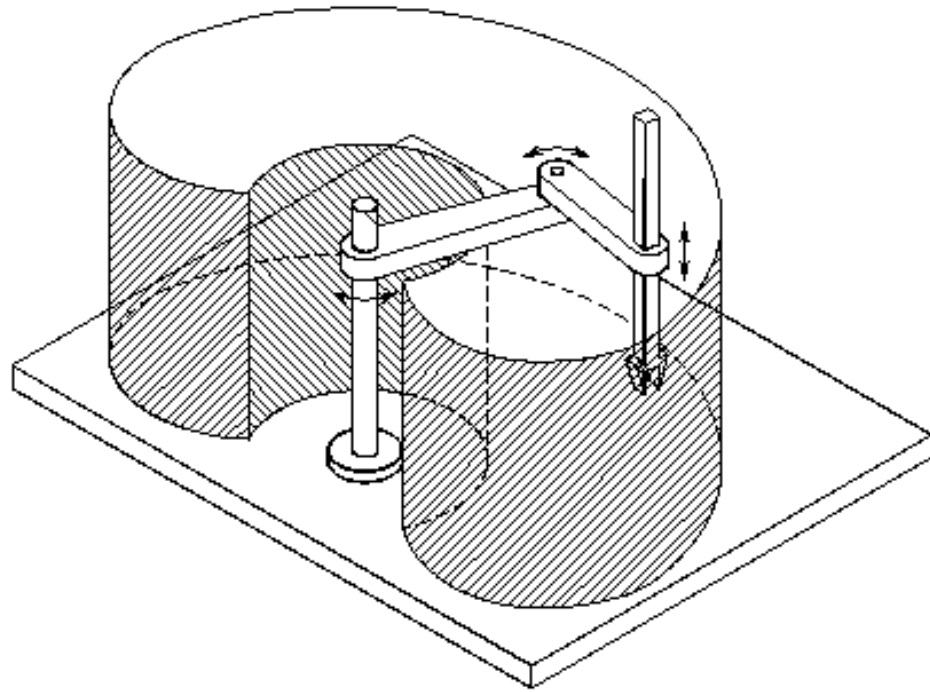


← **Cartesian Robot**



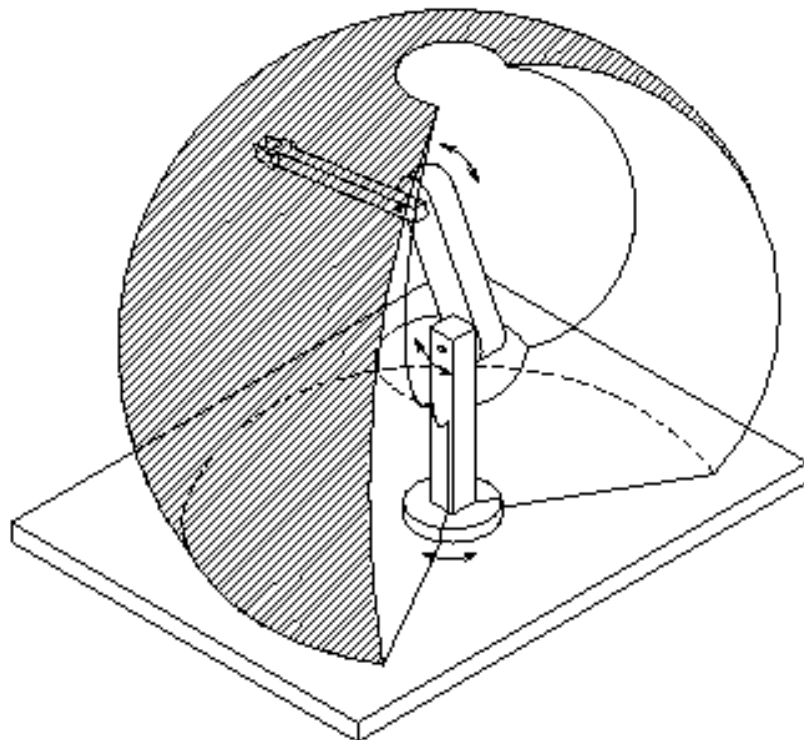
← **Cylindrical Robot**

Types of Robot: Classification



S.C.A.R.A. Robot

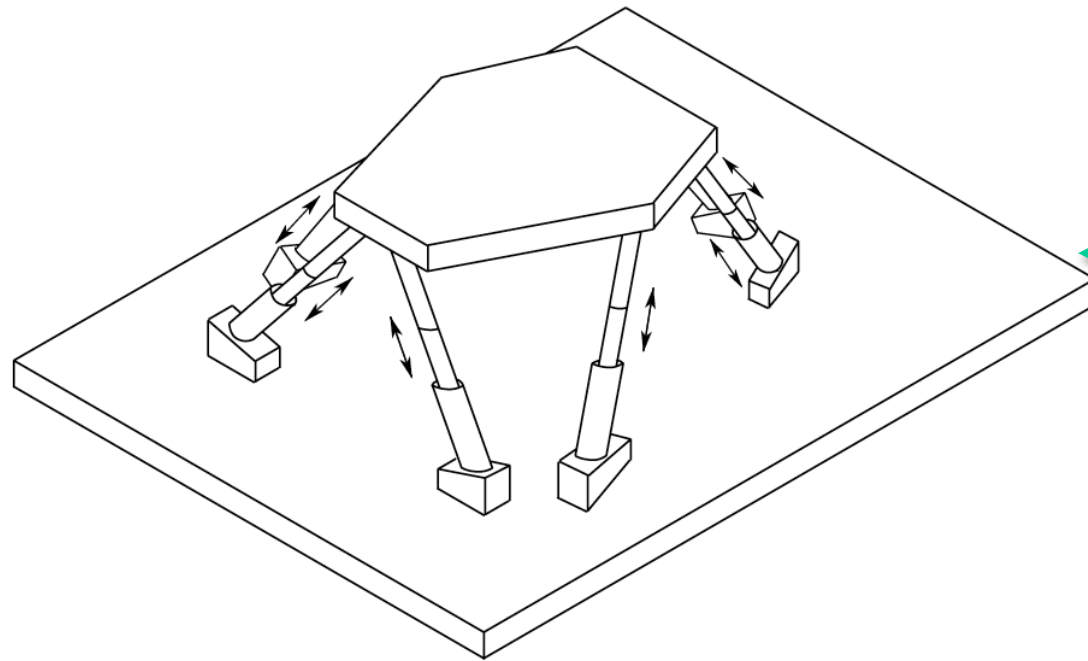
Selective Compliance Assembly Robot Arm



Anthropomorphic Robot



Types of Robot: Classification



Parallel Robot

Closed-chain geometry (multiple kinematic chains connecting the base to the end-effector)

PROS: High structural stiffness, with respect to open-chain manipulators which enables high operational speeds, precision accuracy, short cycle times, and high payload features

CONS: reduced workspace



Parallel Robot ABB IRB 360 Flexpicker

Open and Closed Kinematic Chains and Other terms

Open kinematic chains: each prismatic or revolute joint provides the structure with a single degree of freedom (**DOF**)

Closed kinematic chains: the number of DOFs is less than the number of joints in view of the constraints imposed by the loops

The **degrees of freedom** need to be distributed along the mechanical structure in order to have a sufficient number of them to execute a certain task

General case: for task consisting of arbitrarily positioning and orienting an object in the 3D space 6 DOFs are required

If the robot offers more DOFs than task variables, the manipulator is said to be **redundant** from a kinematic viewpoint.

Workspace: the portion of the environment that the manipulator can “reach” by virtue of its structure and the presence of mechanical joint limits



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Types of Kinematics

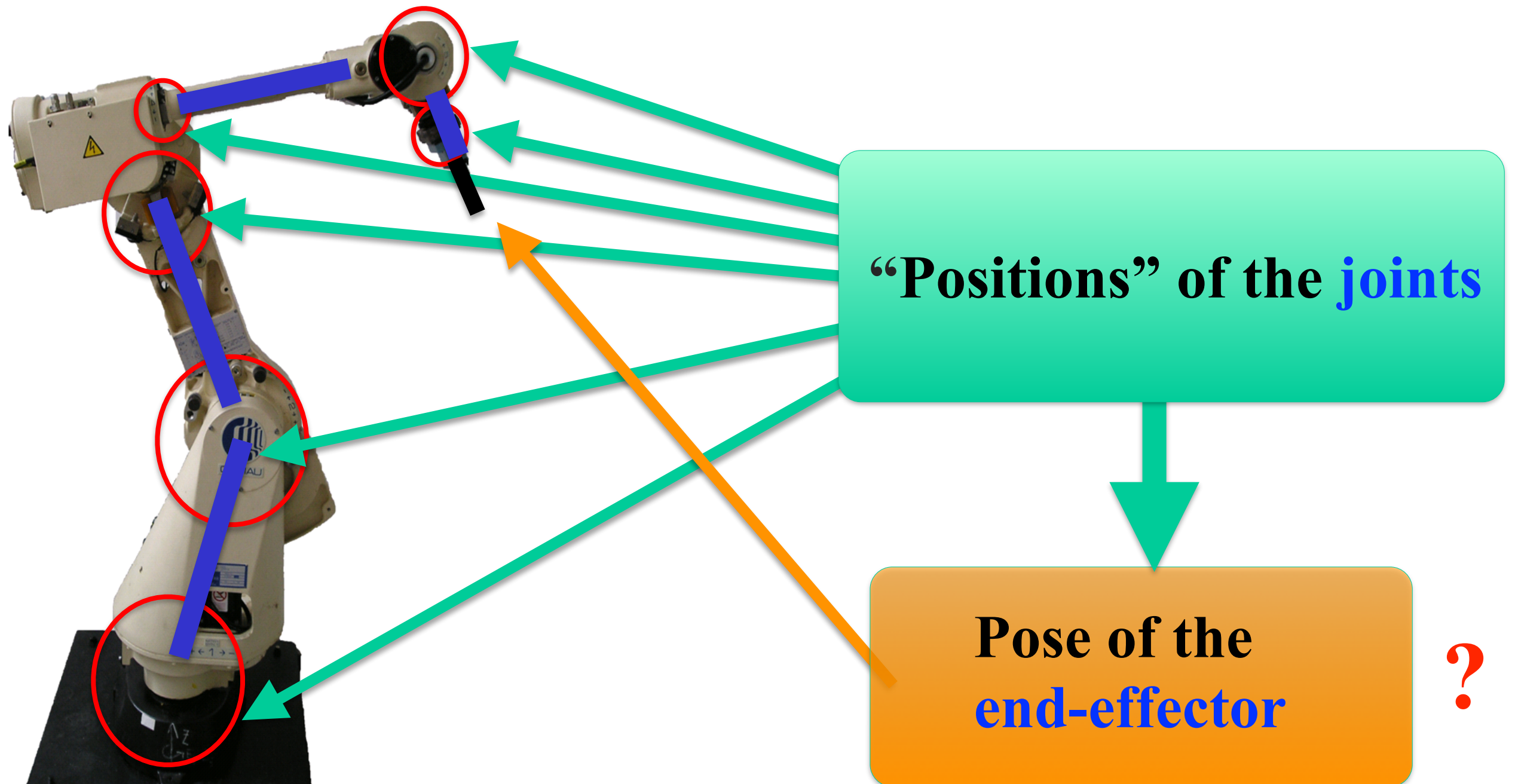
Robot Kinematics

Direct Kinematics

Inverse Kinematics

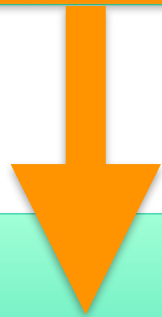
Differential Kinematics

Direct Kinematics

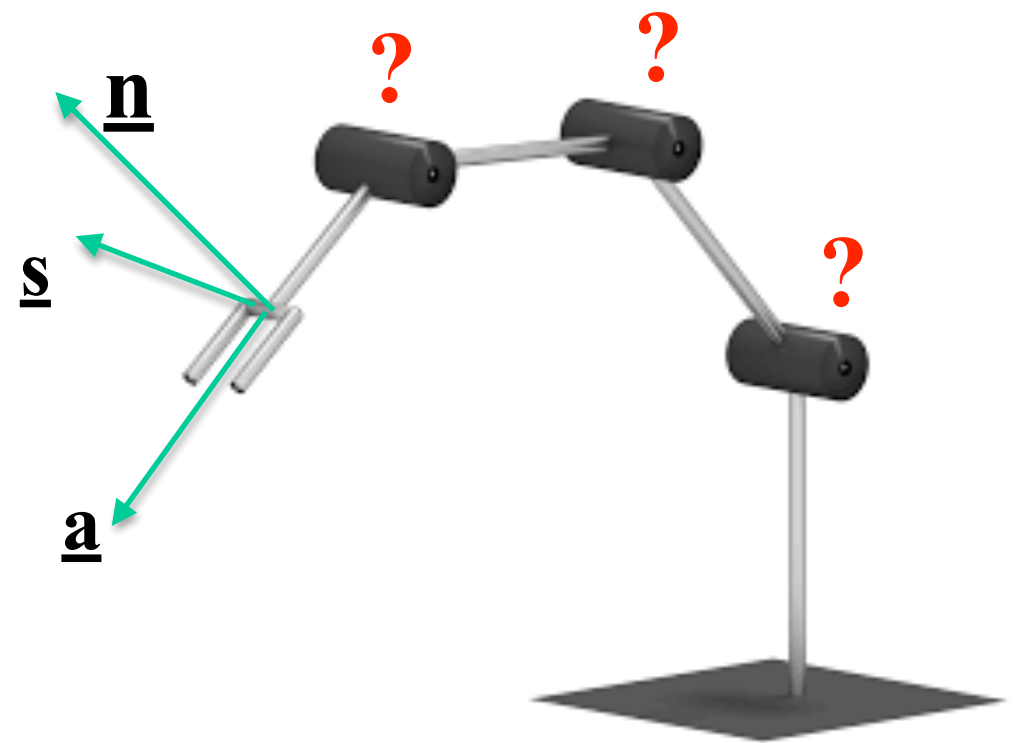


Inverse Kinematics

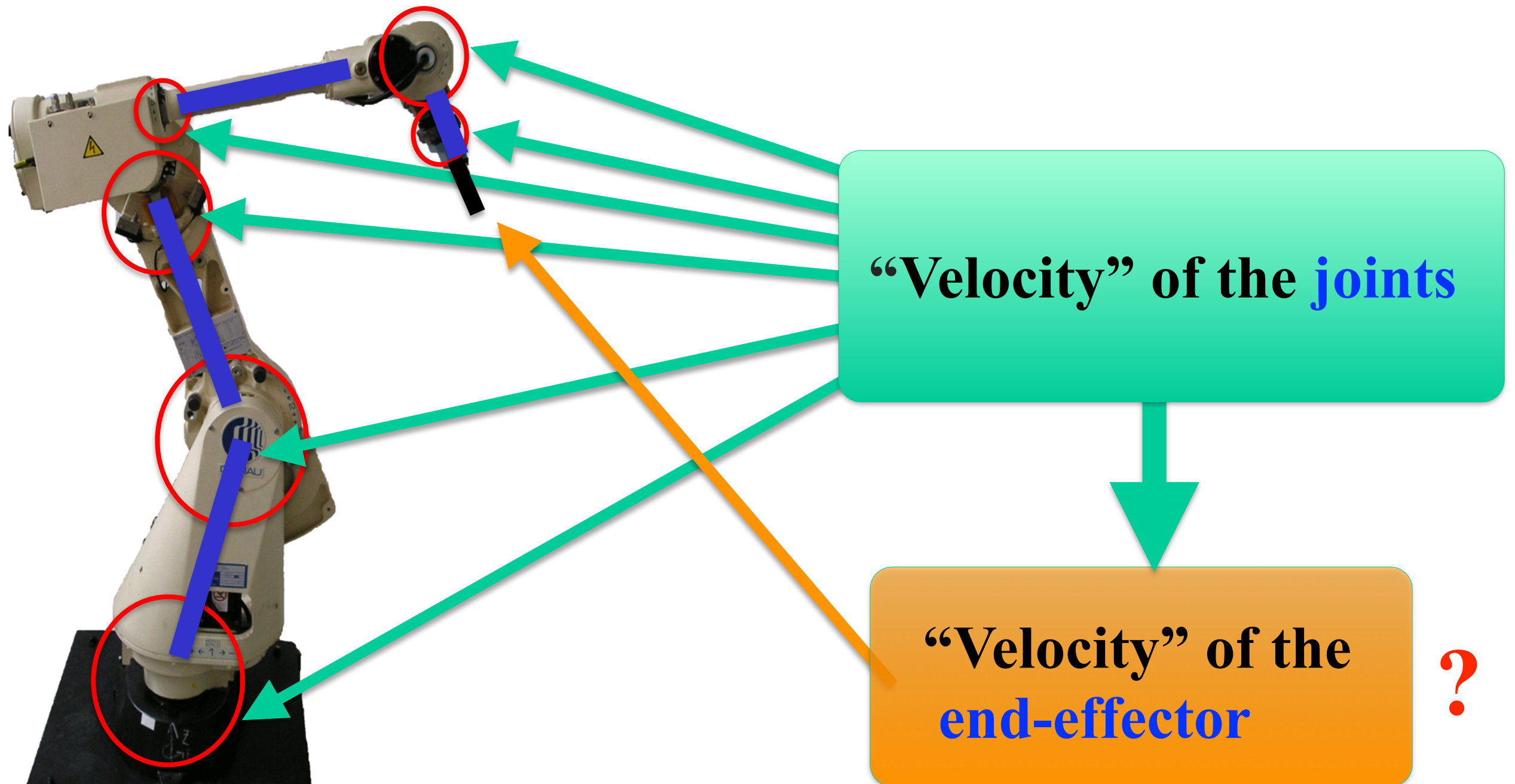
Pose of the
end-effector



“Positions” of the joints



Differential Kinematics





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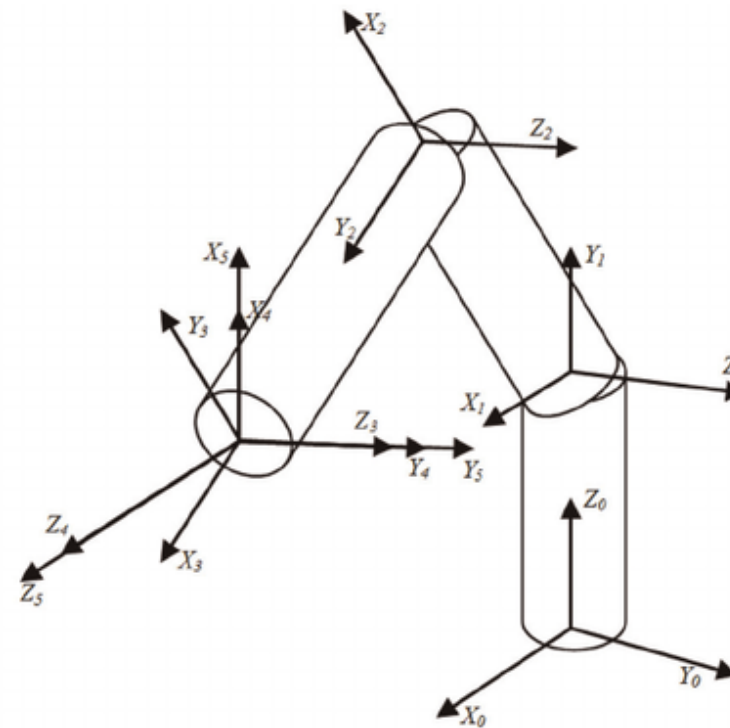
Prof. Antonella Ferrara



Reference Frames and Rotation Matrices

Some Preliminary Considerations On...

✓ Reference Frames and Rotation Matrices



$n+1$ reference frames (n # joints) + base reference frame

Some Preliminary Considerations On...

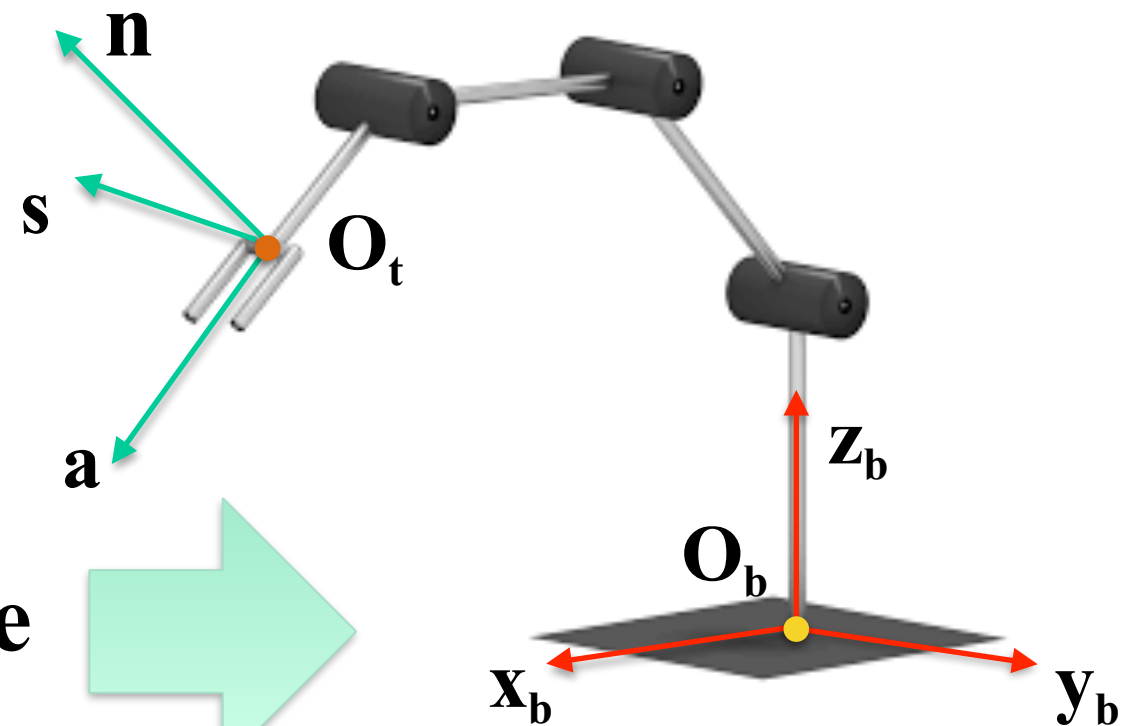
✓ Reference Frames and Rotation Matrices

Tool Reference Frame

$O_t-x_t y_t z_t$ or $O_t-n s a$

Base Reference Frame

$O_b-x_b y_b z_b$

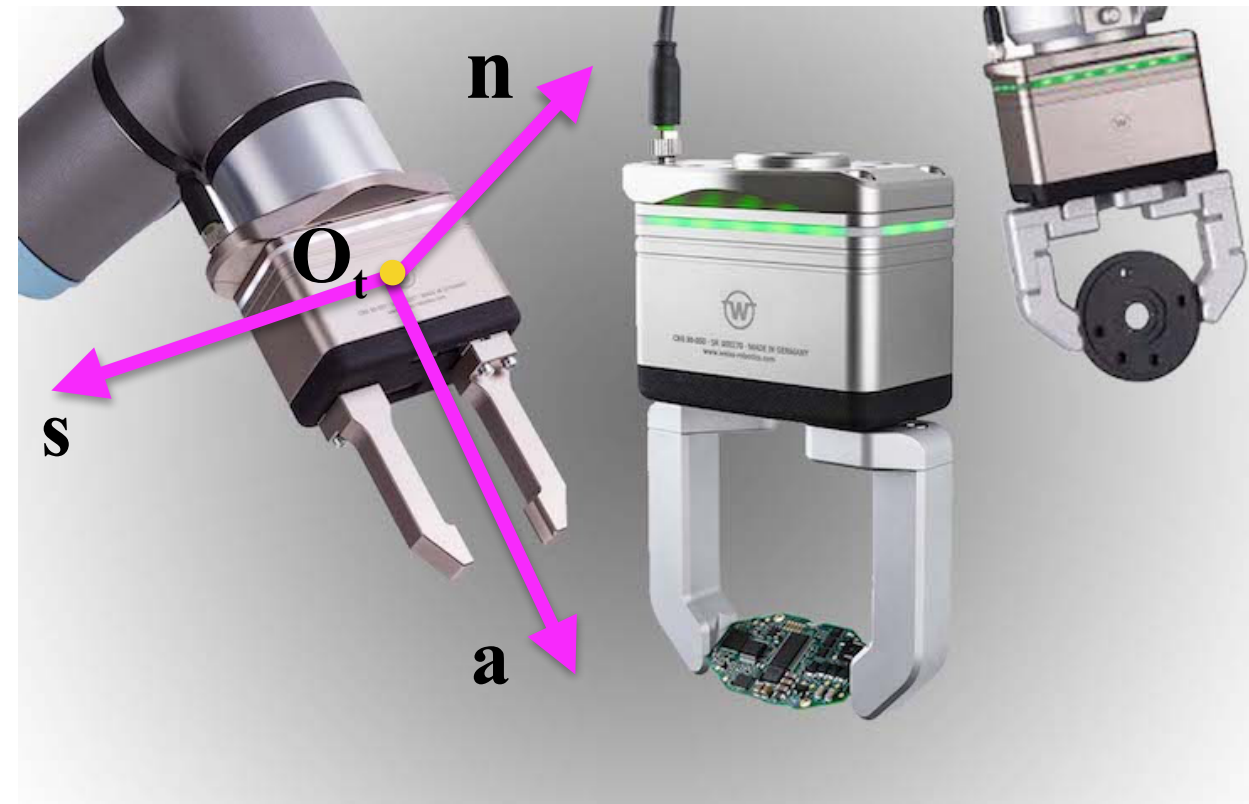


Some Preliminary Considerations On...

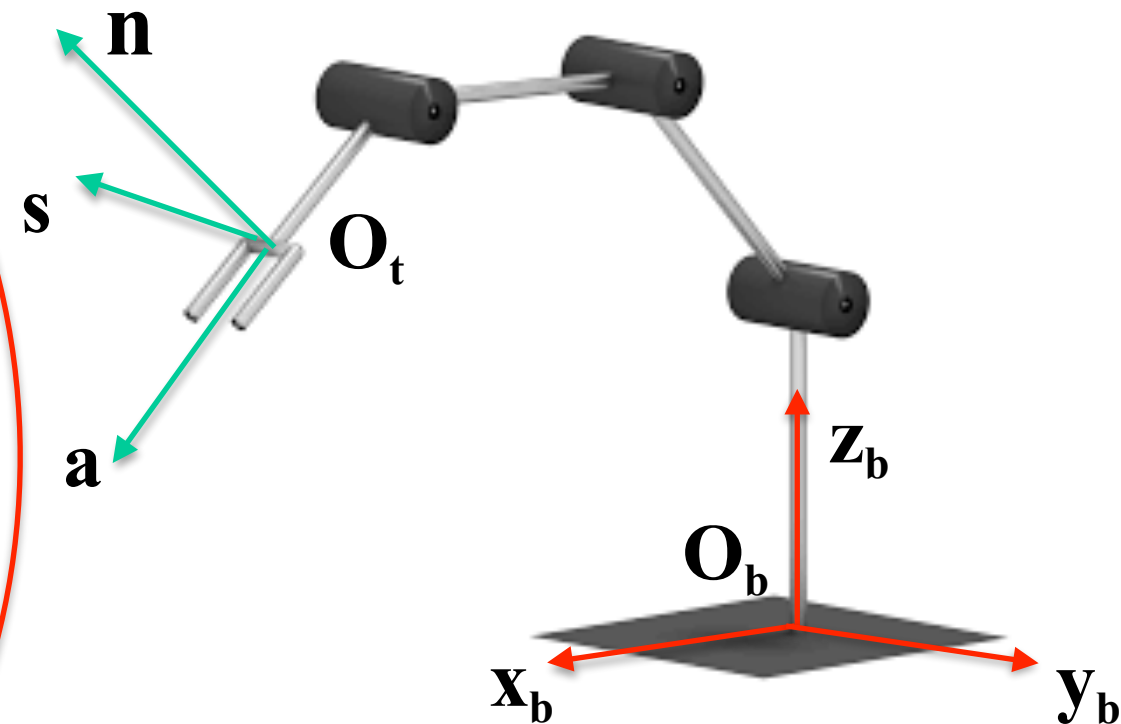
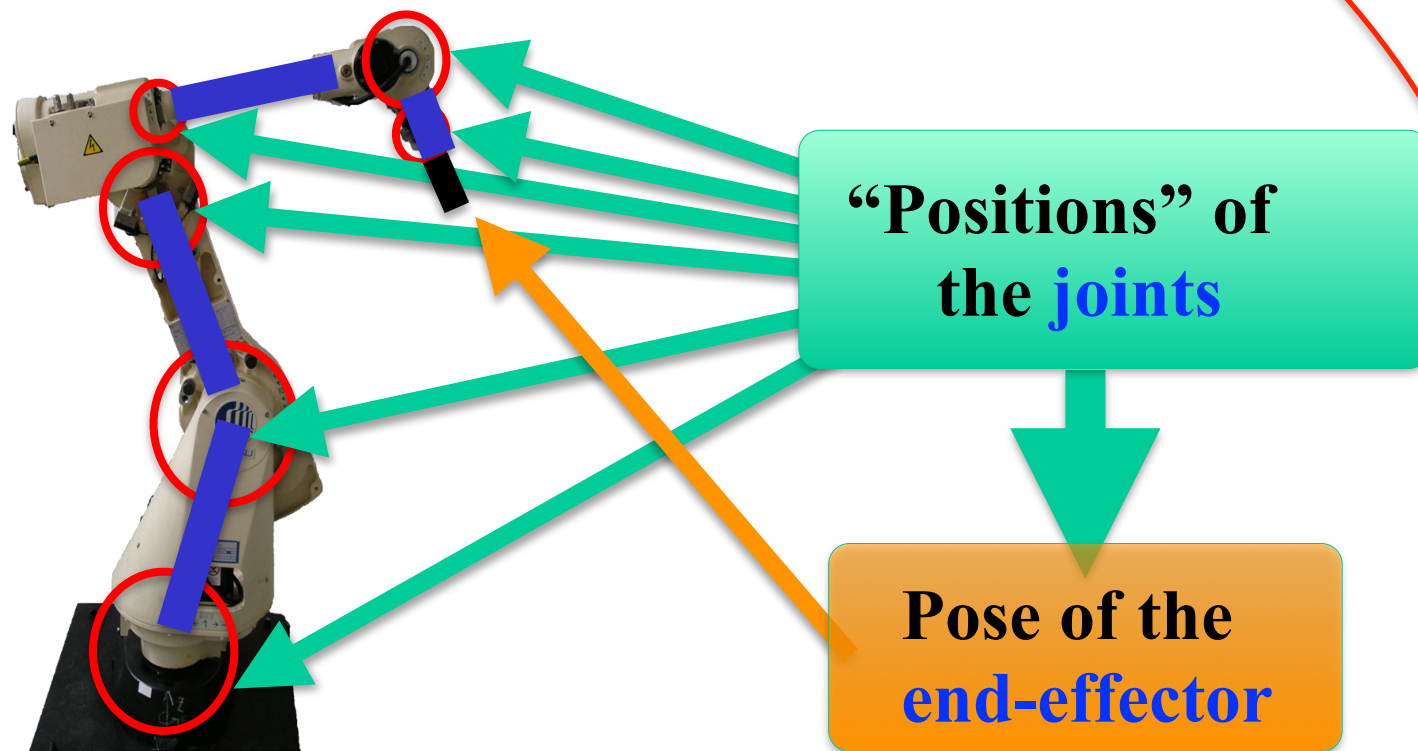
✓ Reference Frames and Rotation Matrices

Tool Reference Frame

O_t - $x_t y_t z_t$ **or** O_t - $n s a$



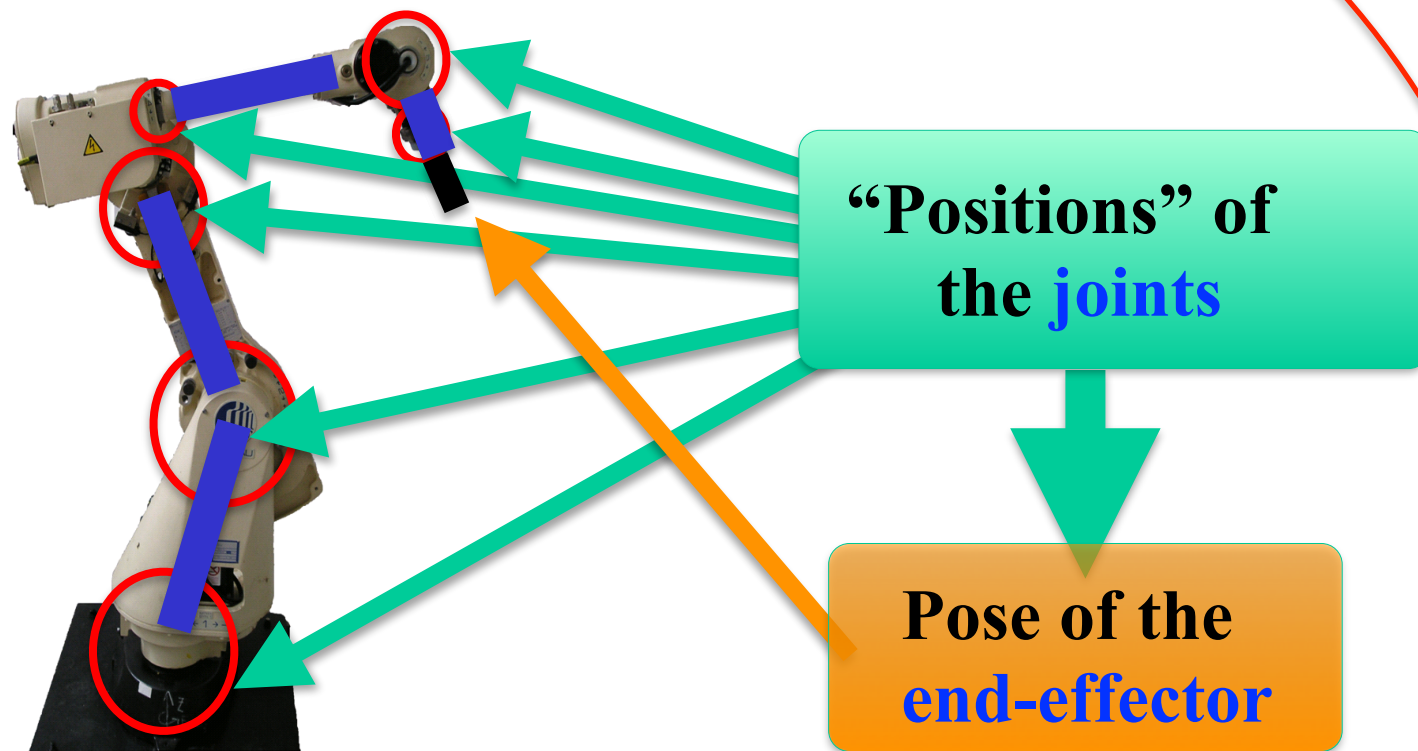
Direct Kinematics Description



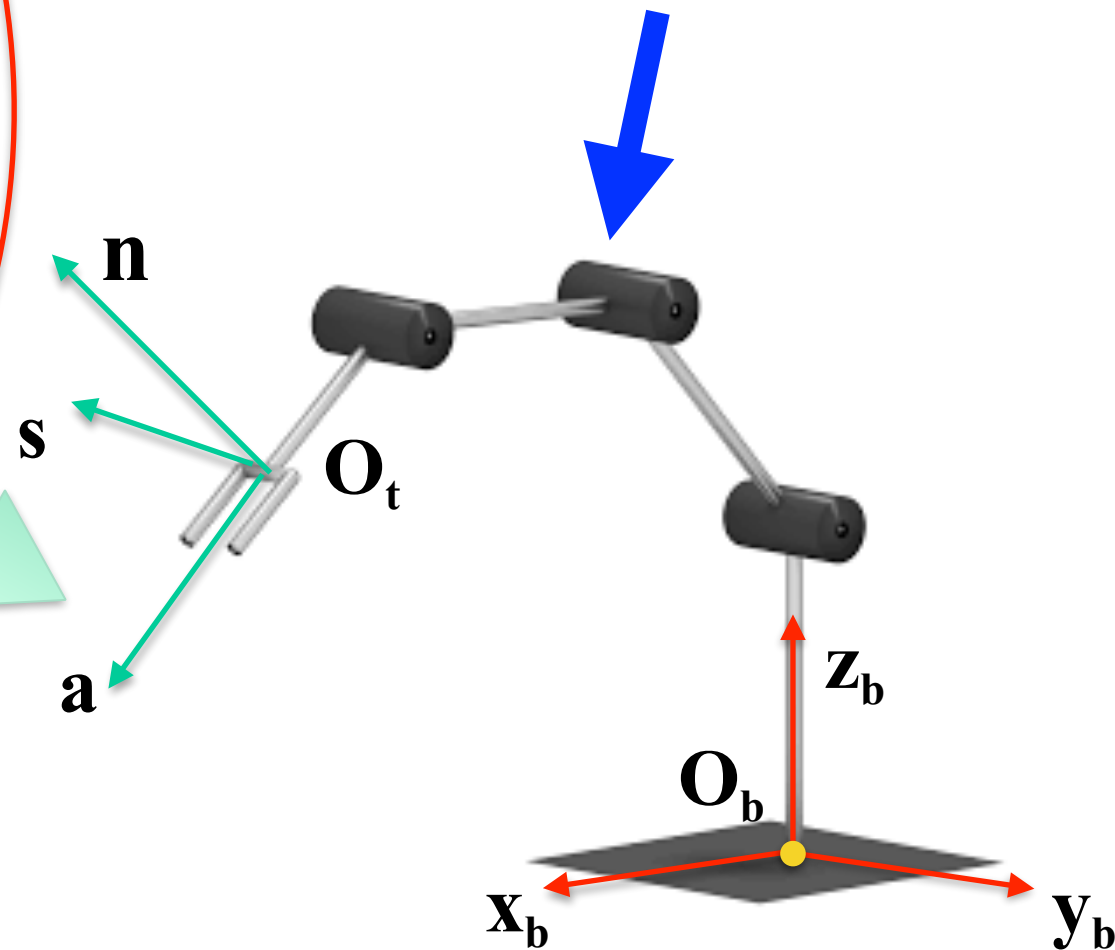
Find the pose of the Tool Reference Frame w.r.t. the Base Reference Frame

Direct Kinematics

Description: first step

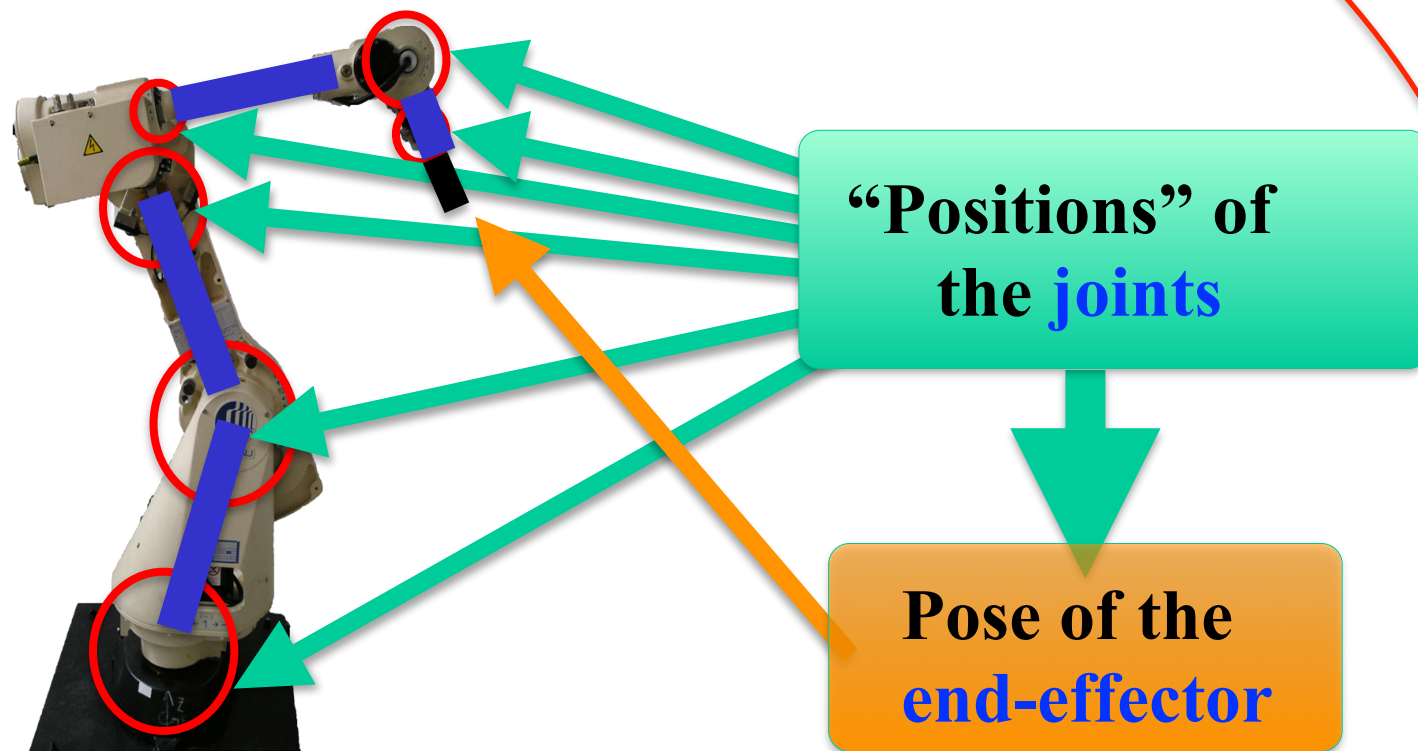


Forget about the kinematic chain

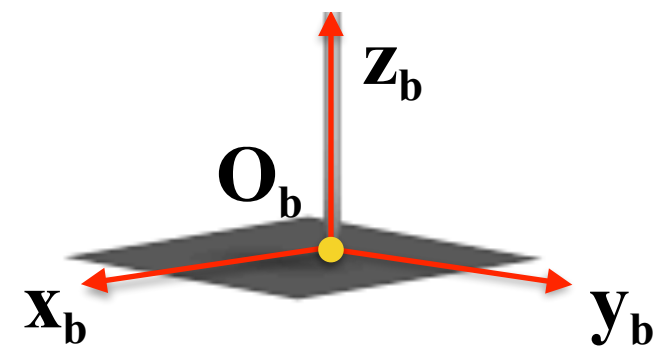
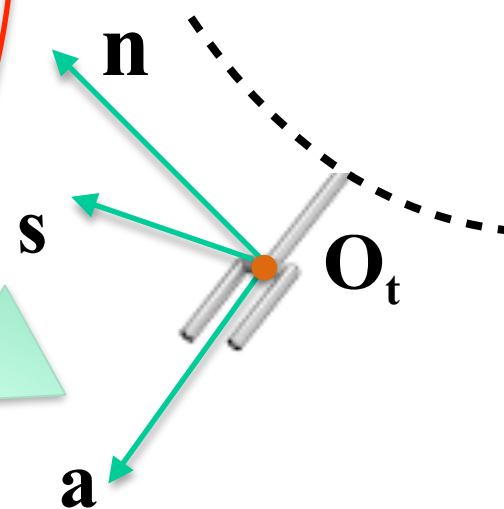


Direct Kinematics

Description: first step



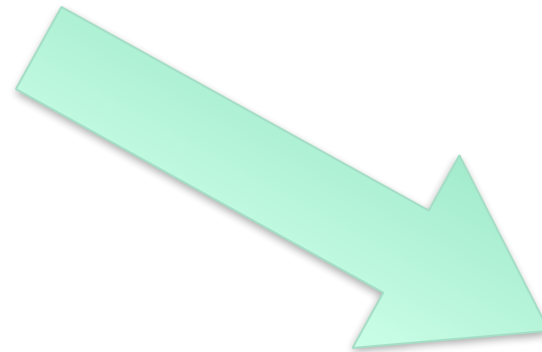
Forget about the kinematic chain



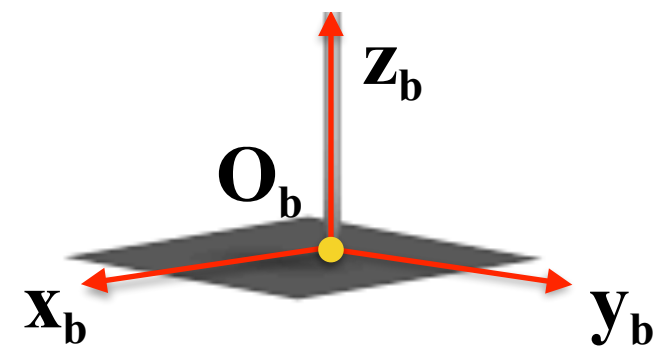
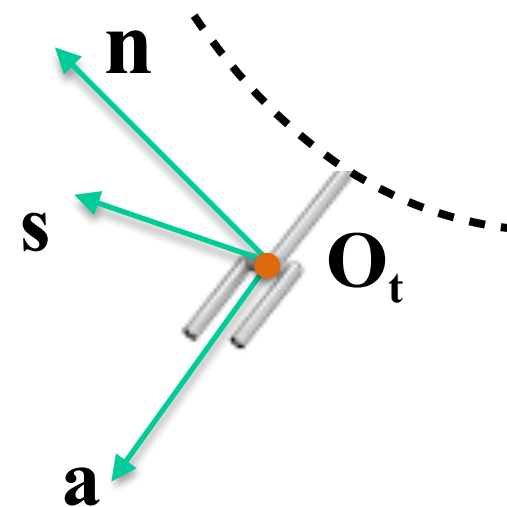
Direct Kinematics

Description: first step

The End-Effector is a rigid body in the 3D space



Forget about the kinematic chain



Direct Kinematics

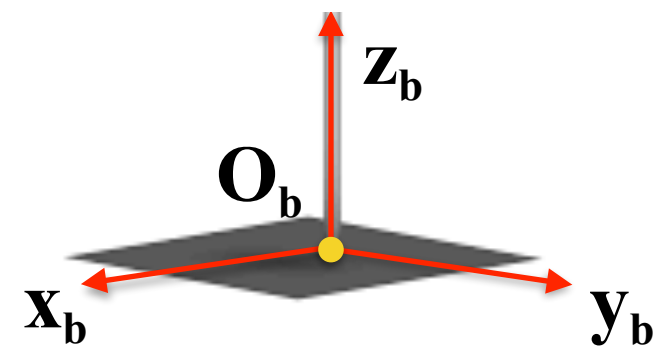
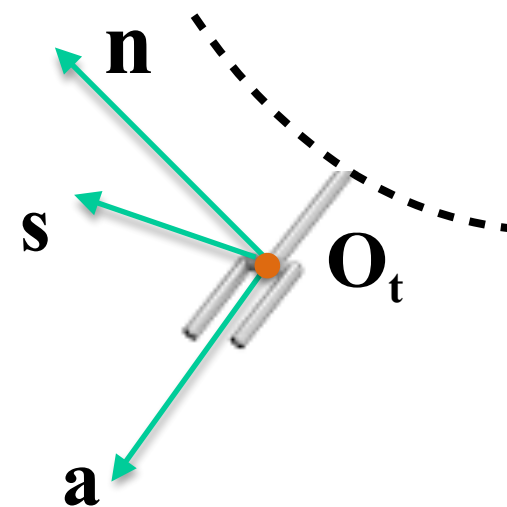
Description: first step

The End-Effector is a rigid body in the 3D space

Forget about the kinematic chain

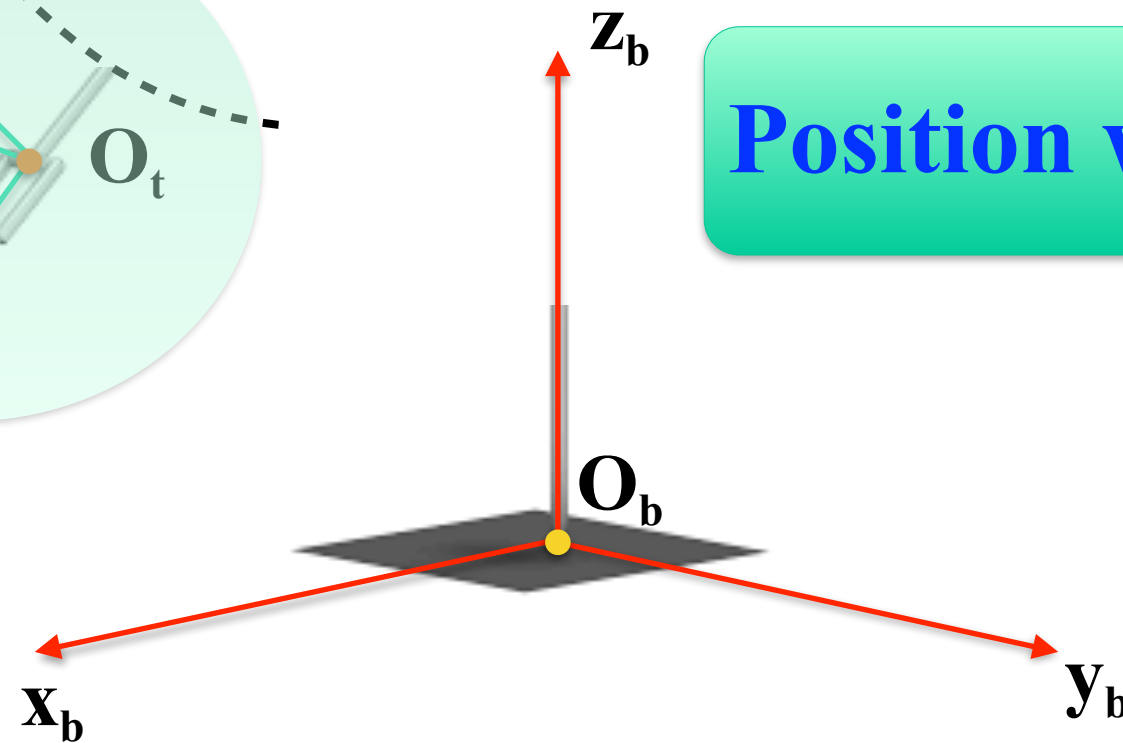
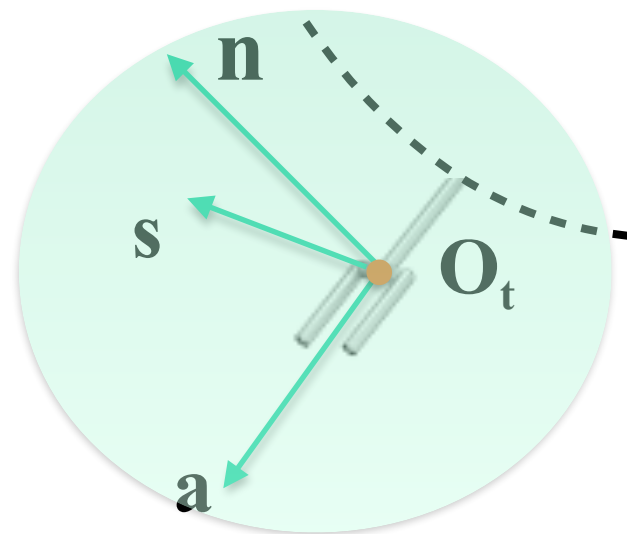
Problem to solve:

Find a description of the pose of a rigid body in w.r.t. the Base Reference Frame



“Pose” of a Rigid Object in 3D Space

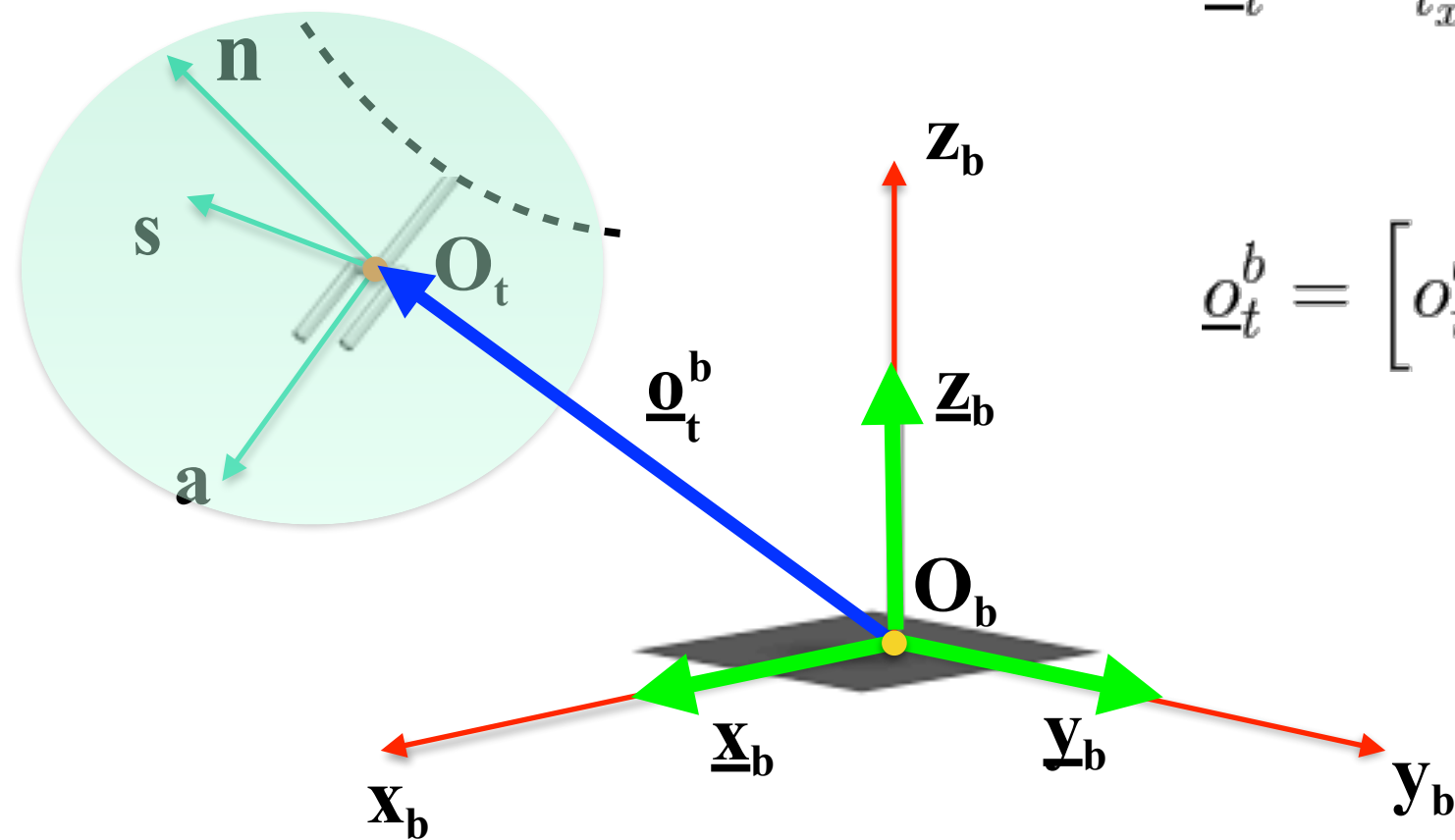
Find a description of the Pose of a Rigid Object in 3D Space means



Position w.r.t. O_b - x_b y_b z_b

Orientation w.r.t.
 O_b - x_b y_b z_b

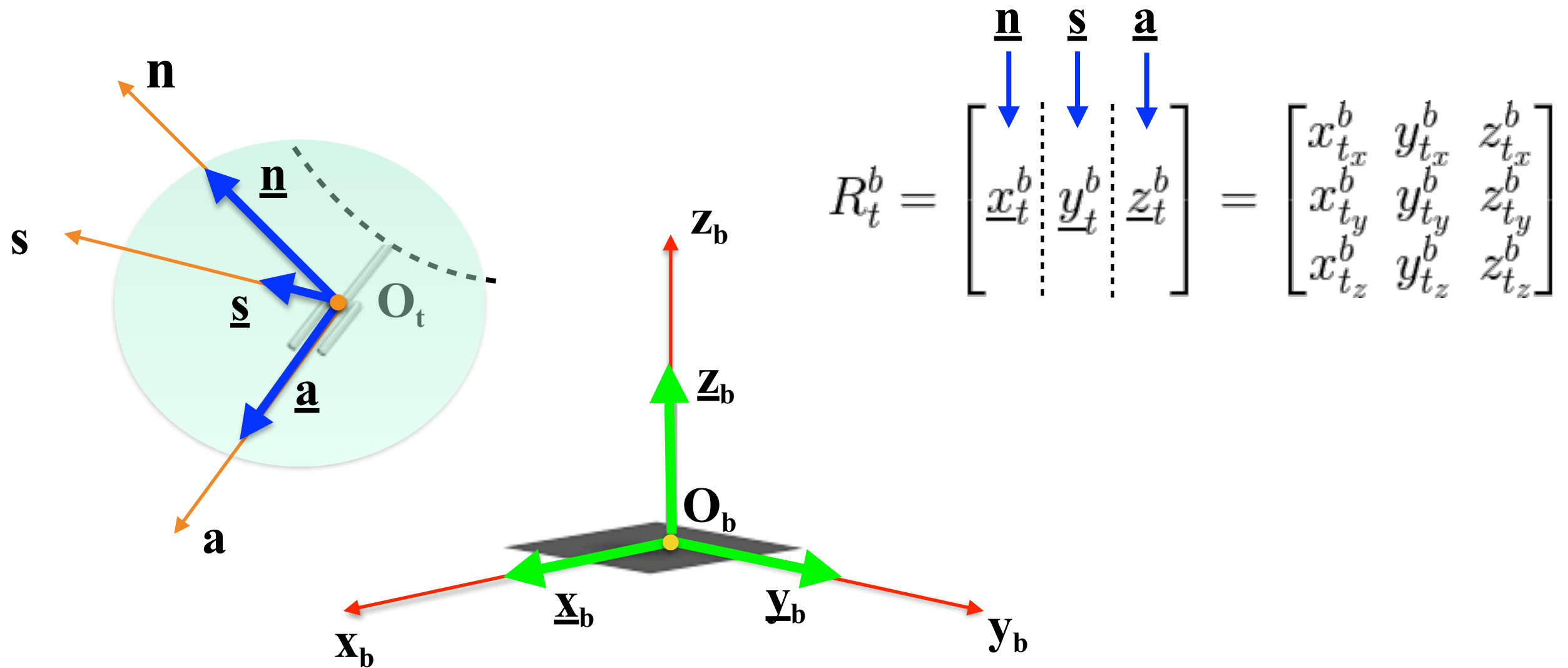
Position of a Rigid Object in 3D Space



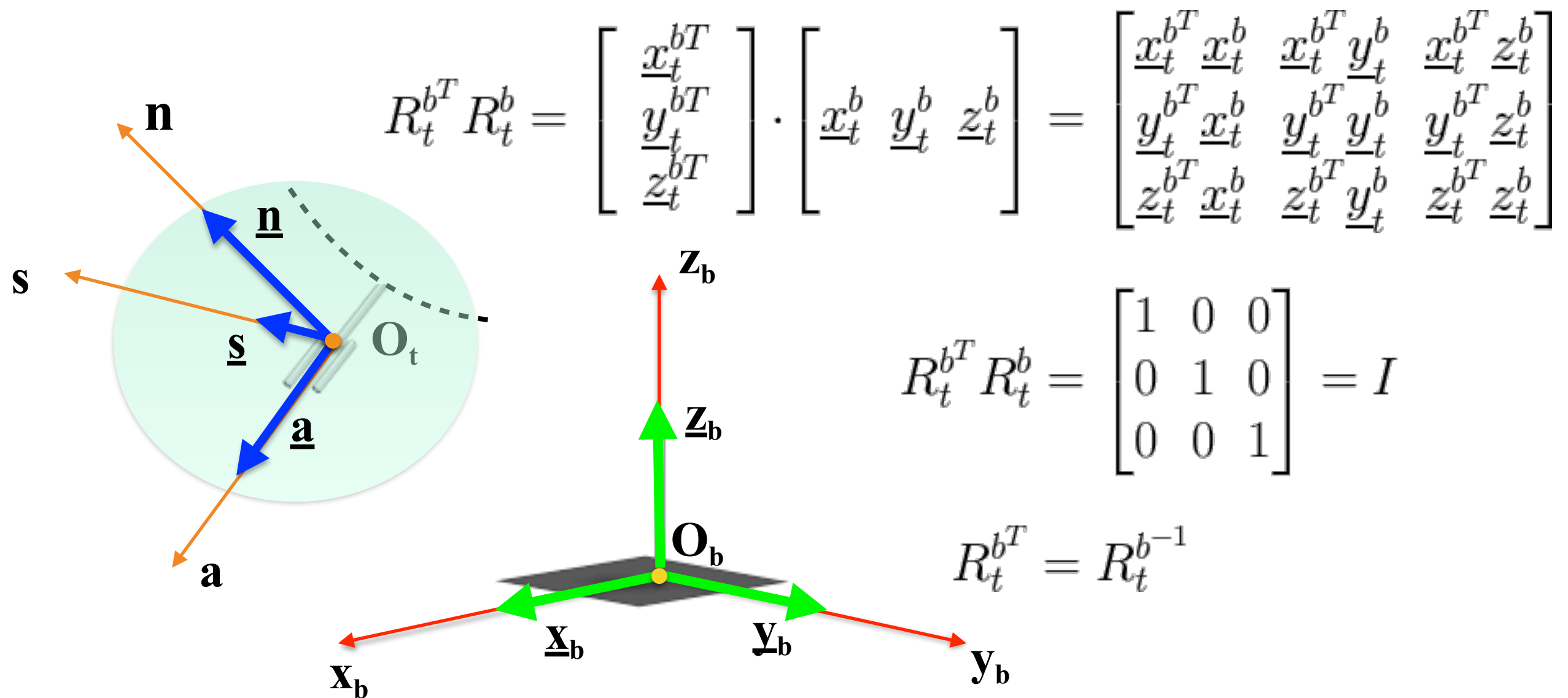
$$\underline{o}_t^b = o_{t_x}^b \underline{x}_b + o_{t_y}^b \underline{y}_b + o_{t_z}^b \underline{z}_b$$

$$\underline{o}_t^b = \begin{bmatrix} o_{t_x}^b & o_{t_y}^b & o_{t_z}^b \end{bmatrix}^T$$

Orientation of a Rigid Object in 3D Space



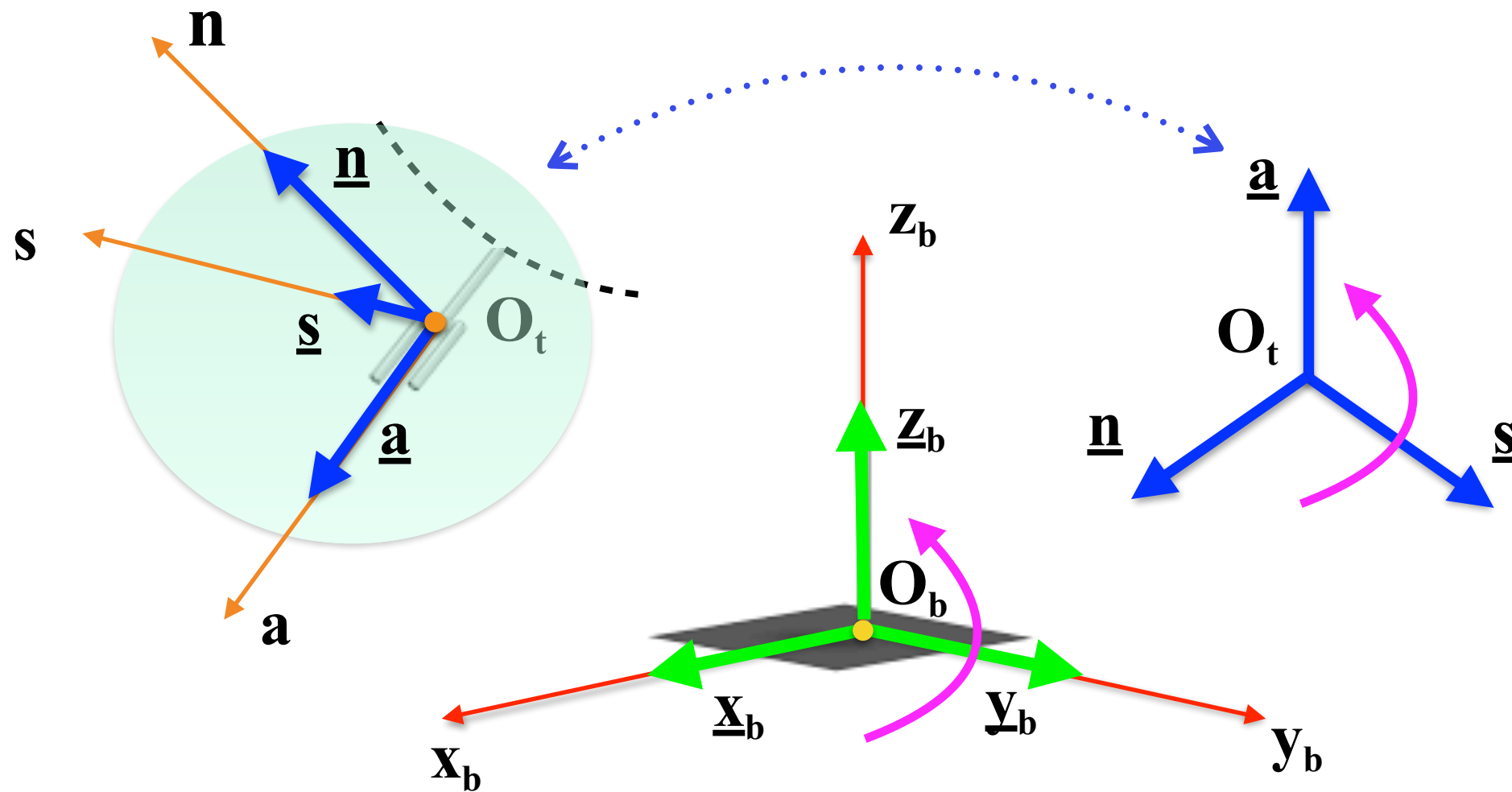
Rotation Matrices are Orthogonal Matrices



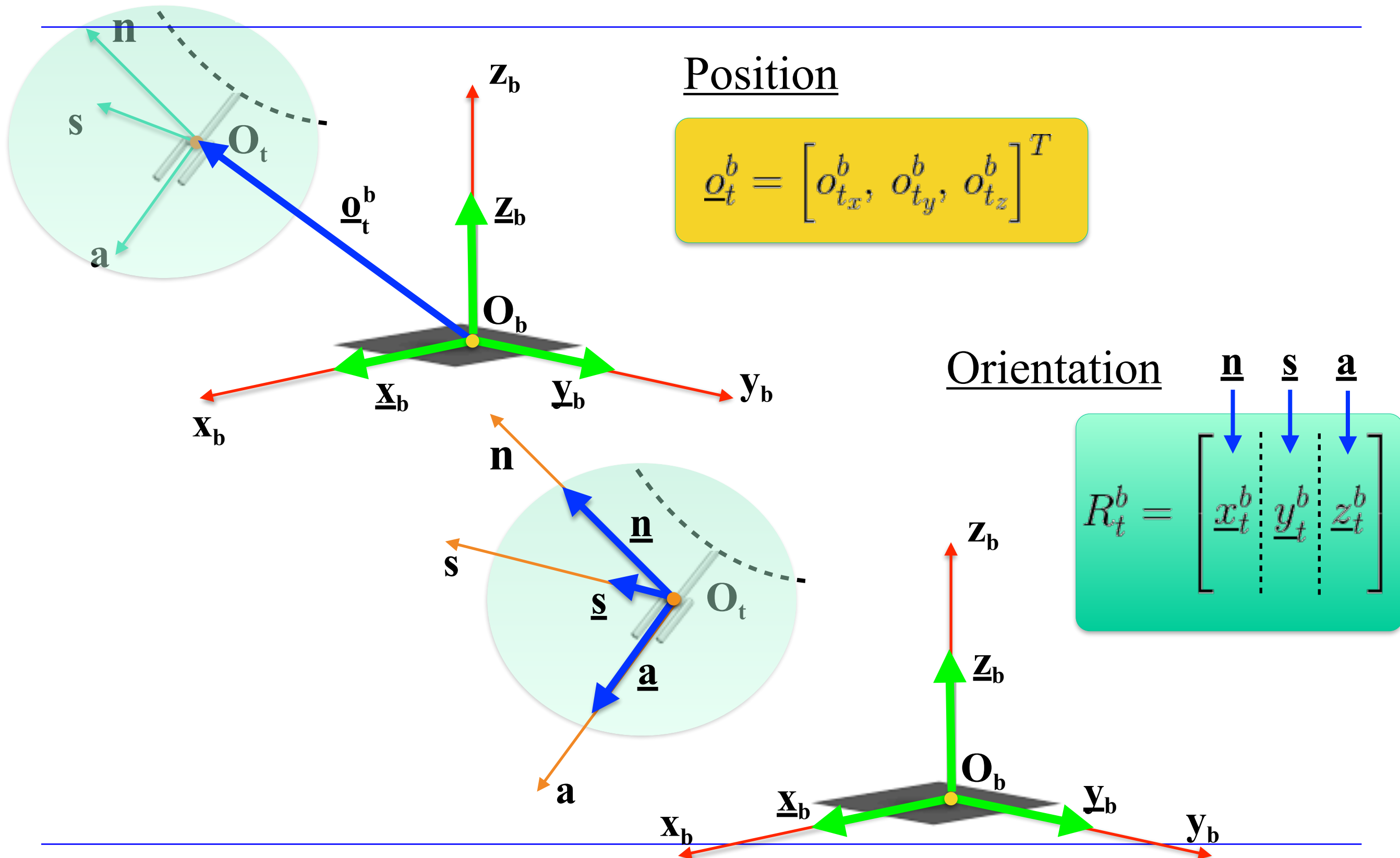
Rotation matrices are orthogonal square matrices ($\det = 1$ or -1 depending on the direction of rotation to name the axes from x to z : counterclockwise or clockwise)

Rotation Matrices Determinant

In the course: determinant is equal to 1 since the reference frames we use have axes taken in the counterclockwise rotation sense



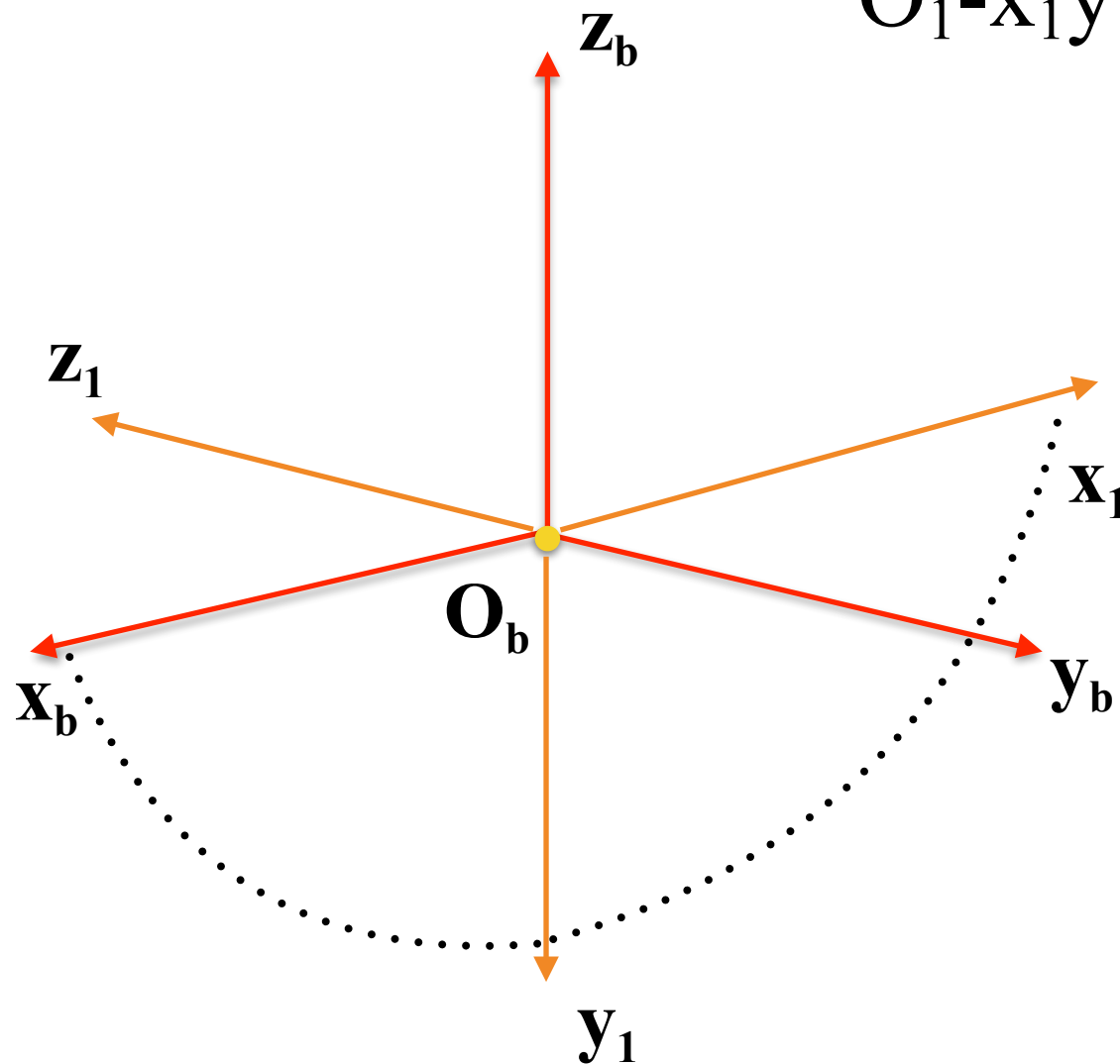
Summarizing



Examples of Determination of Simple Rotation Matrices

Ex. 1:

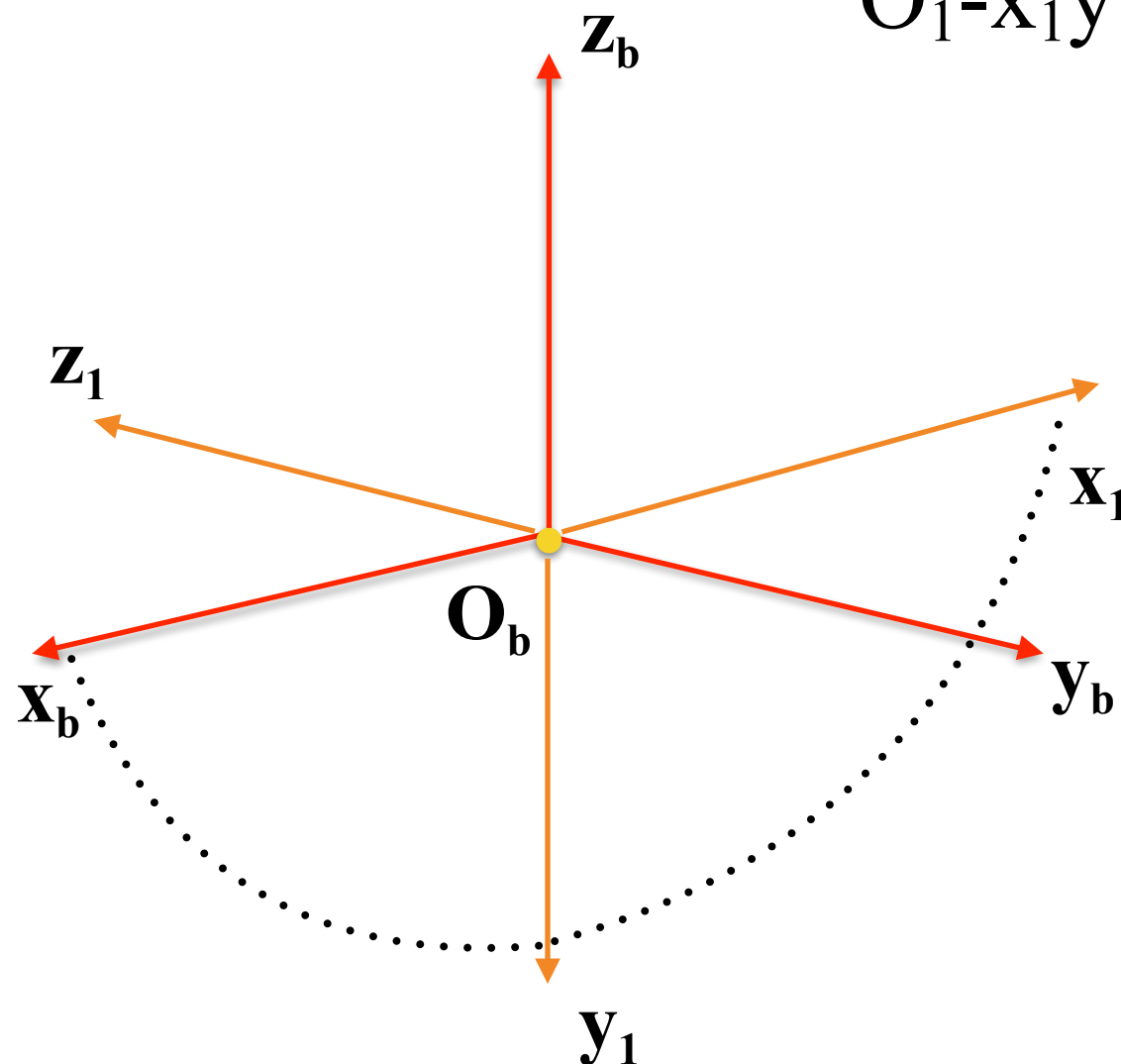
Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$



Examples of Determination of Simple Rotation Matrices

Ex. 1:

Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$

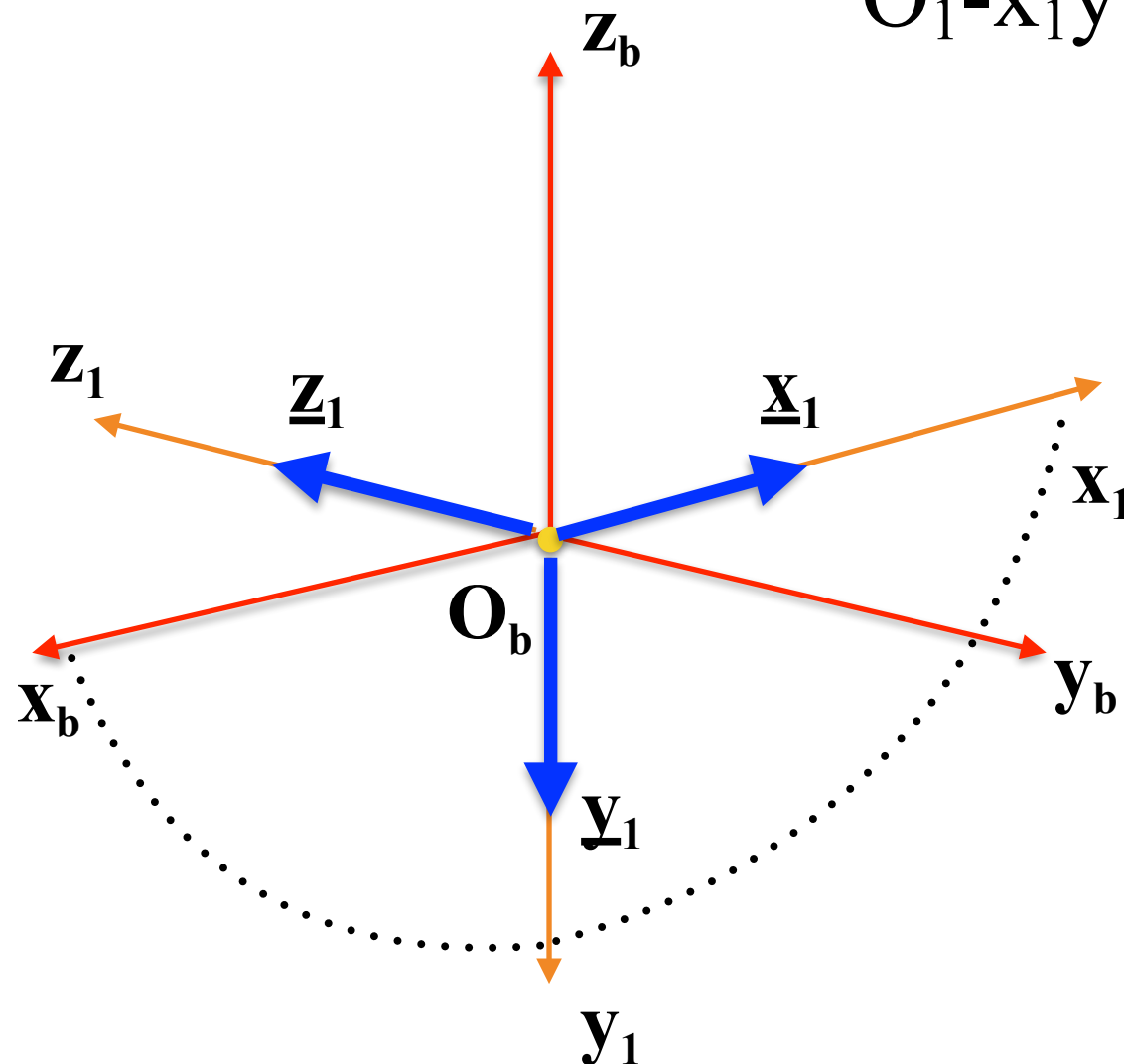


Solution:

Examples of Determination of Simple Rotation Matrices

Ex. 1:

Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$

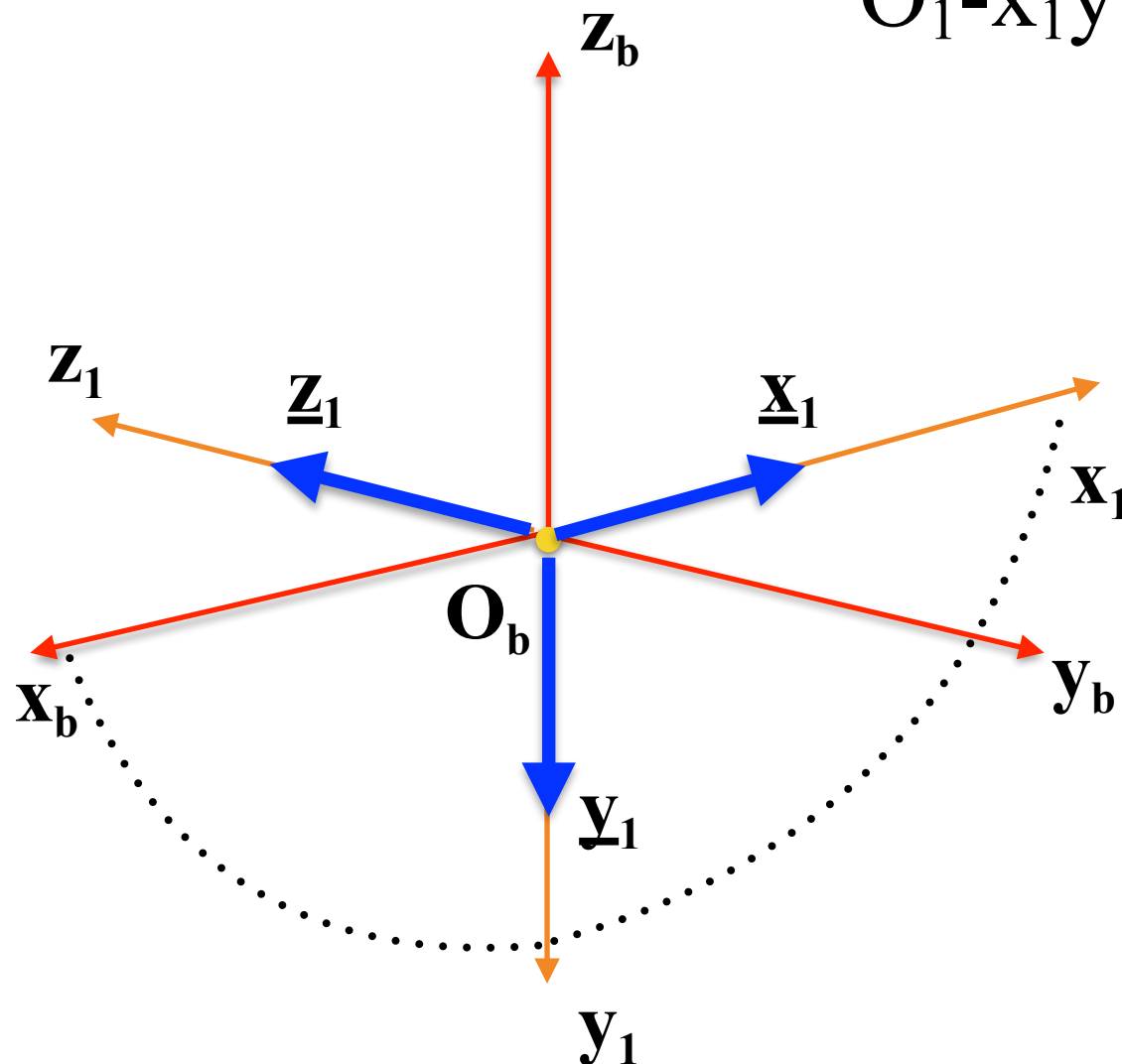


Solution:

Examples of Determination of Simple Rotation Matrices

Ex. 1:

Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$



Solution:

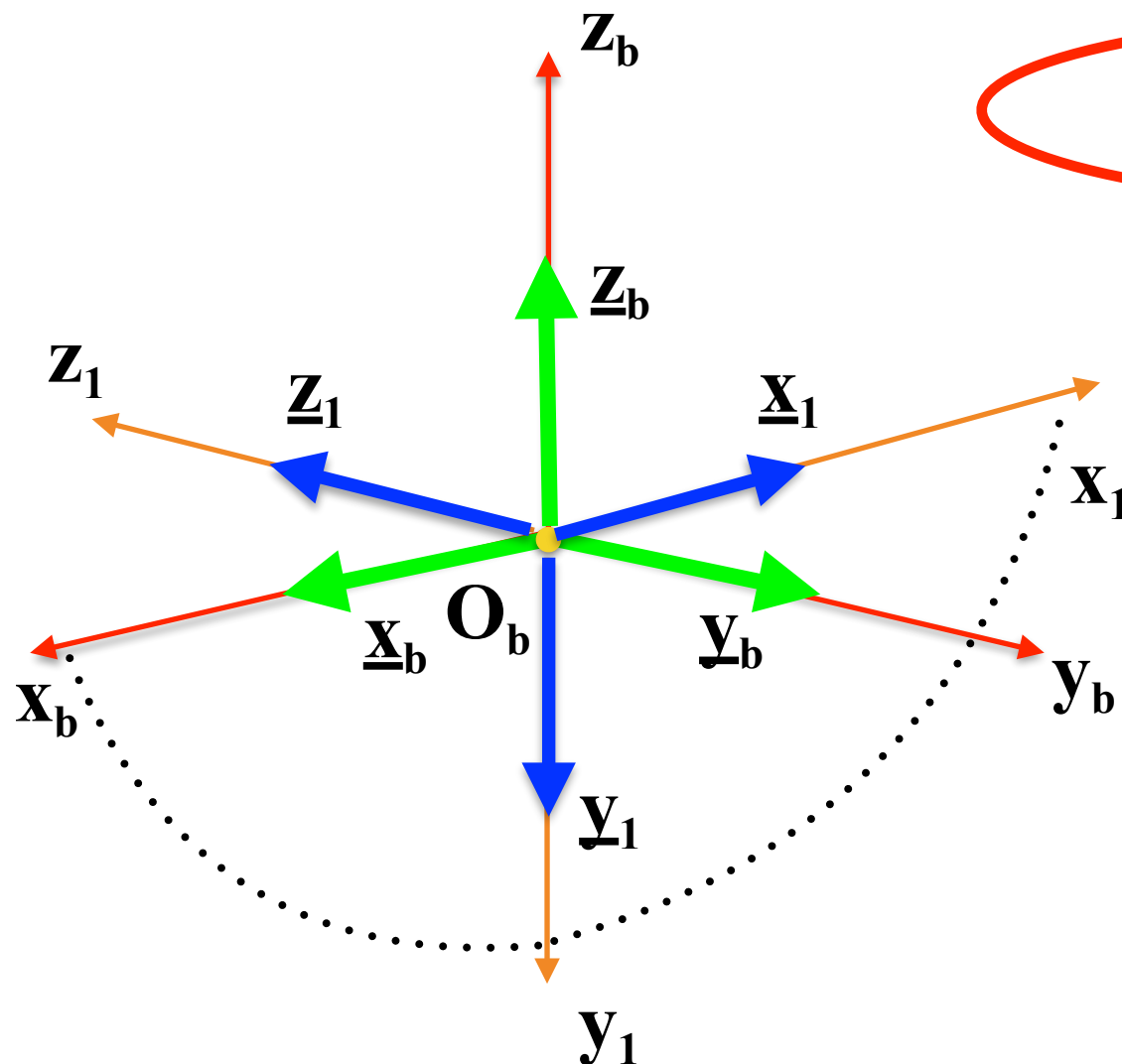
$$R_1^b = \begin{bmatrix} x_1^b & y_1^b & z_1^b \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Examples of Determination of Simple Rotation Matrices

Ex. 1:

Note that:

$$\underline{x}_1^b = -1 \underline{x}_b + 0 \underline{y}_b + 0 \underline{z}_b$$



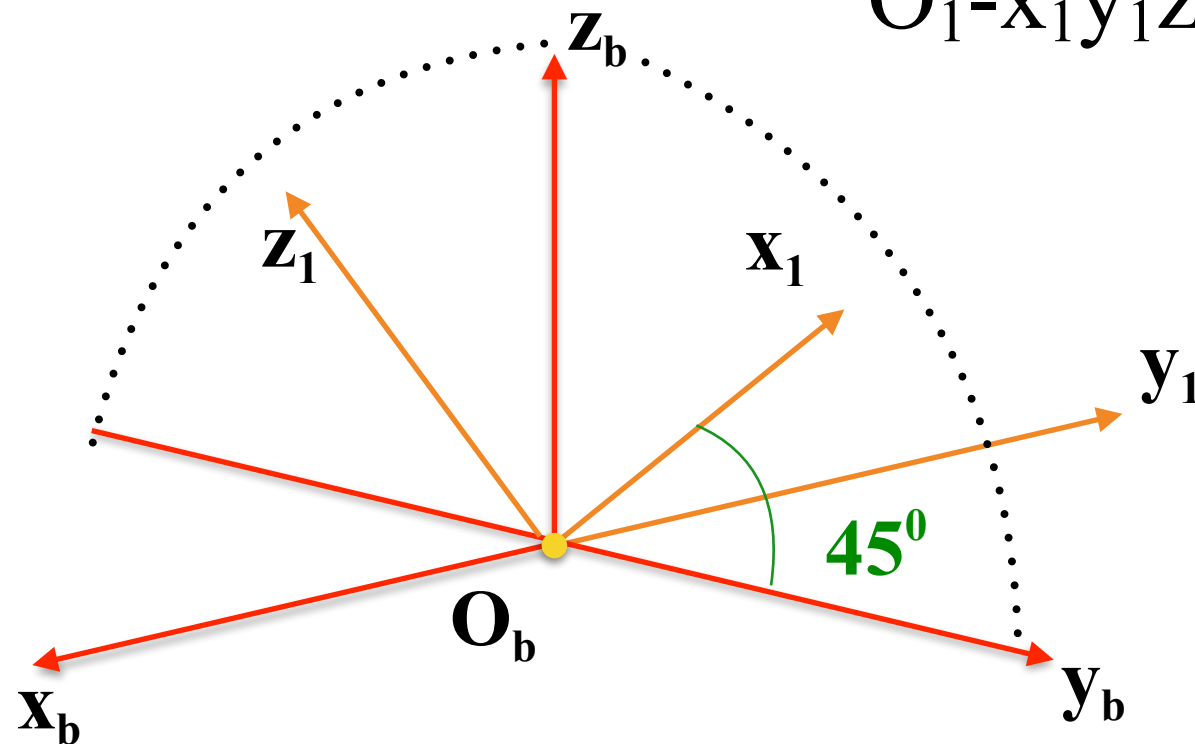
Solution:

$$R_1^b = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Examples of Determination of Simple Rotation Matrices

Ex. 2:

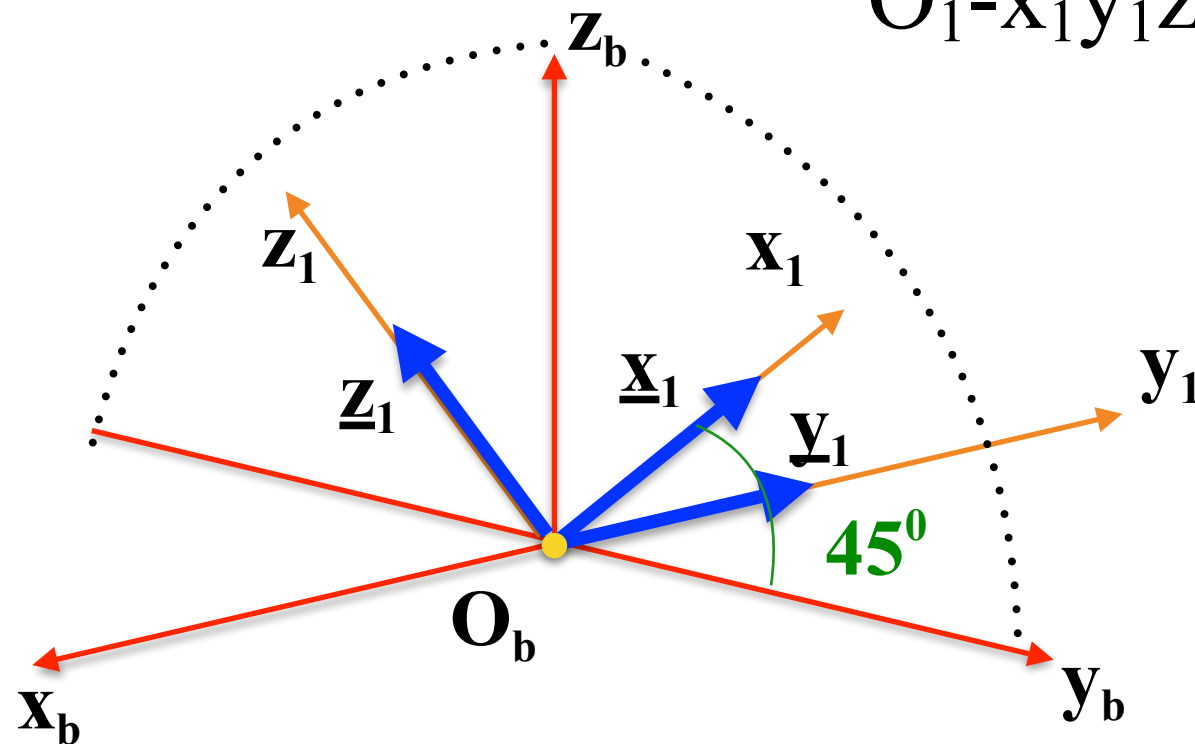
Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$



Examples of Determination of Simple Rotation Matrices

Ex. 2:

Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$

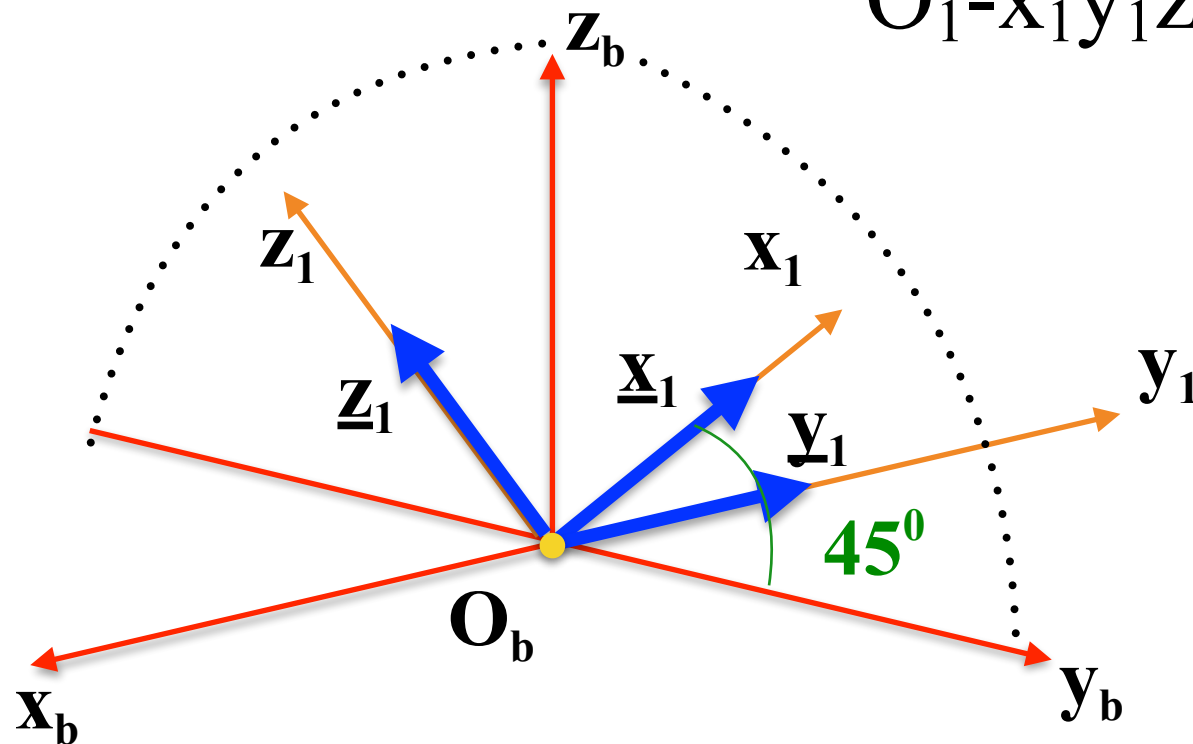


Solution:

Examples of Determination of Simple Rotation Matrices

Ex. 2:

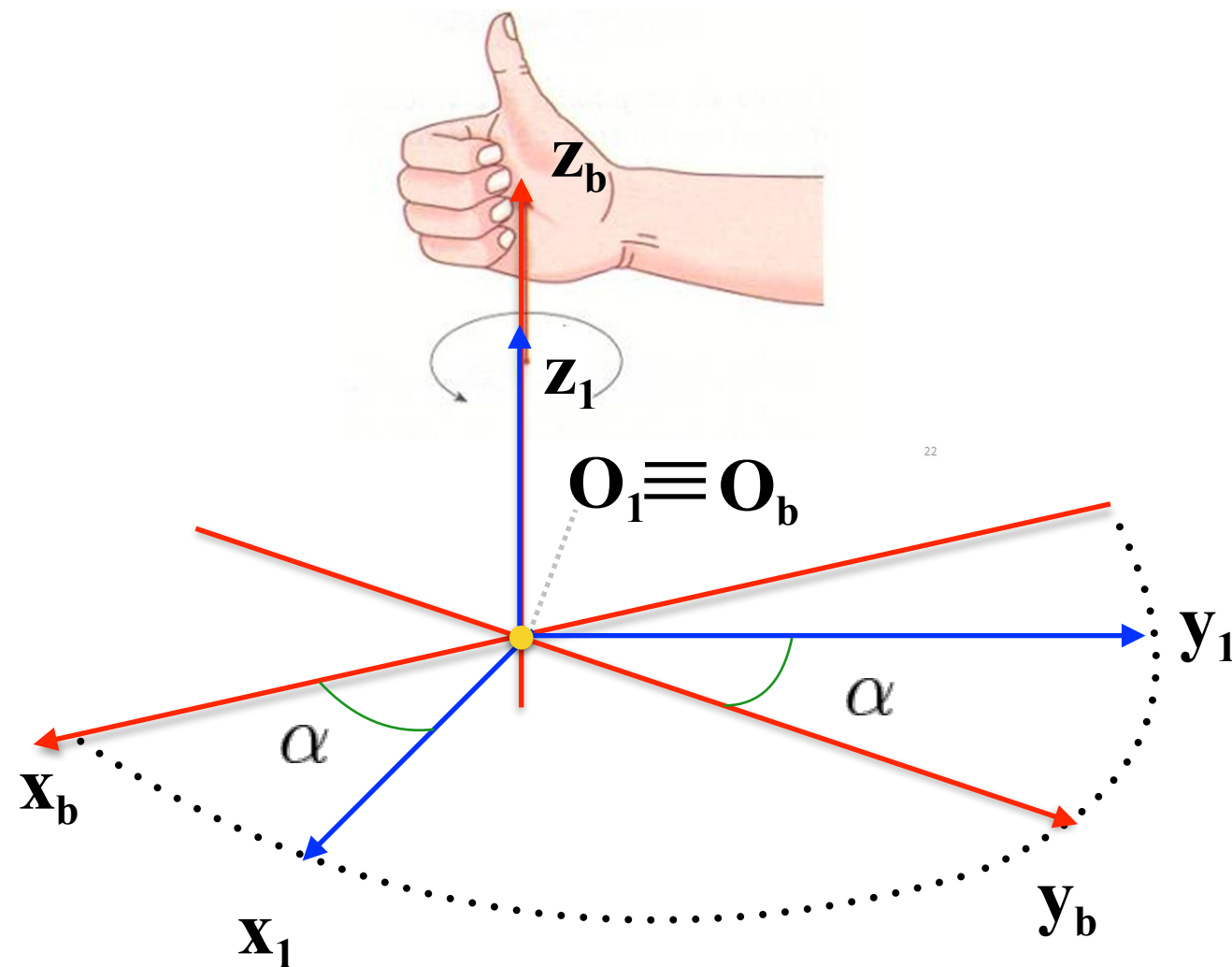
Find the Rotation Matrix describing $O_1-x_1y_1z_1$ w.r.t. $O_b-x_by_bz_b$



Solution:

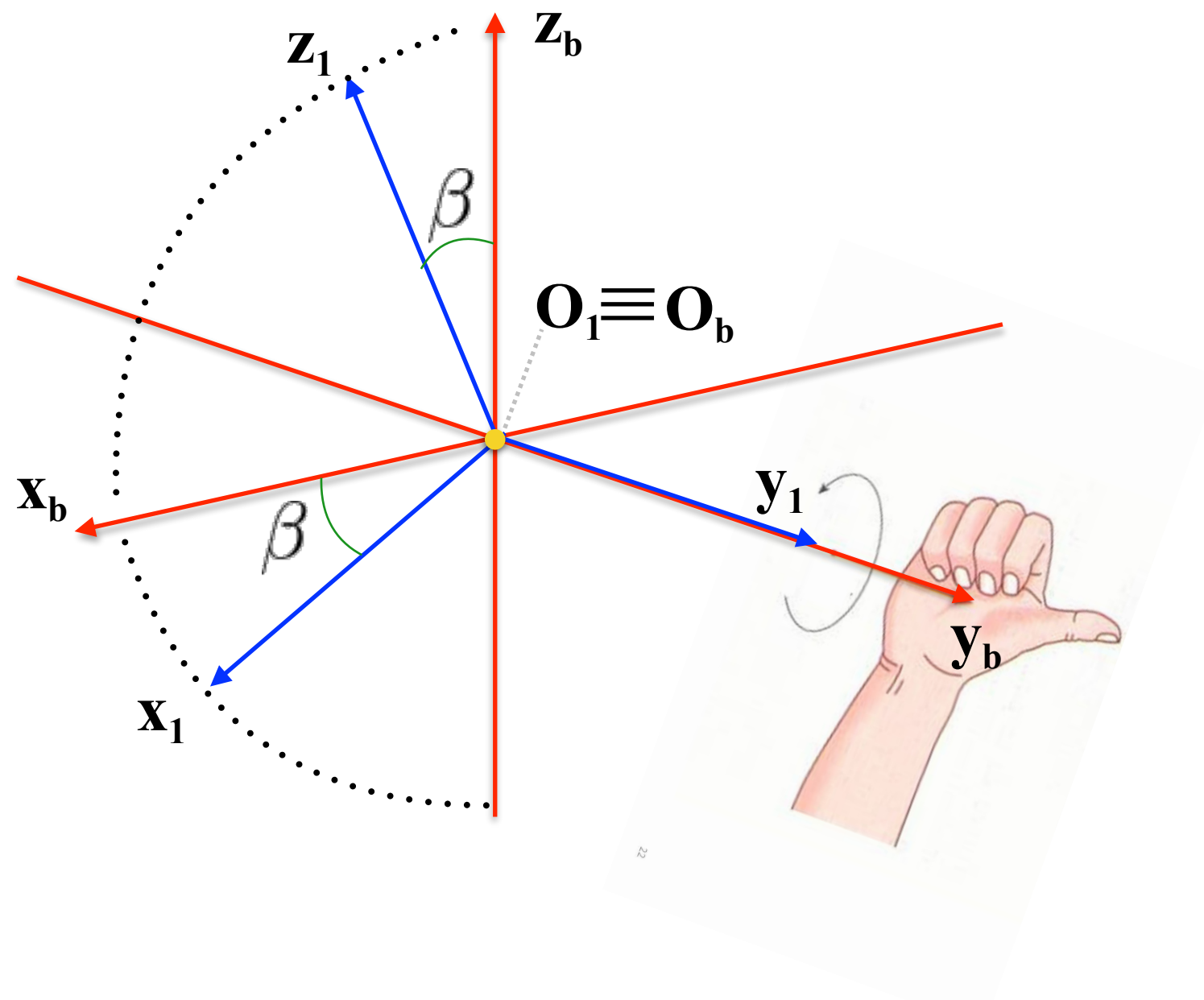
$$R_1^b = \begin{bmatrix} x_1^b & y_1^b & z_1^b \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

Elementary Rotations (counterclockwise)



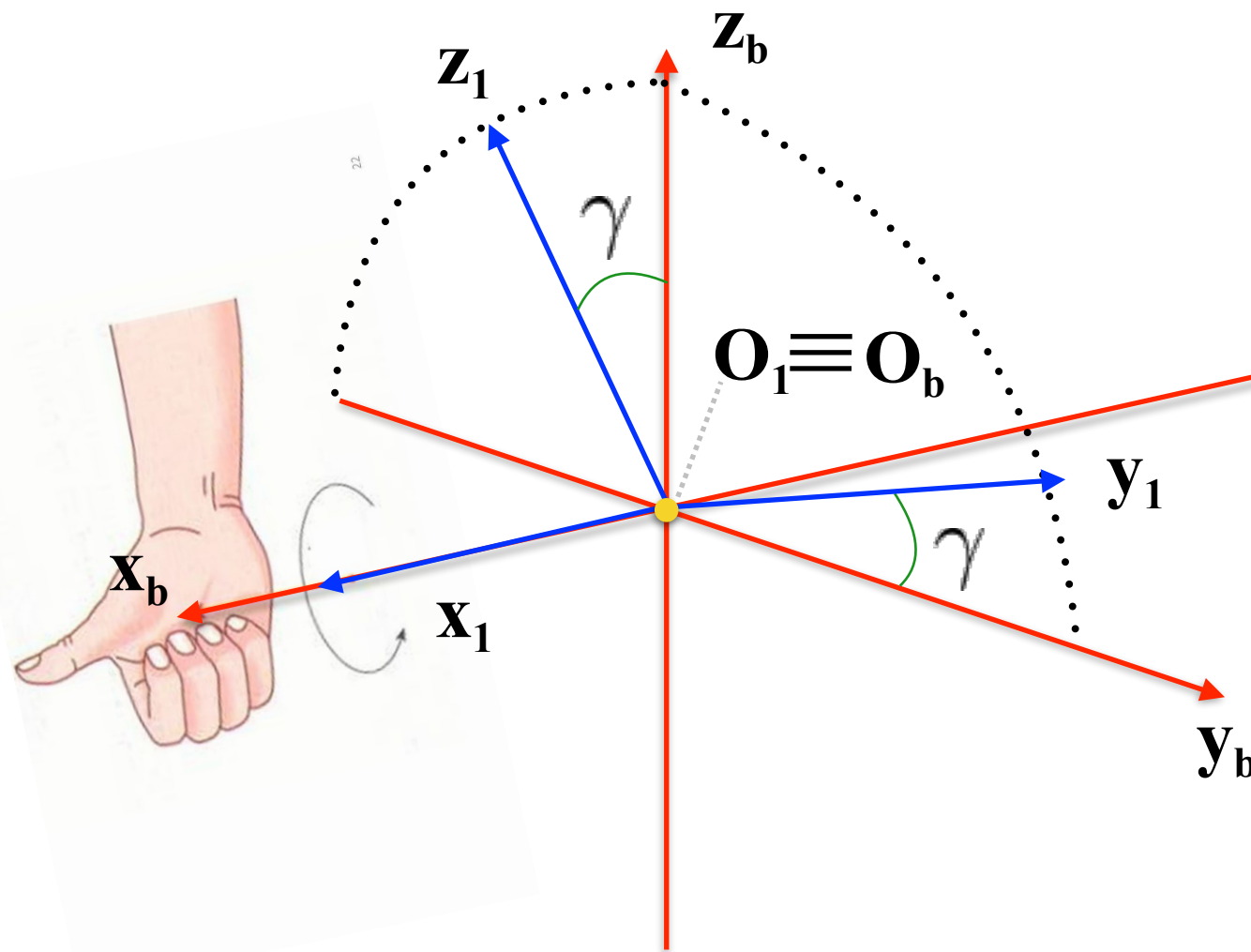
$$R_1^b = R_z(\alpha)$$

Elementary Rotations (counterclockwise)



$$R_1^b = R_y(\beta)$$

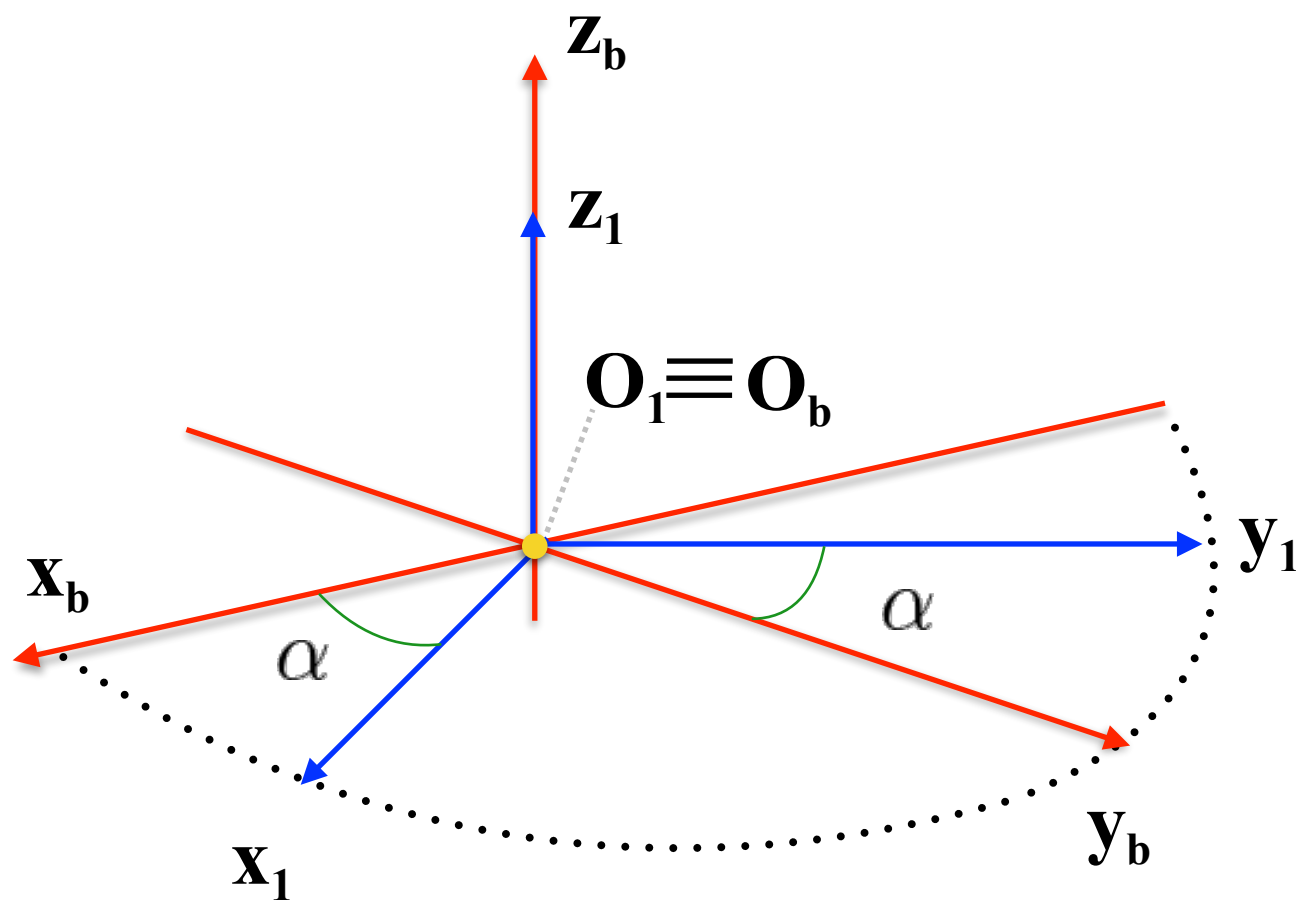
Elementary Rotations (counterclockwise)



$$R_1^b = R_x(\gamma)$$

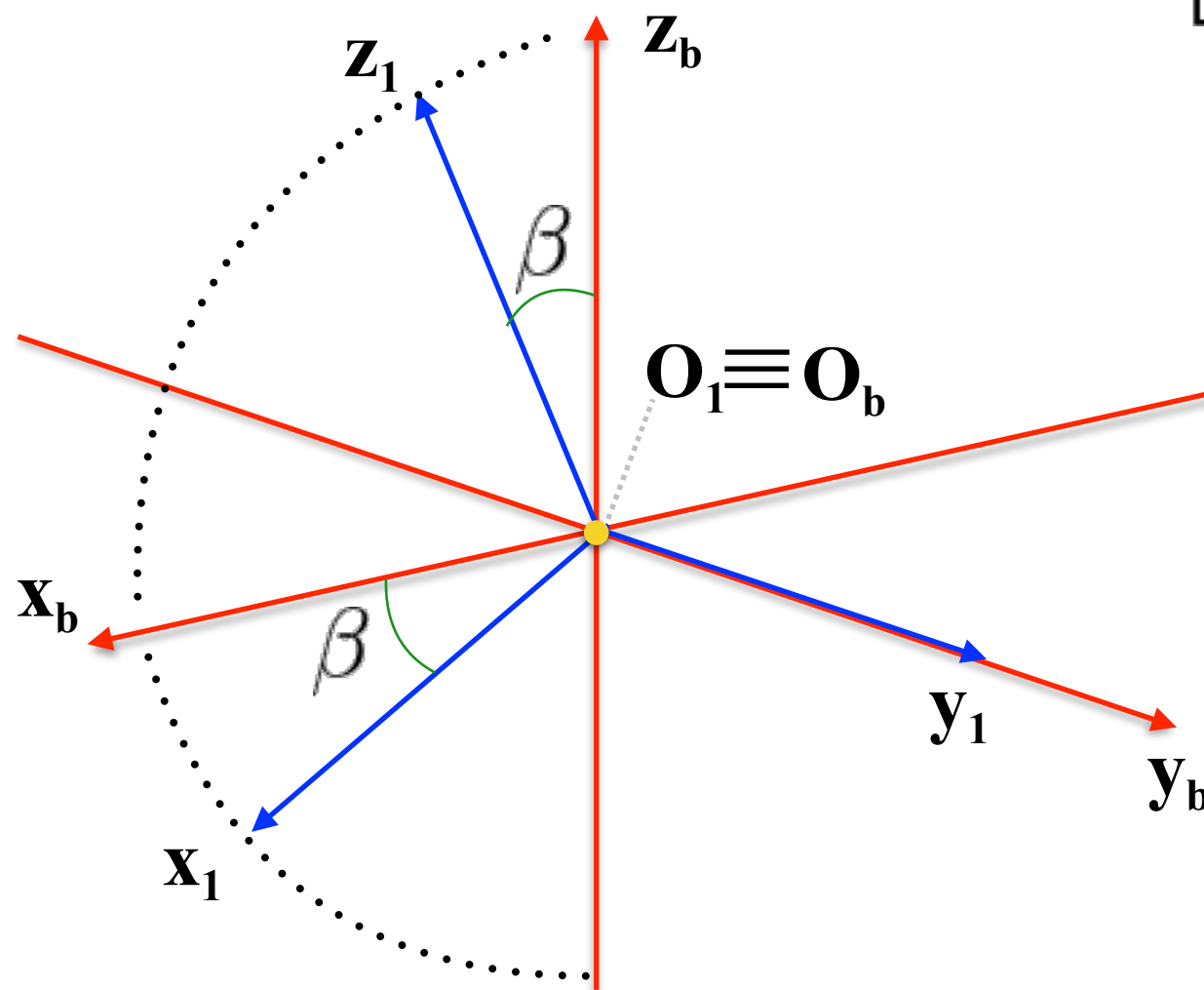
Elementary Rotations (counterclockwise)

$$R_1^b = R_z(\alpha) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



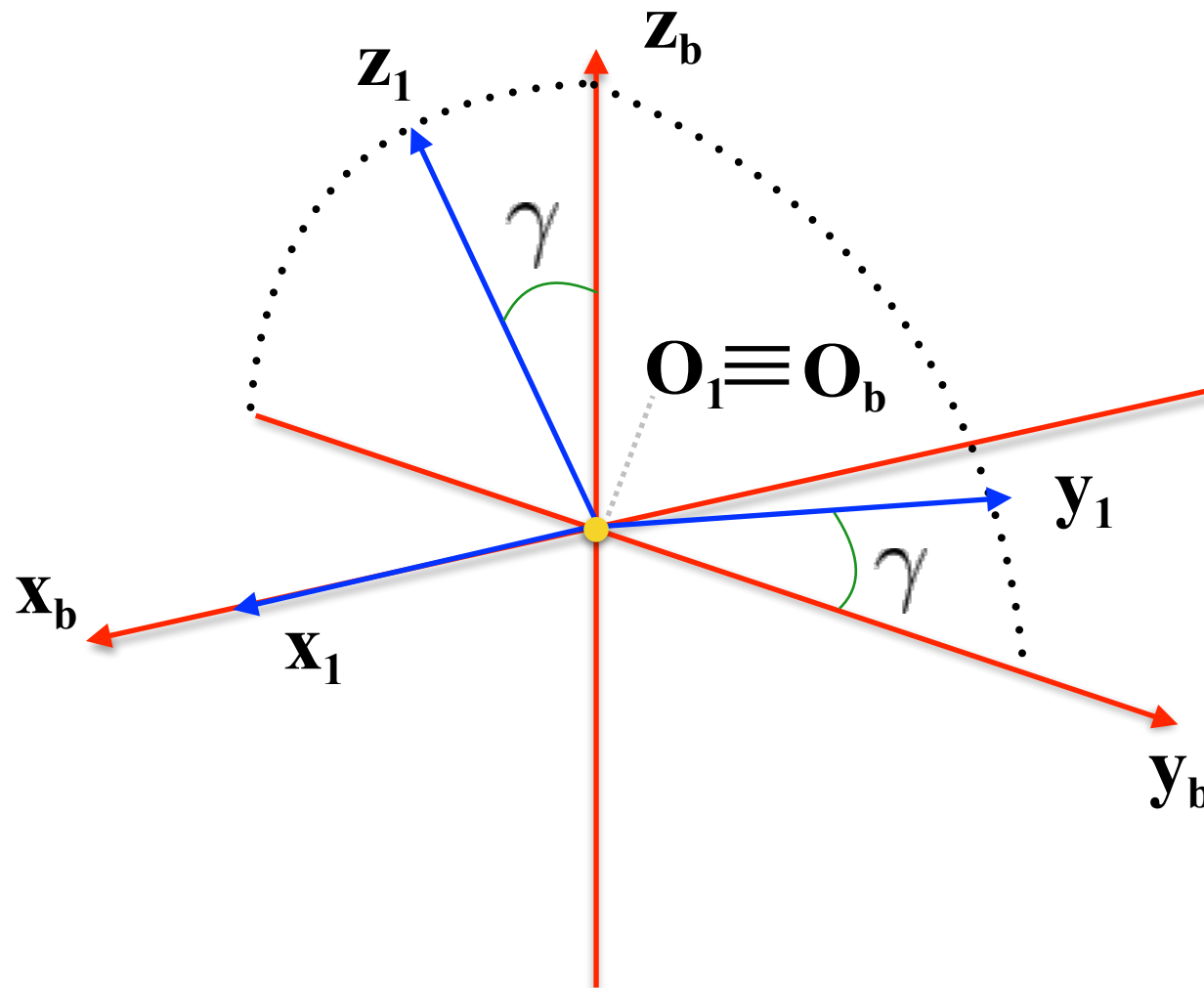
Elementary Rotations (counterclockwise)

$$R_1^b = R_y(\beta) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$



Elementary Rotations (counterclockwise)

$$R_1^b = R_x(\gamma) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$



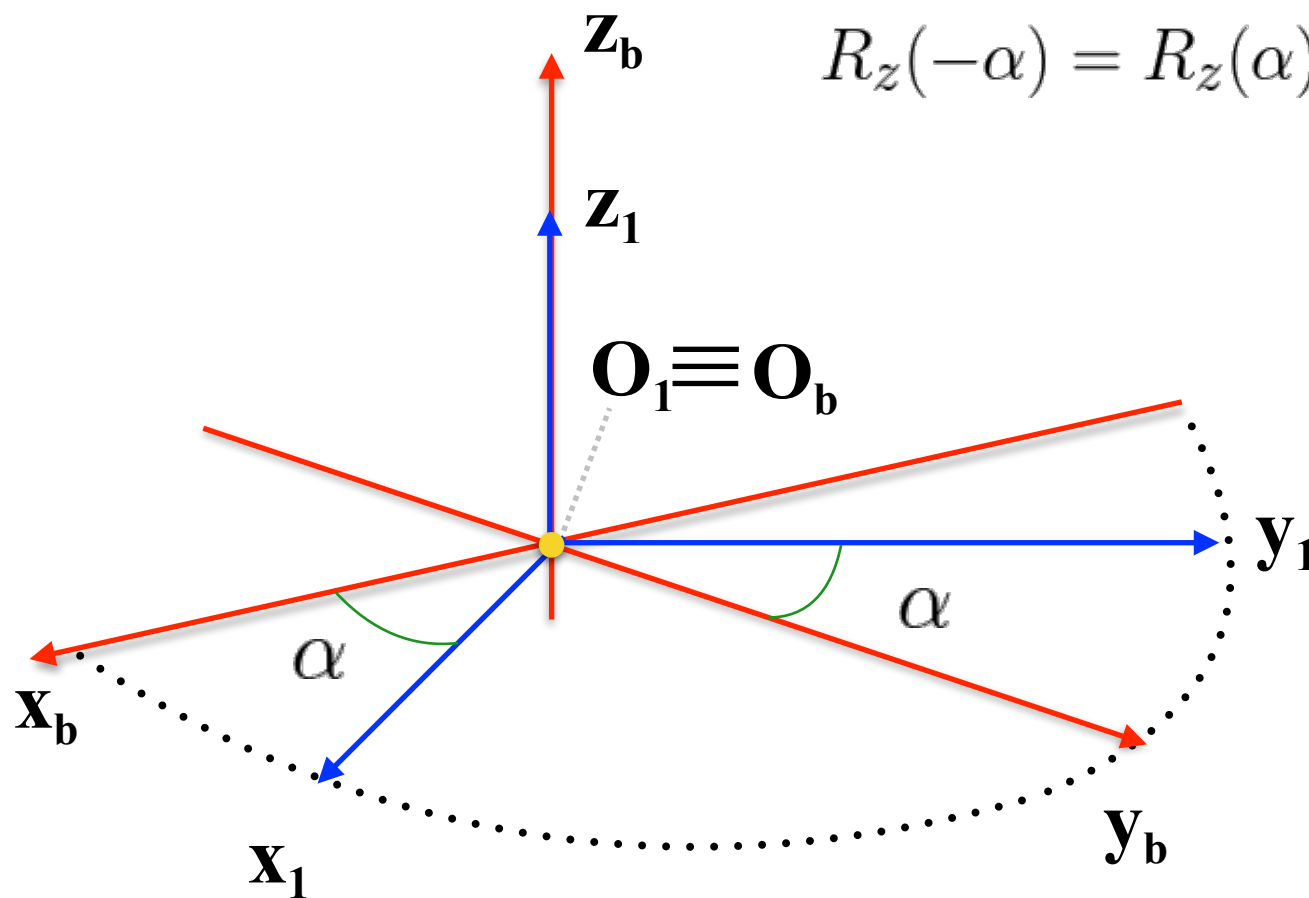
Elementary Rotations (clockwise)



$$R_1^b = R_z(\alpha) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1^b = R_z(-\alpha) = \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) & 0 \\ \sin(-\alpha) & \cos(-\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

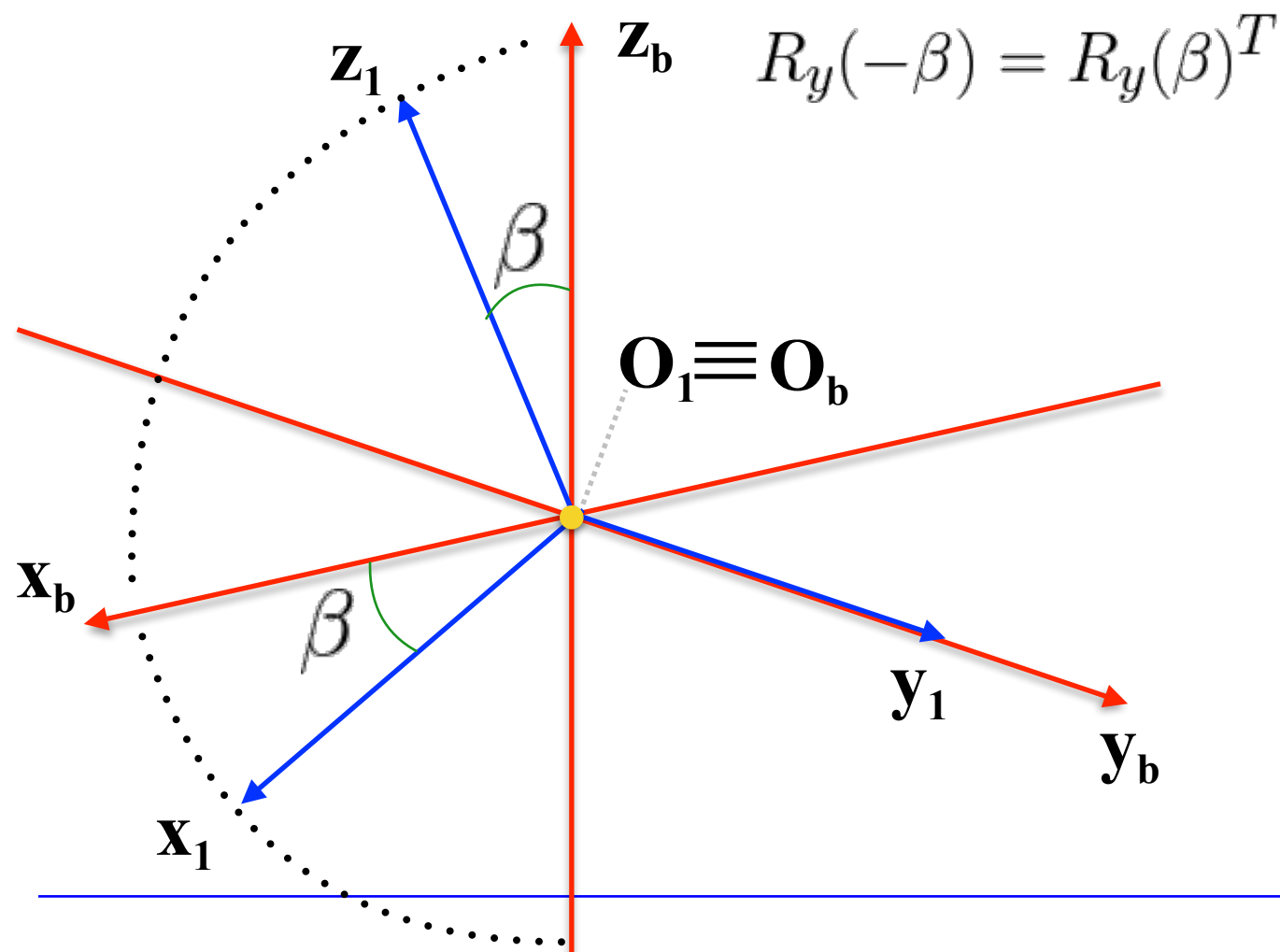
$$R_z(-\alpha) = R_z(\alpha)^T$$



Elementary Rotations (clockwise)

$$R_1^b = R_y(\beta) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$R_1^b = R_y(-\beta) = \begin{bmatrix} \cos(-\beta) & 0 & \sin(-\beta) \\ 0 & 1 & 0 \\ -\sin(-\beta) & 0 & \cos(-\beta) \end{bmatrix} = \begin{bmatrix} \cos(\beta) & 0 & -\sin(\beta) \\ 0 & 1 & 0 \\ \sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$



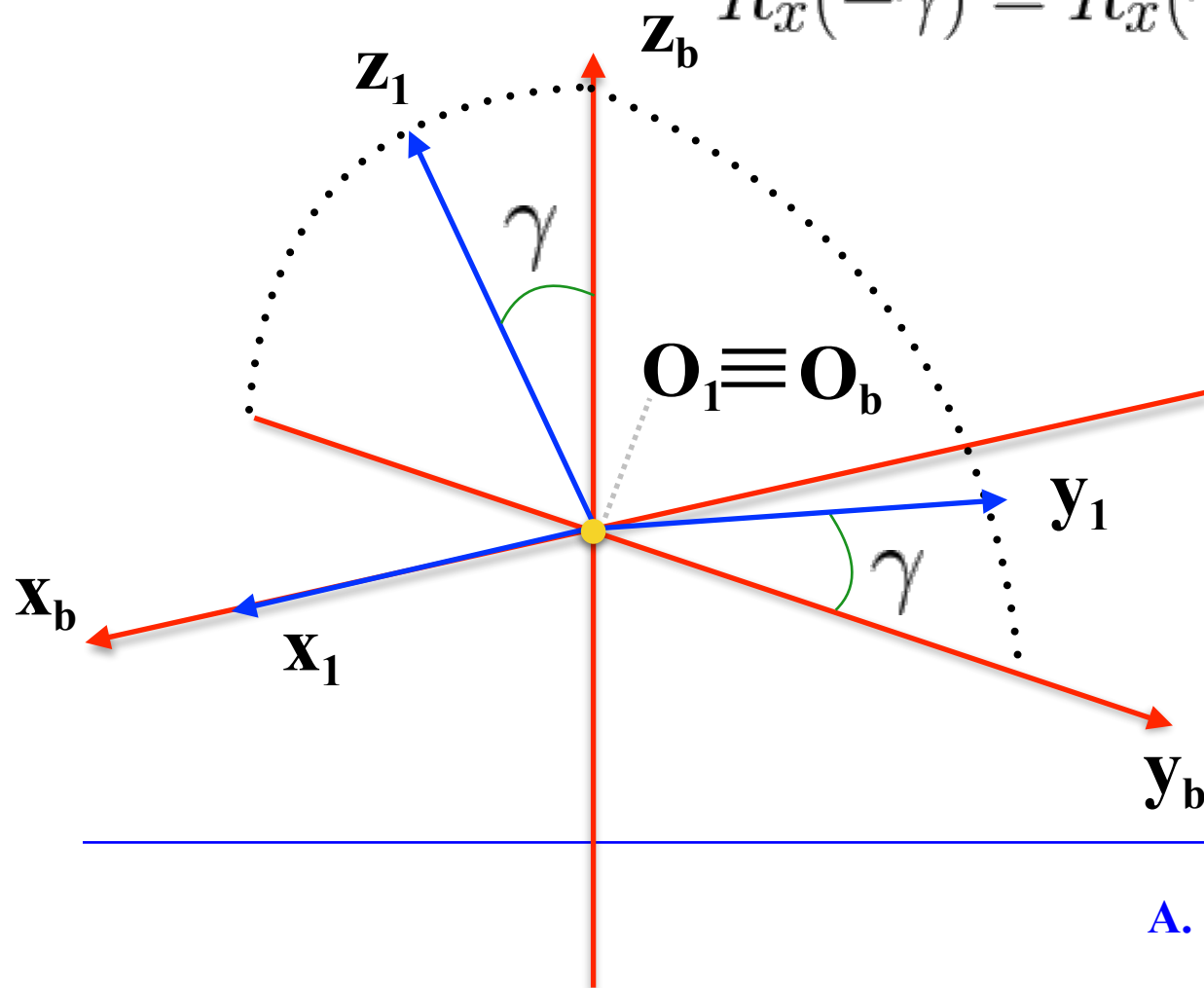
Elementary Rotations (clockwise)



$$R_1^b = R_x(\gamma) = \begin{bmatrix} \underline{x}_1^b & \underline{y}_1^b & \underline{z}_1^b \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

$$R_1^b = R_x(-\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-\gamma) & -\sin(-\gamma) \\ 0 & \sin(-\gamma) & \cos(-\gamma) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & \sin(\gamma) \\ 0 & -\sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

$$R_x(-\gamma) = R_x(\gamma)^T$$





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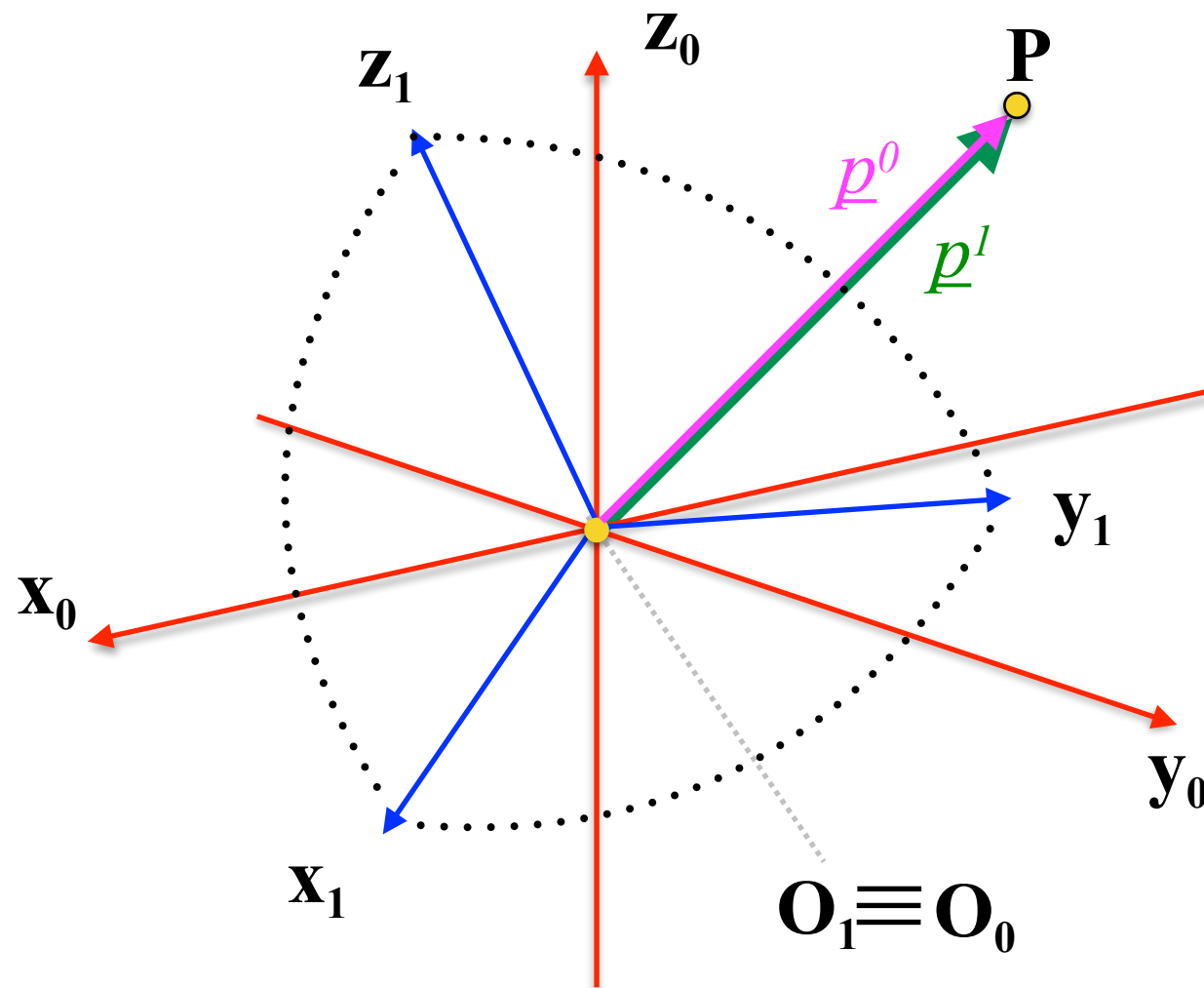
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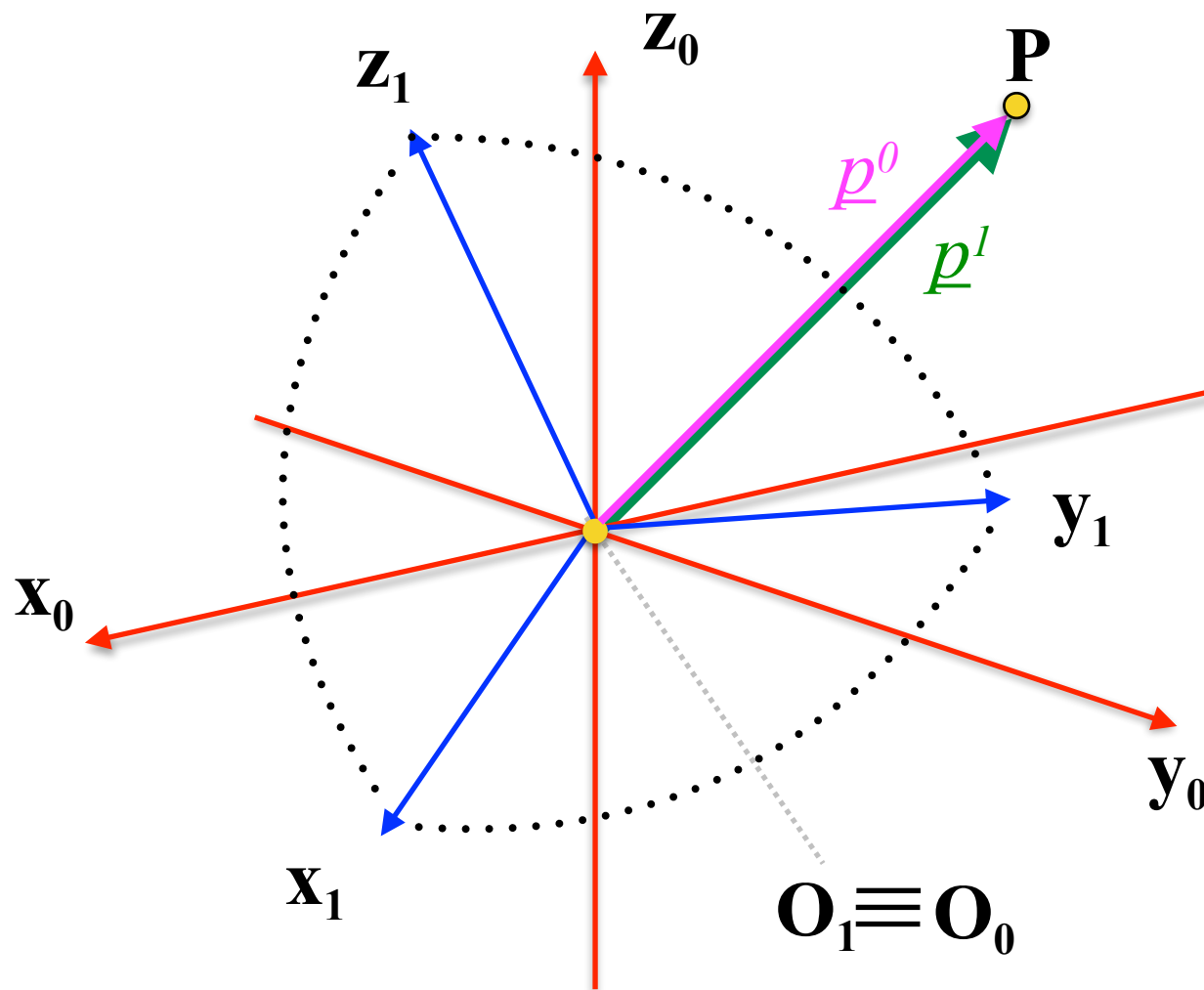
Interpretation of Rotation Matrices

Rotation Matrices are Coordinates Transformation Operators



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$
$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

Rotation Matrices are Coordinates Transformation Operators

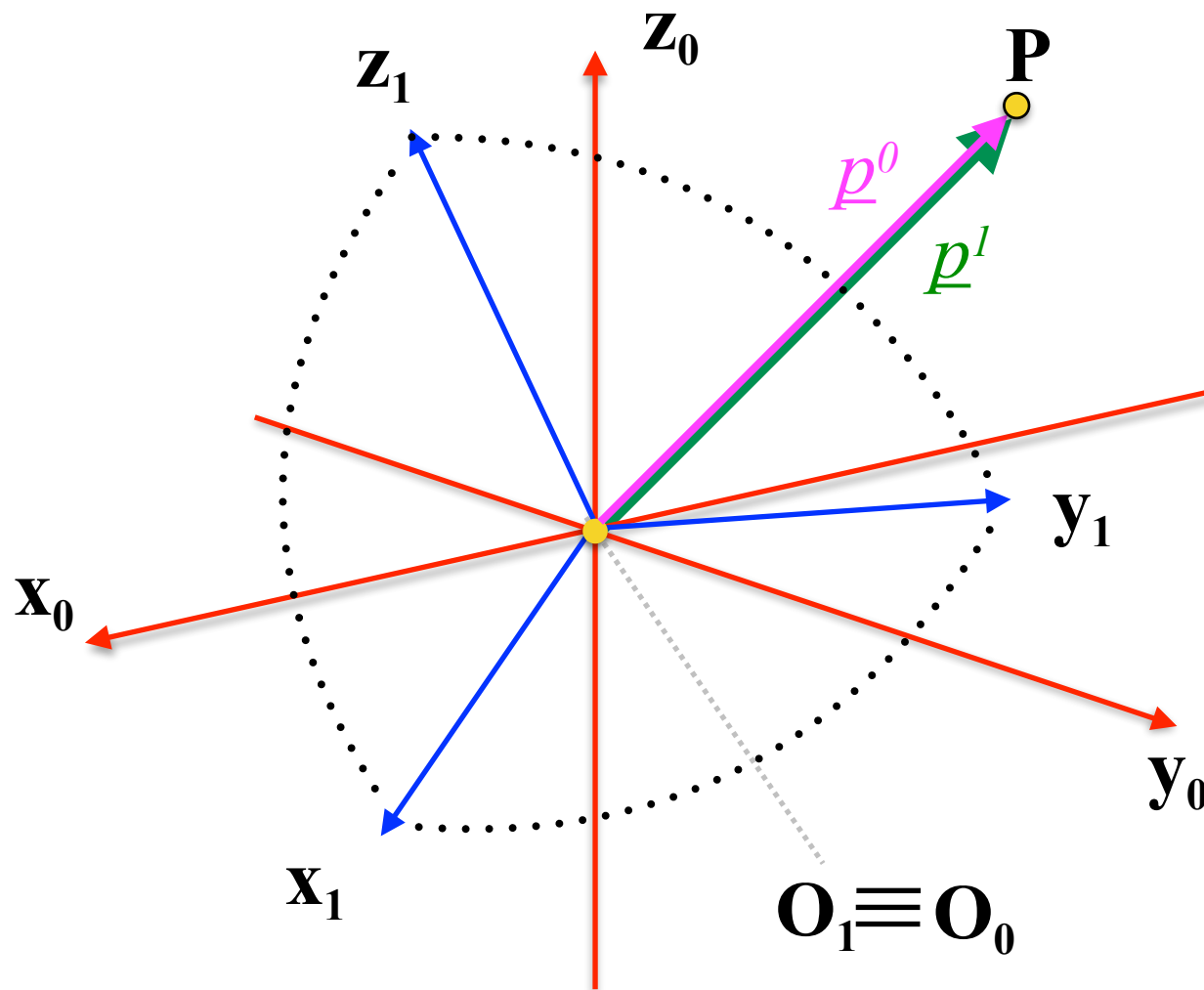


$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\begin{aligned} \underline{p}^0 &= p_x^0 \underline{x}_0 + p_y^0 \underline{y}_0 + p_z^0 \underline{z}_0 \\ &= p_x^1 \underline{x}_1^0 + p_y^1 \underline{y}_1^0 + p_z^1 \underline{z}_1^0 \\ &= \begin{bmatrix} \vdots & \vdots & \vdots \\ \underline{x}_1^0 & \underline{y}_1^0 & \underline{z}_1^0 \\ \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix} \end{aligned}$$

Rotation Matrices are Coordinates Transformation Operators



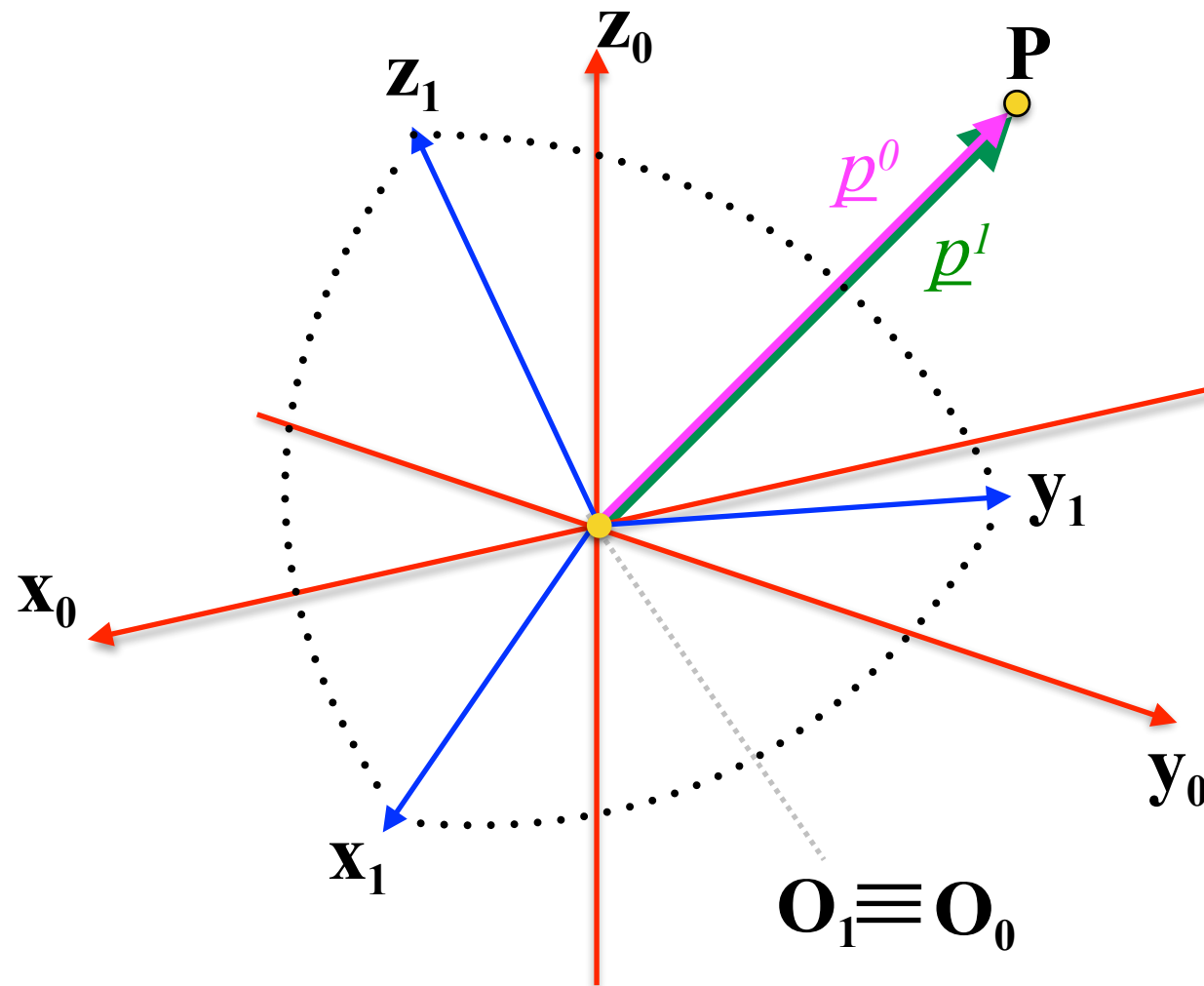
$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\begin{aligned}\underline{p}^0 &= p_x^0 \underline{x}_0 + p_y^0 \underline{y}_0 + p_z^0 \underline{z}_0 \\ &= p_x^1 \underline{x}_1 + p_y^1 \underline{y}_1 + p_z^1 \underline{z}_1 \\ &= \begin{bmatrix} \vdots & \vdots & \vdots \\ \underline{x}_1^0 & \underline{y}_1^0 & \underline{z}_1^0 \\ \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix}\end{aligned}$$

$$\underline{p}^0 = R_1^0 \cdot \underline{p}^1$$

Rotation Matrices are Coordinates Transformation Operators



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$
$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\underline{p}^0 = R_1^0 \cdot \underline{p}^1$$

$$R_1^{0^{-1}} \cdot \underline{p}^0 = R_1^{0^{-1}} \cdot R_1^0 \cdot \underline{p}^1$$

$$\underline{p}^1 = R_1^{0^{-1}} \cdot \underline{p}^0 = R_1^{0T} \cdot \underline{p}^0 = R_0^1 \cdot \underline{p}^0$$



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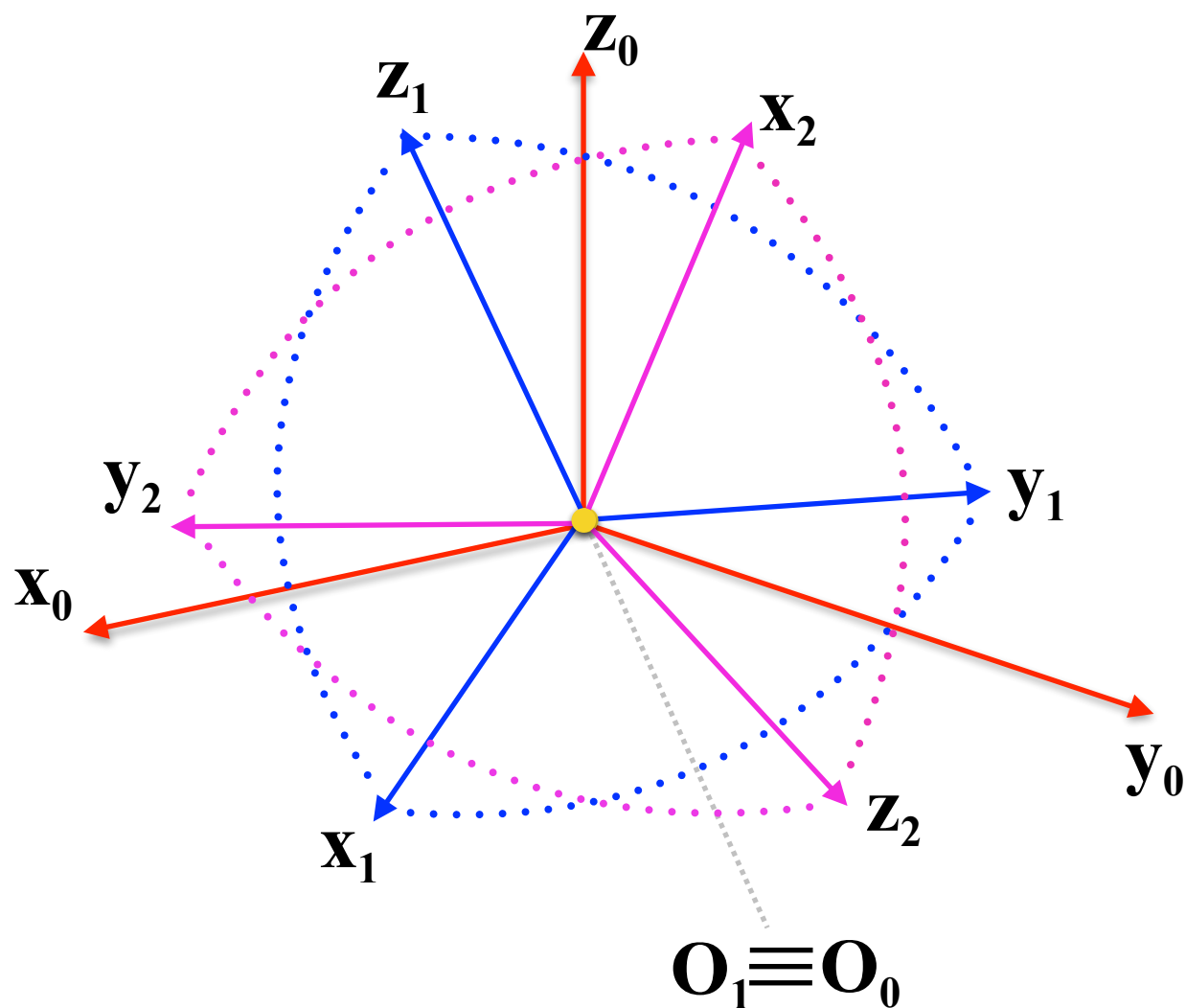
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Composition of Rotations

Composition of Rotations

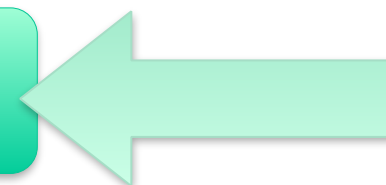


$$\underline{p}^0 = R_1^0 \cdot \underline{p}^1$$

$$\underline{p}^1 = R_2^1 \cdot \underline{p}^2$$

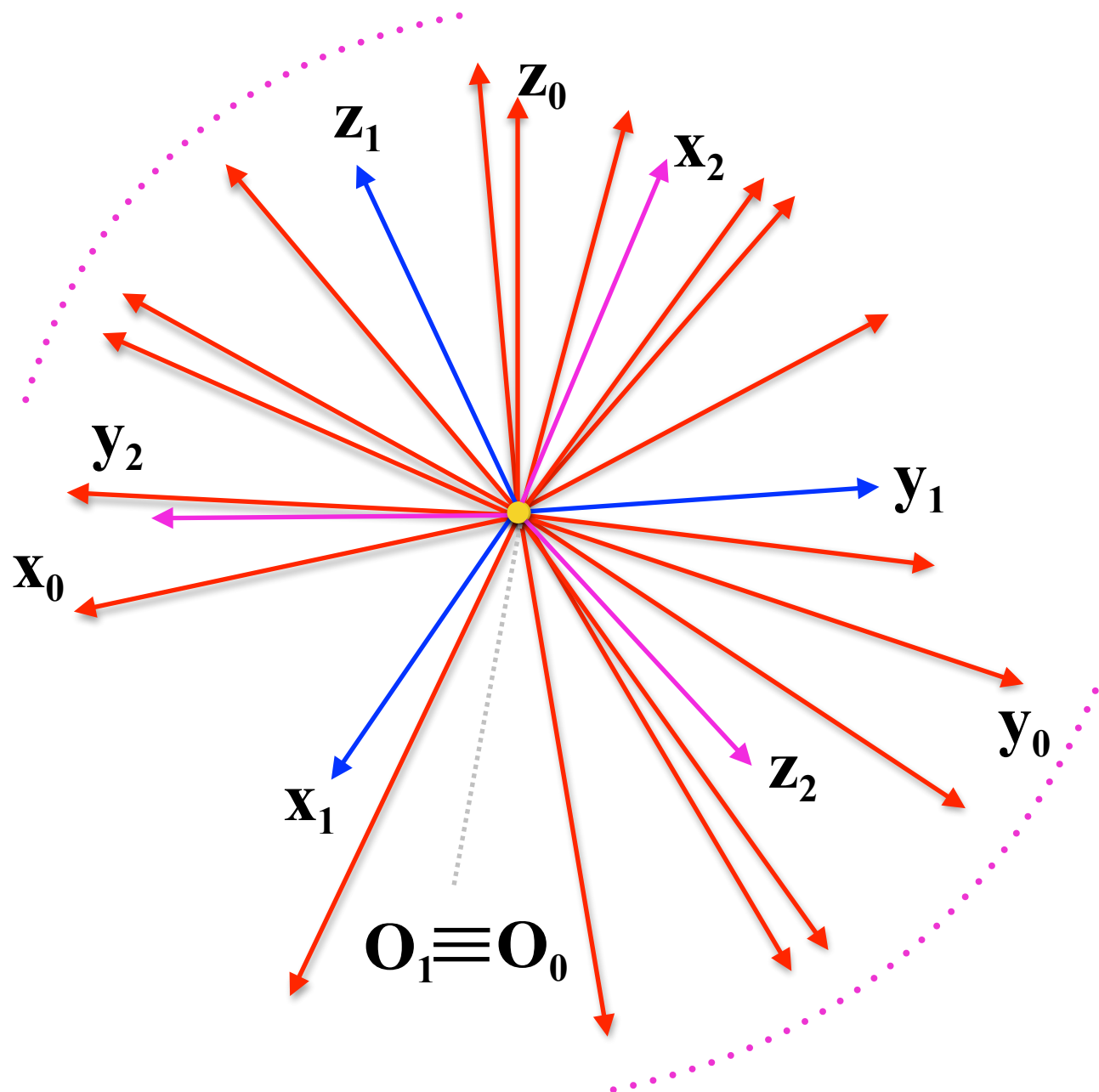
$$\underline{p}^0 = R_1^0 \cdot R_2^1 \cdot \underline{p}^2 = R_2^0 \cdot \underline{p}^2$$

$$R_2^0 = R_1^0 \cdot R_2^1$$



Composition of Rotations

n reference frames



$$R_2^0 = R_1^0 \cdot R_2^1$$

$$R_n^0 = R_1^0 \cdot R_2^1 \dots R_{n-1}^{n-2} \cdot R_n^{n-1}$$



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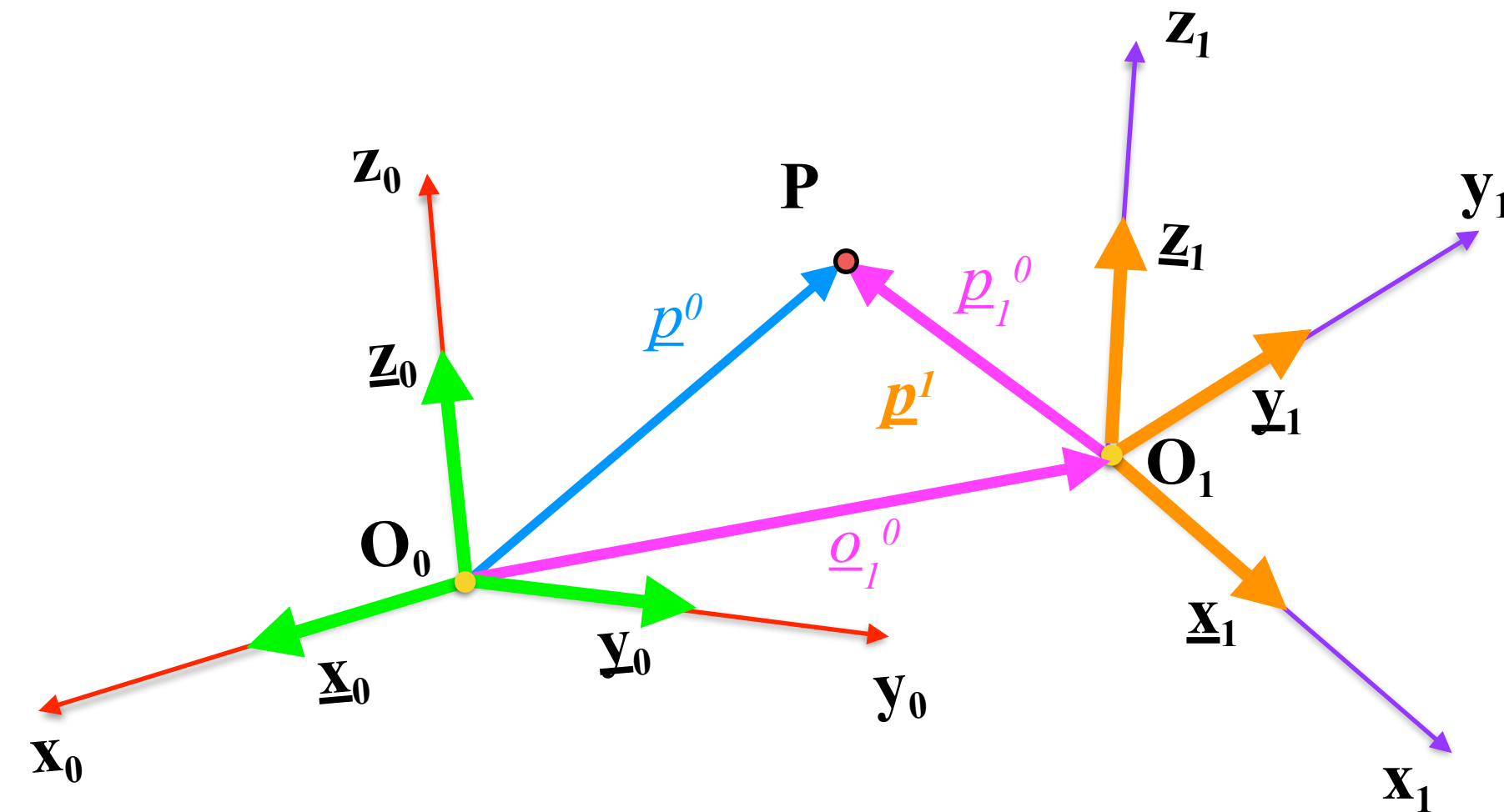
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Roto-Translations

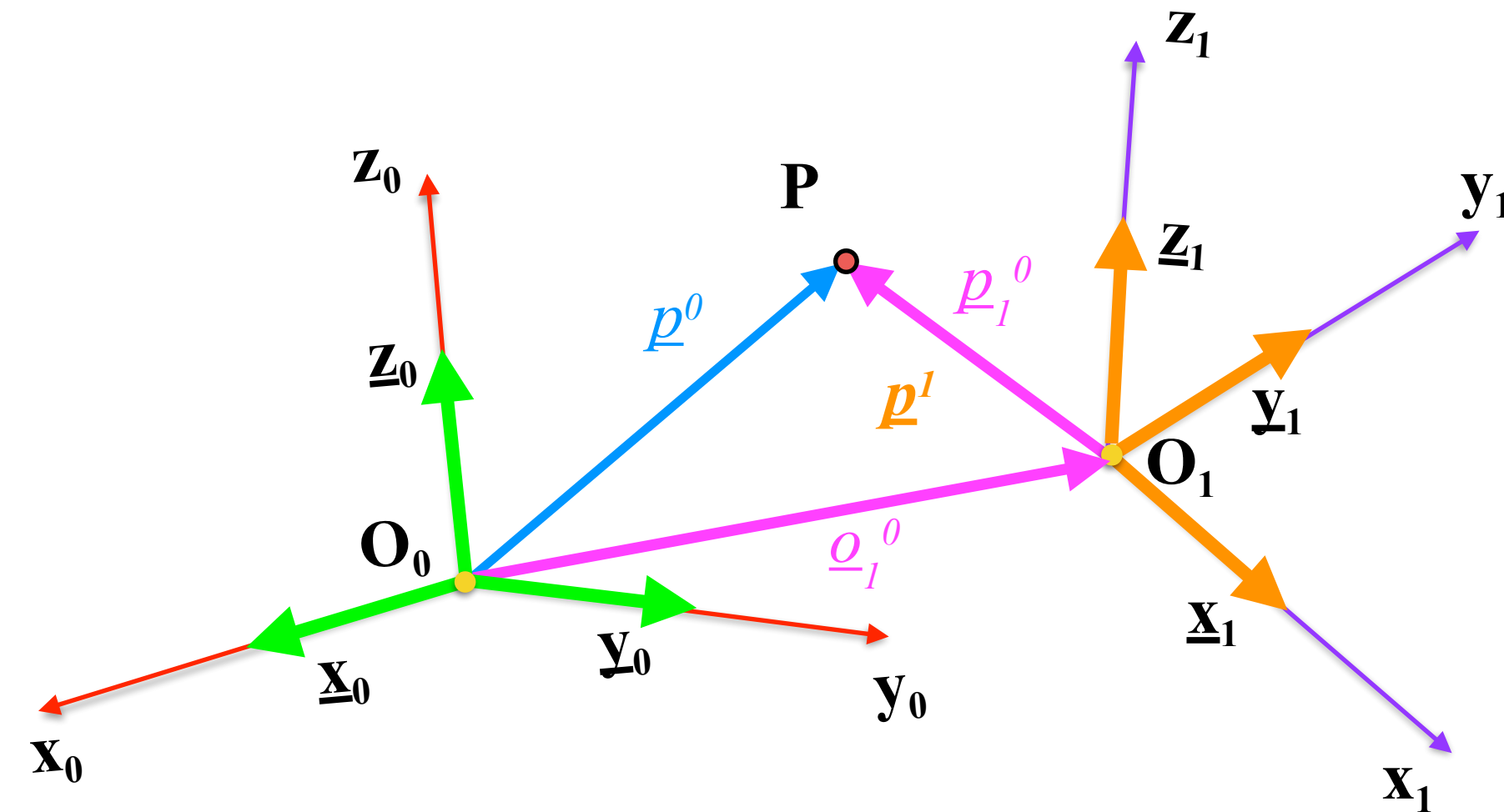
How to Describe Roto-Translations of Frames



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

How to Describe Roto-Translations of Frames

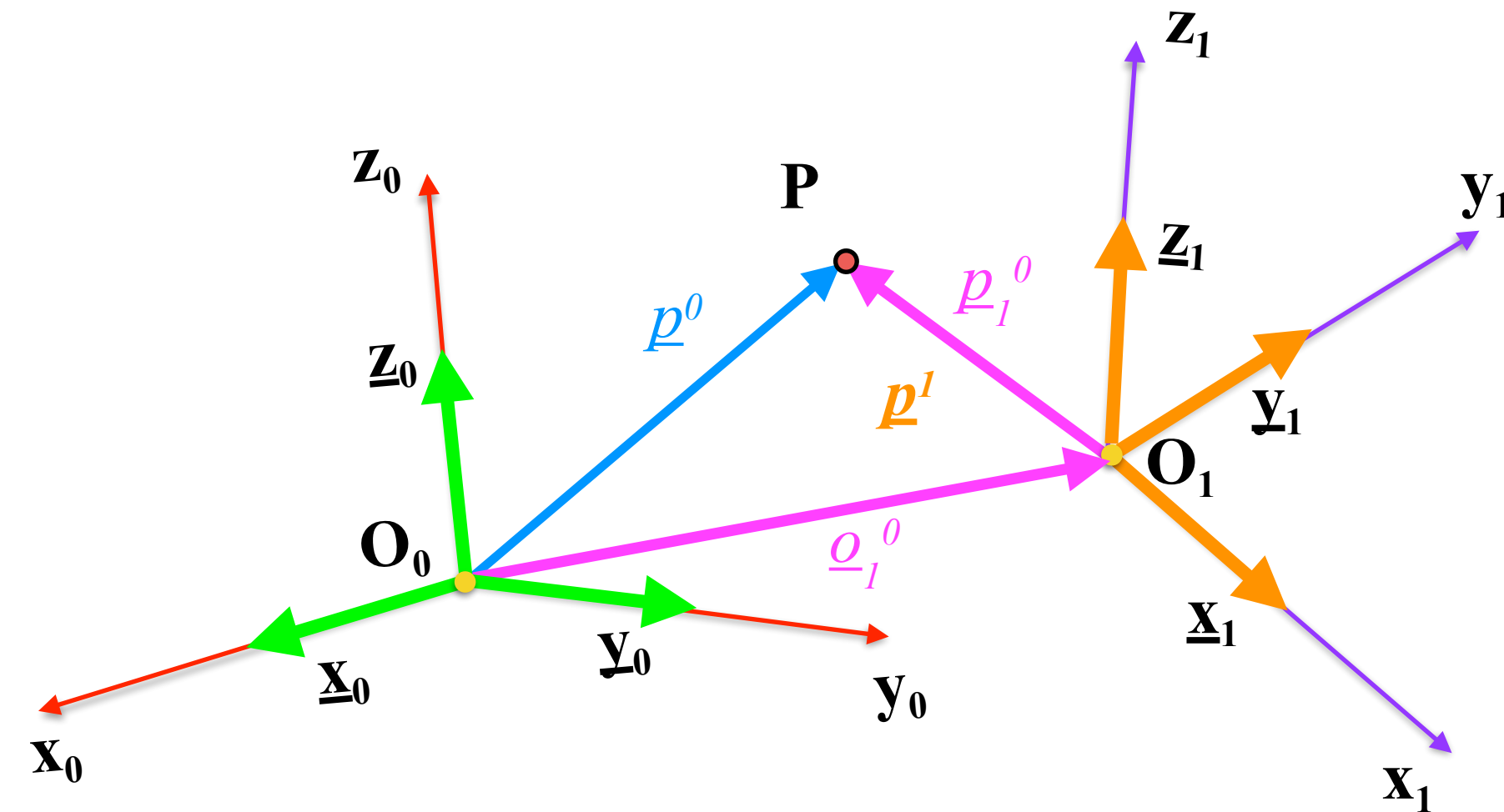


$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

Problem: Find the coordinates of P w.r.t. O_0 - $x_0y_0z_0$ knowing its coordinates in the reference frame O_1 - $x_1y_1z_1$

How to Describe Roto-Translations of Frames

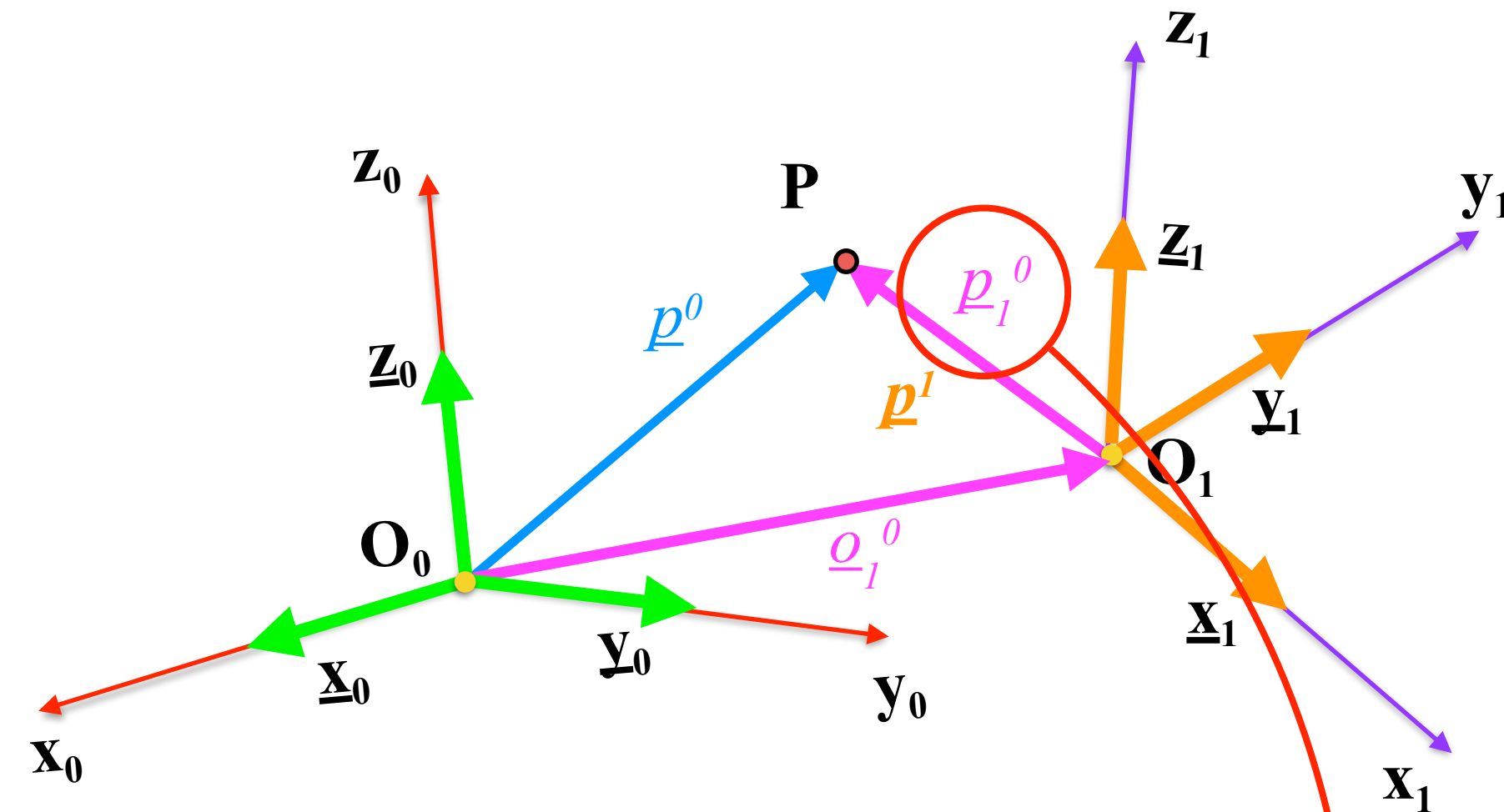


$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

How to Describe Roto-Translations of Frames

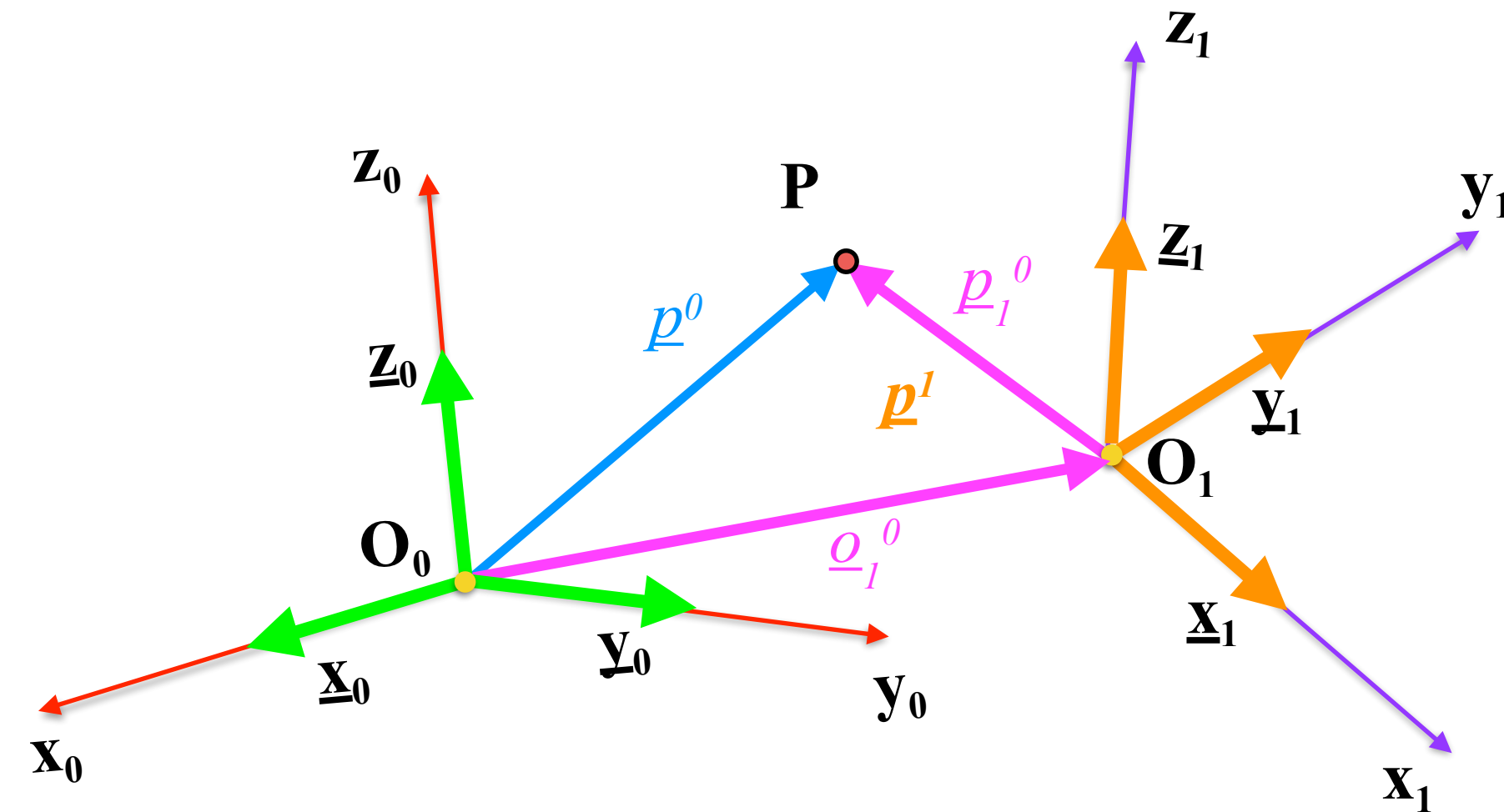


$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

How to Describe Roto-Translations of Frames



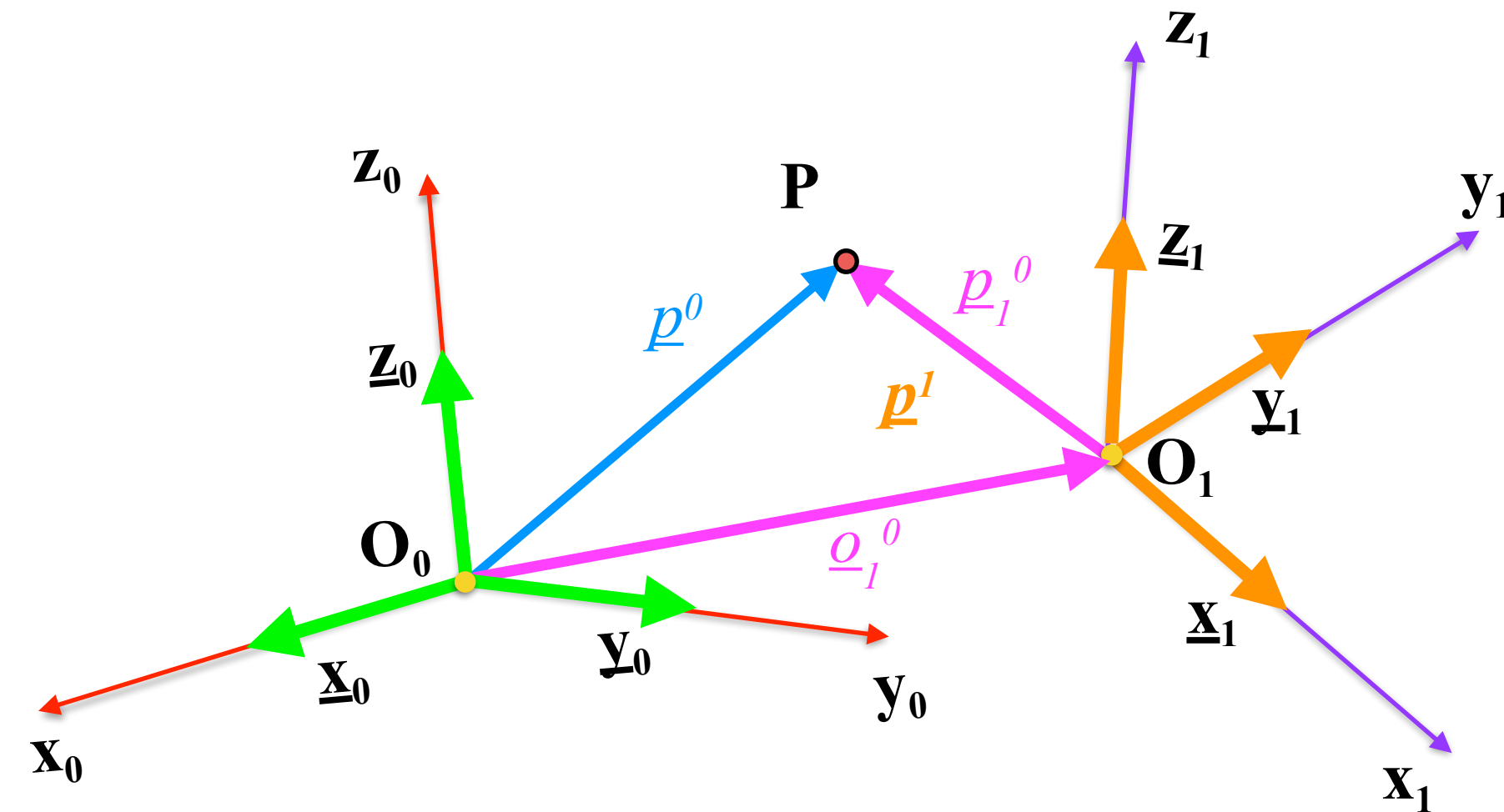
$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

Direct Formula

How to Describe Roto-Translations of Frames



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

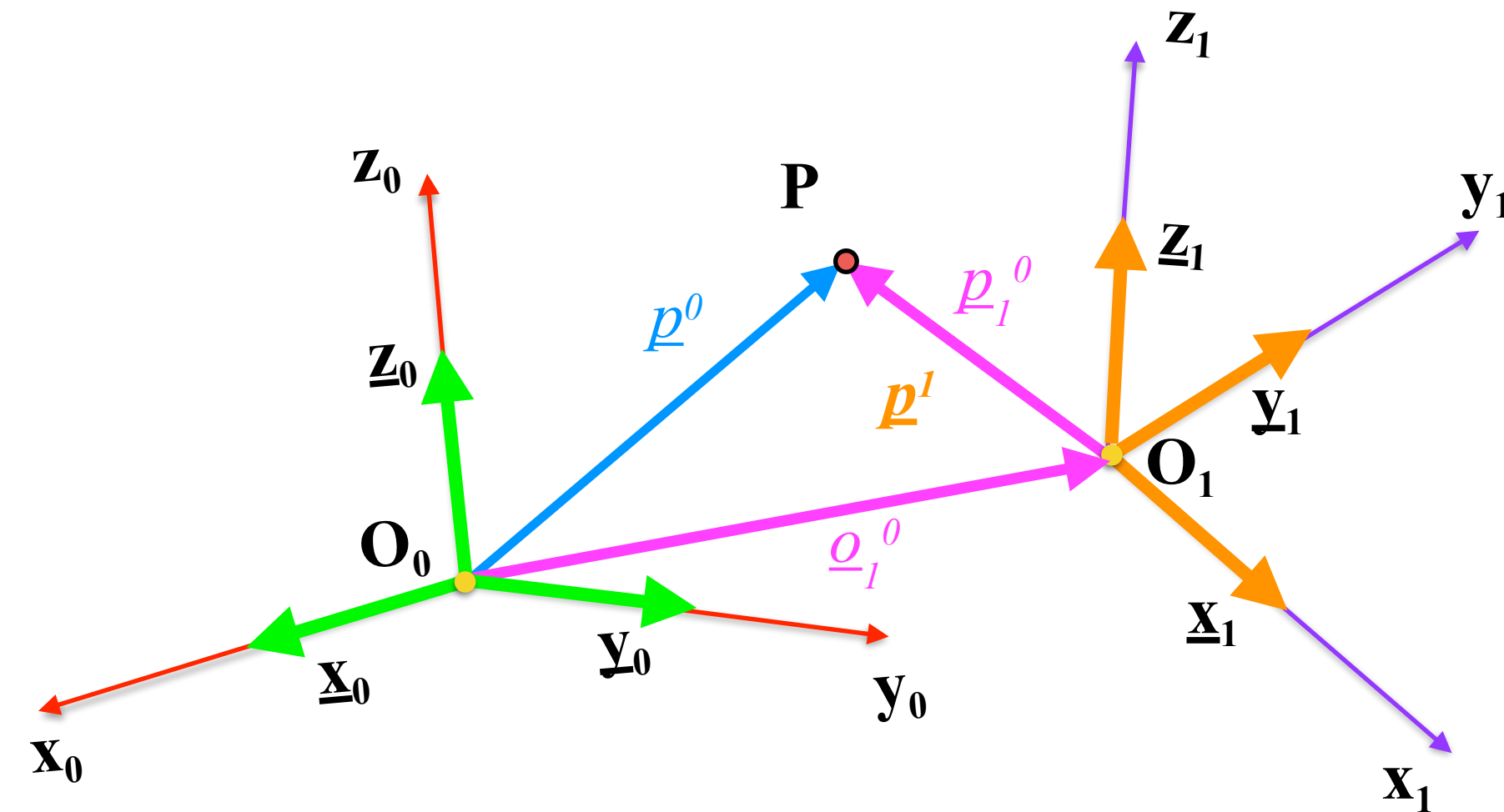
$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

Direct Formula:

$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

$$\underline{R}_1^{0^{-1}} \underline{p}^0 = \underline{R}_1^{0^{-1}} \underline{o}_1^0 + \underline{R}_1^{0^{-1}} R_1^0 \underline{p}^1$$

How to Describe Roto-Translations of Frames



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

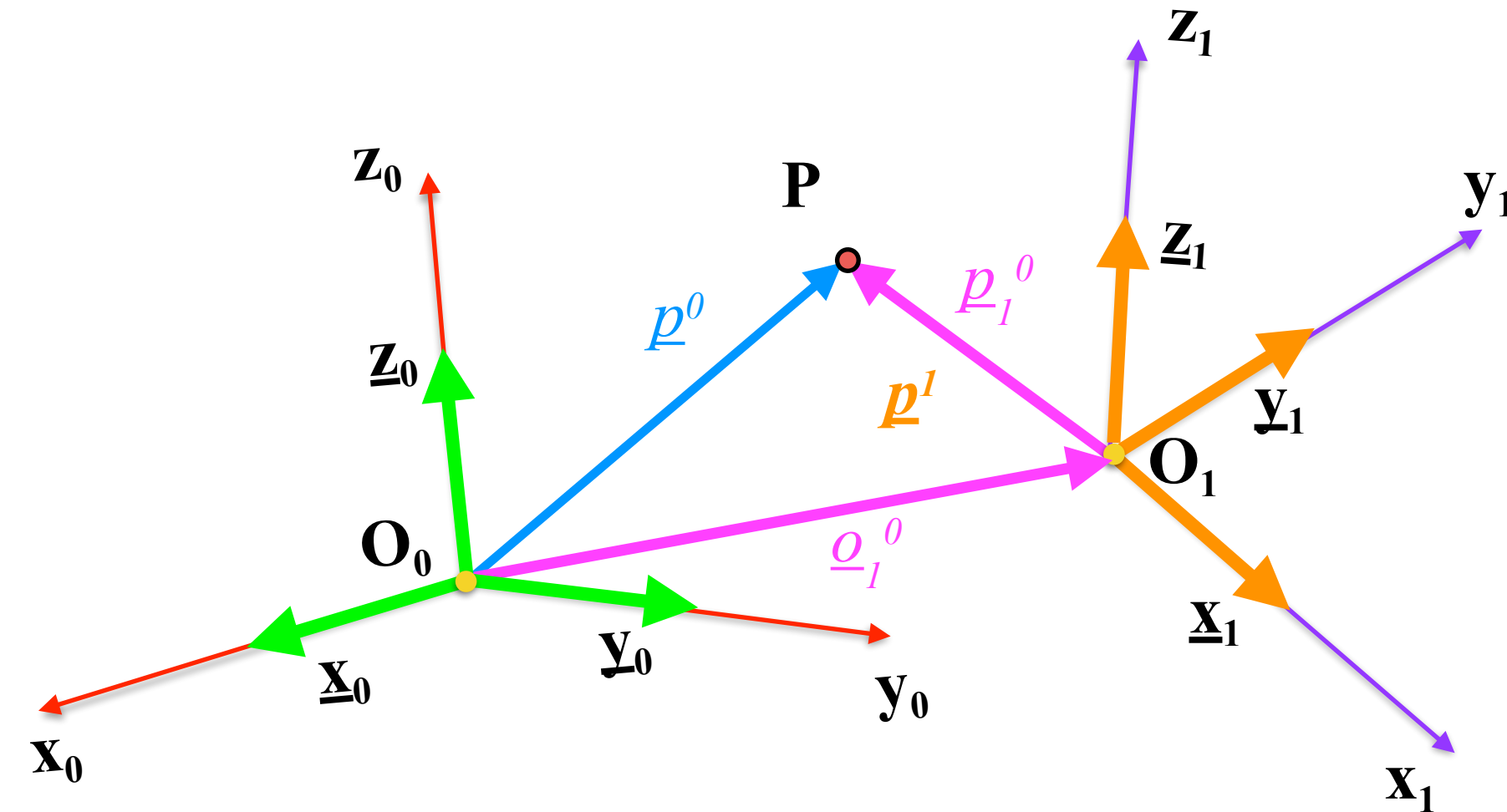
Direct Formula:

$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

$$R_1^{0^{-1}} \underline{p}^0 = R_1^{0^{-1}} \underline{o}_1^0 + R_1^{0^{-1}} R_1^0 \underline{p}^1$$

$\mathbf{I}_{3 \times 3}$

How to Describe Roto-Translations of Frames



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

Direct Formula:

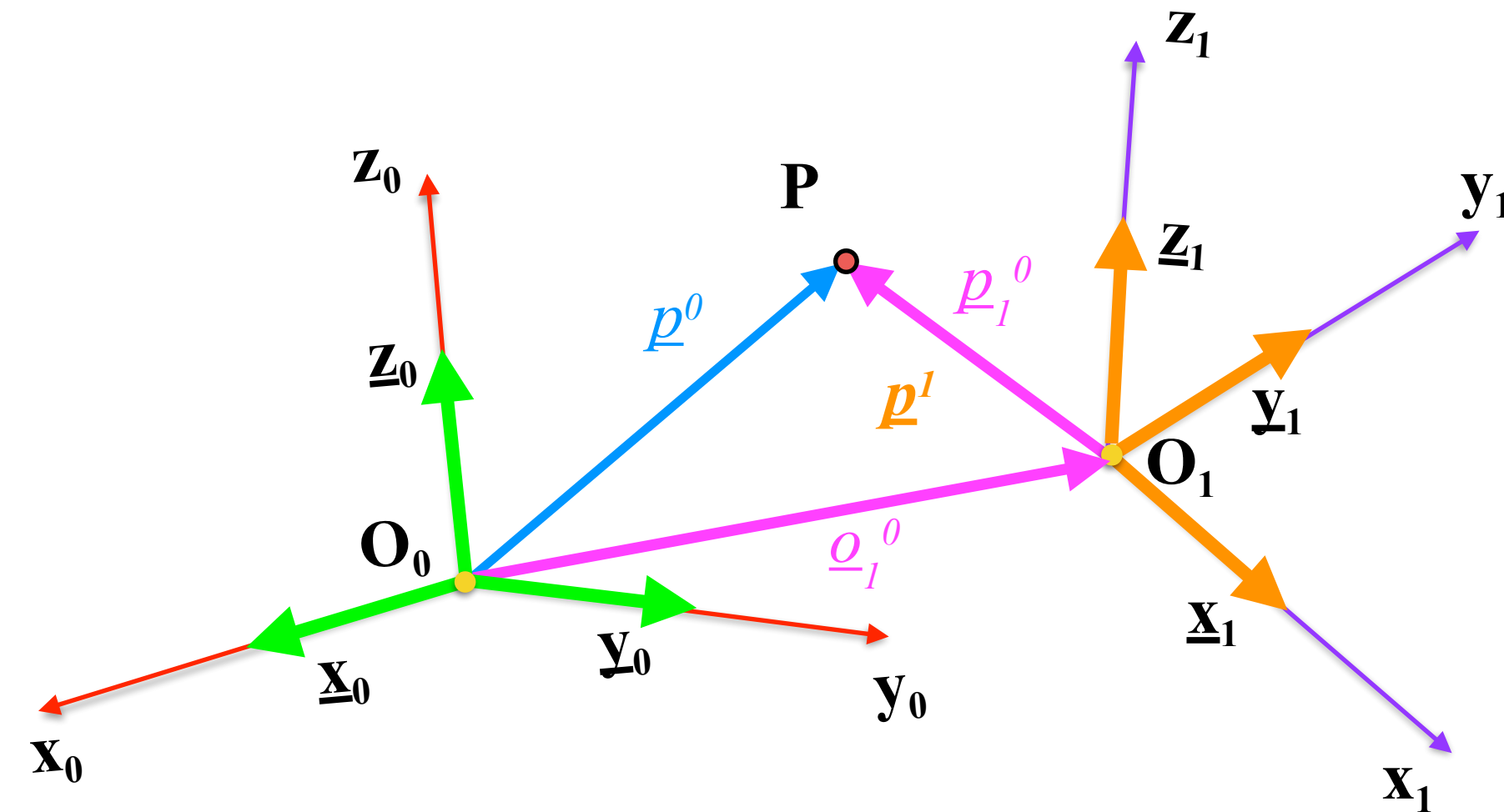
$$\underline{p}^0 = \underline{o}_1^0 + \underline{p}_1^0 = \underline{o}_1^0 + R_1^0 \underline{p}^1$$

Indirect Formula:

$$R_1^{0^{-1}} \underline{p}^0 = R_1^{0^{-1}} \underline{o}_1^0 + R_1^{0^{-1}} R_1^0 \underline{p}^1$$

$$\underline{p}^1 = R_1^{0^{-1}} \underline{p}^0 - R_1^{0^{-1}} \underline{o}_1^0$$

How to Describe Roto-Translations of Frames



$$\underline{p}^0 = [p_x^0, p_y^0, p_z^0]^T$$

$$\underline{p}^1 = [p_x^1, p_y^1, p_z^1]^T$$

Three different versions of the Indirect Formula:

$$\underline{p}^1 = R_1^{0^{-1}} \underline{p}^0 - R_1^{0^{-1}} \underline{o}_1^0 = R_1^{0^T} \underline{p}^0 - R_1^{0^T} \underline{o}_1^0 = R_0^1 \underline{p}^0 - R_0^1 \underline{o}_1^0$$



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Homogeneous Transformations

How to Re-write the Previous Formulas in Compact Form: Homogeneous Coordinates

Homogeneous Vectors

$$\underline{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \underline{\tilde{p}} = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} \underline{p} \\ \dots \\ 1 \end{bmatrix}$$

Homogeneous Transformation Matrices

$$A = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

Homogeneous Vectors

$$\underline{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \underline{\tilde{p}} = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} \underline{p} \\ \cdots \\ 1 \end{bmatrix}$$

Homogeneous Transformation Matrices

4x4

$$A = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

Homogeneous Vectors

$$\underline{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \underline{\tilde{p}} = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} \underline{p} \\ \cdots \\ 1 \end{bmatrix}$$

Homogeneous Transformation Matrices

$$A = \begin{bmatrix} \begin{matrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{matrix} & \cdot \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3x3

Homogeneous Coordinates

Homogeneous Vectors

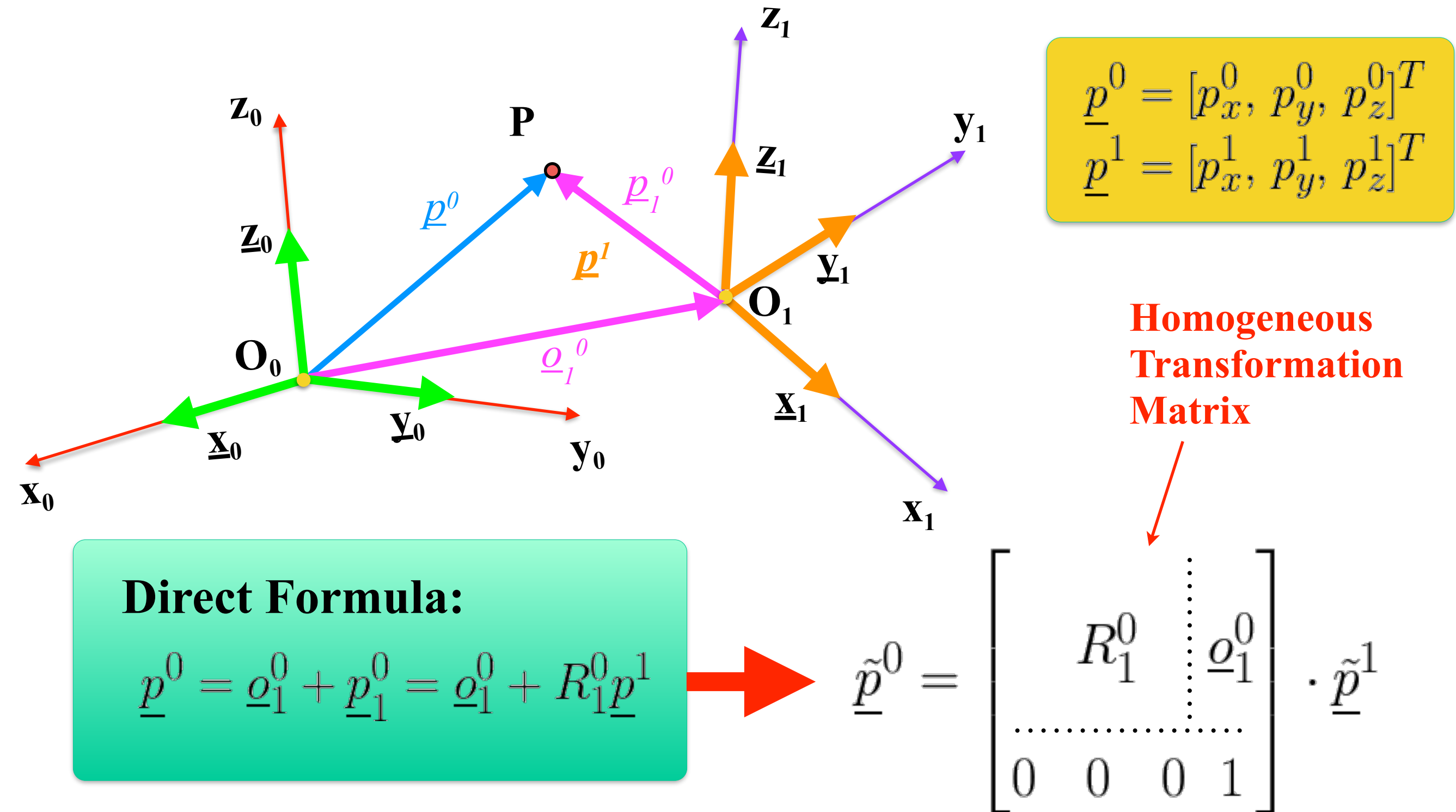
$$\underline{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \underline{\tilde{p}} = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} \underline{p} \\ \cdots \\ 1 \end{bmatrix}$$

Homogeneous Transformation Matrices

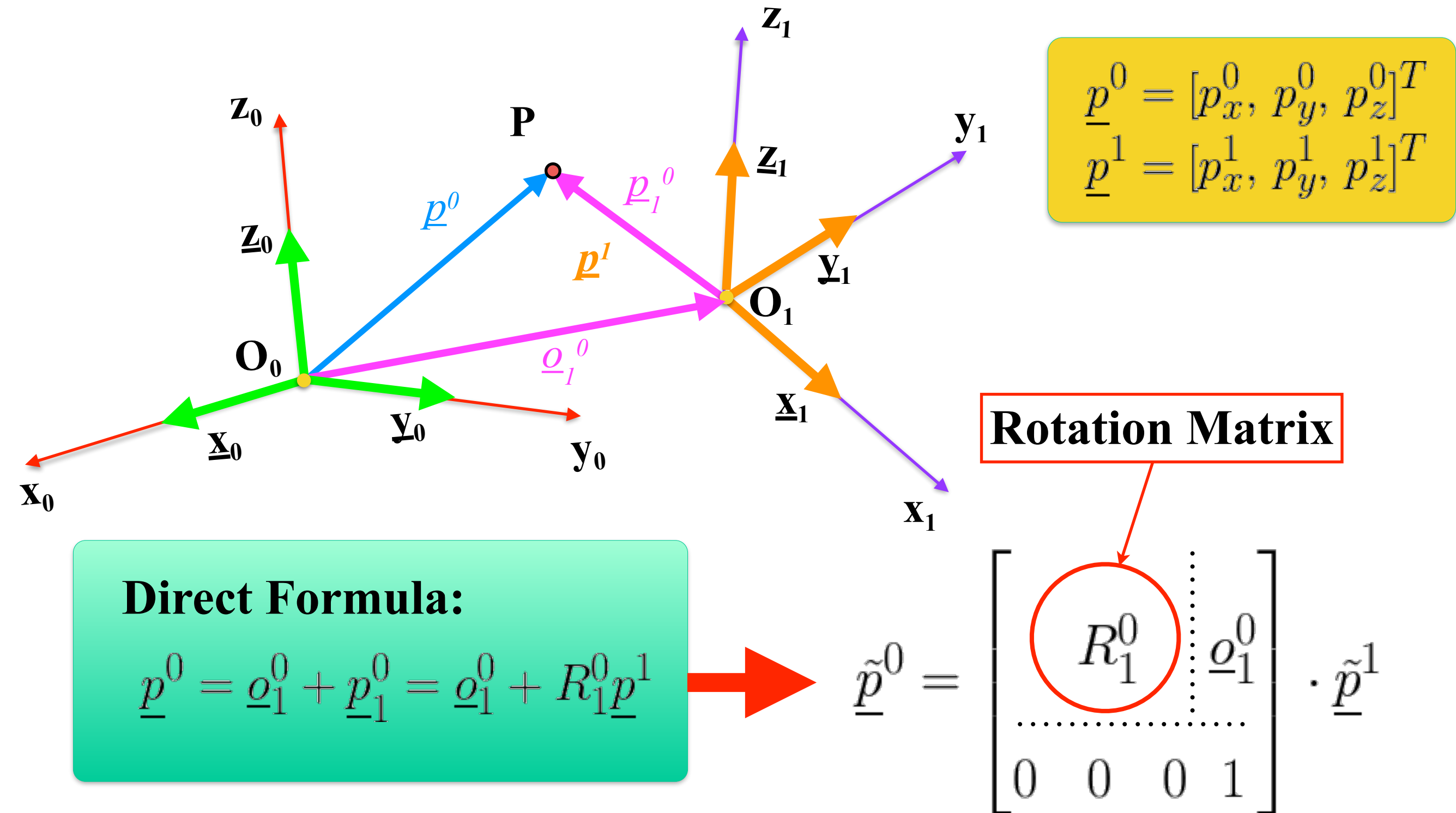
$$A = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3x1

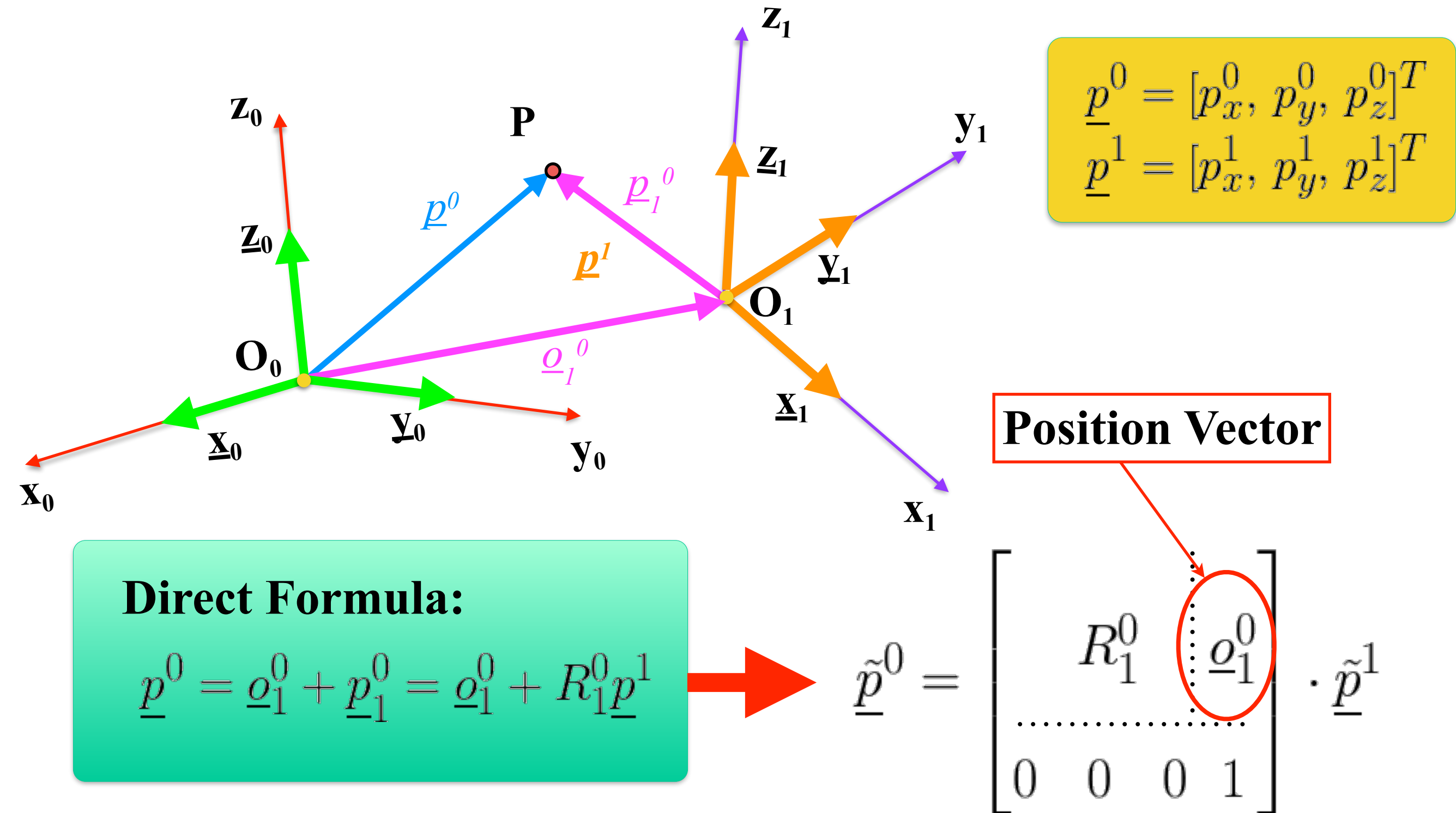
How to Describe Roto-Translations of Frames via Homogeneous Coordinates



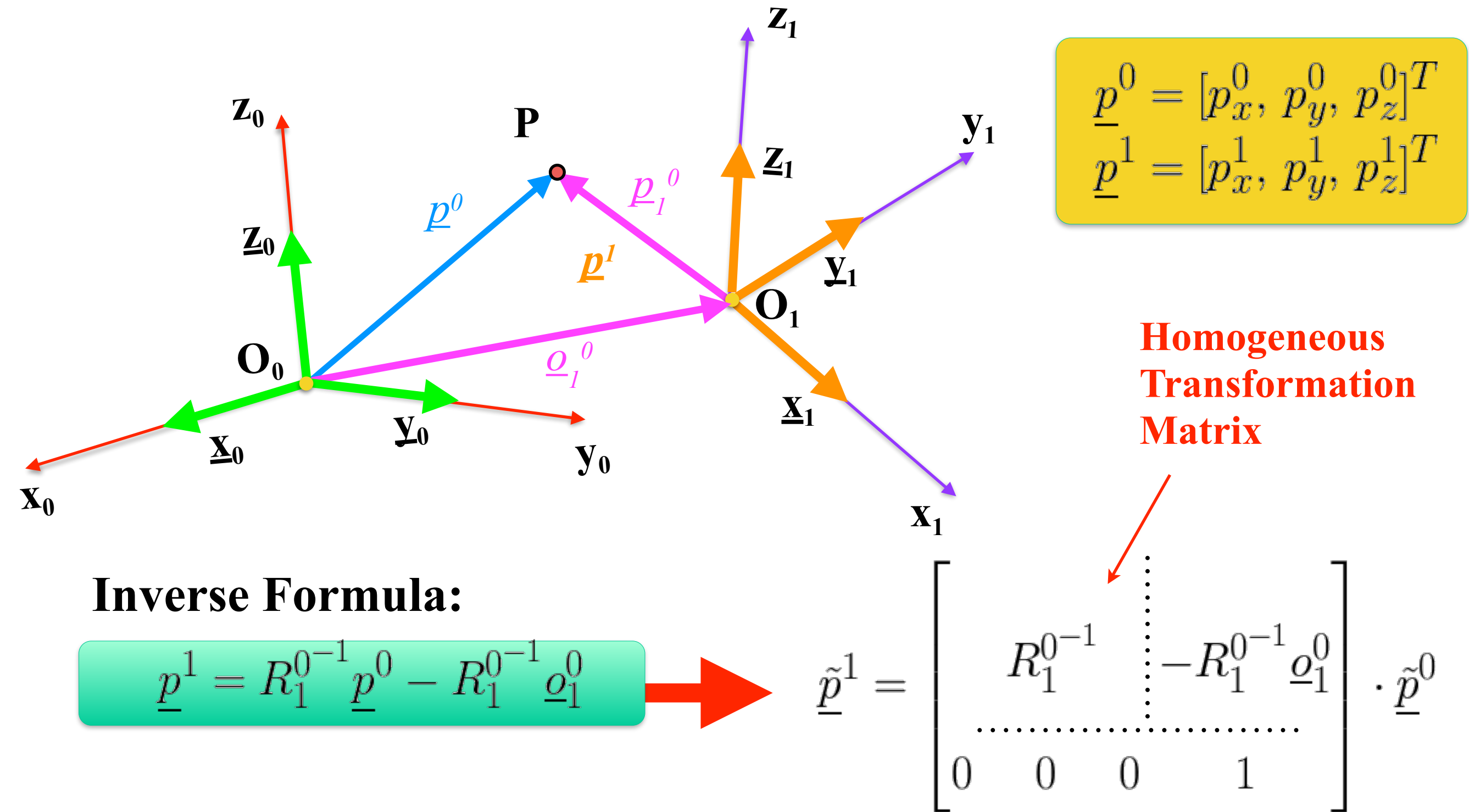
How to Describe Roto-Translations of Frames via Homogeneous Coordinates



How to Describe Roto-Translations of Frames via Homogeneous Coordinates



How to Describe Roto-Translations of Frames via Homogeneous Coordinates

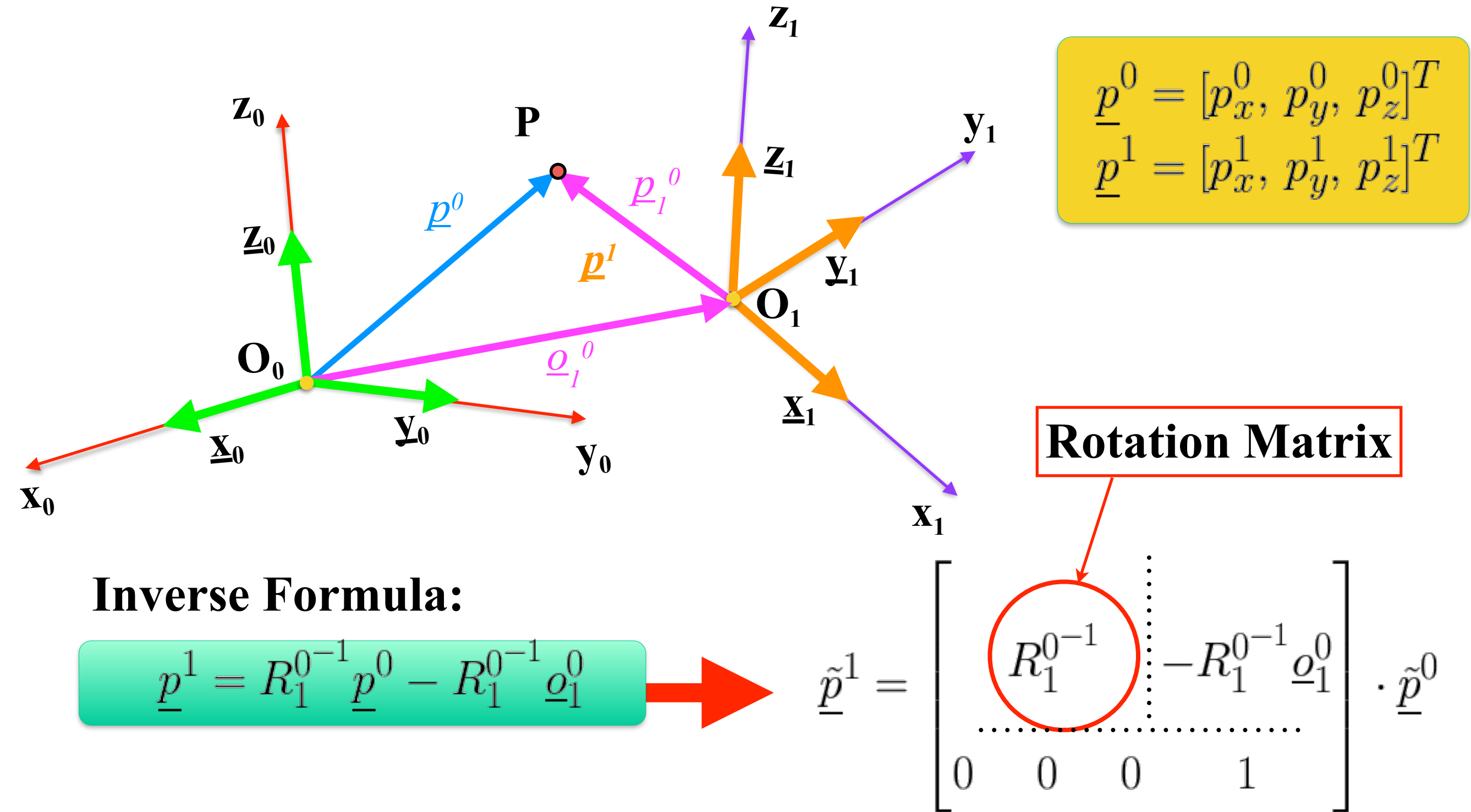


Inverse Formula:

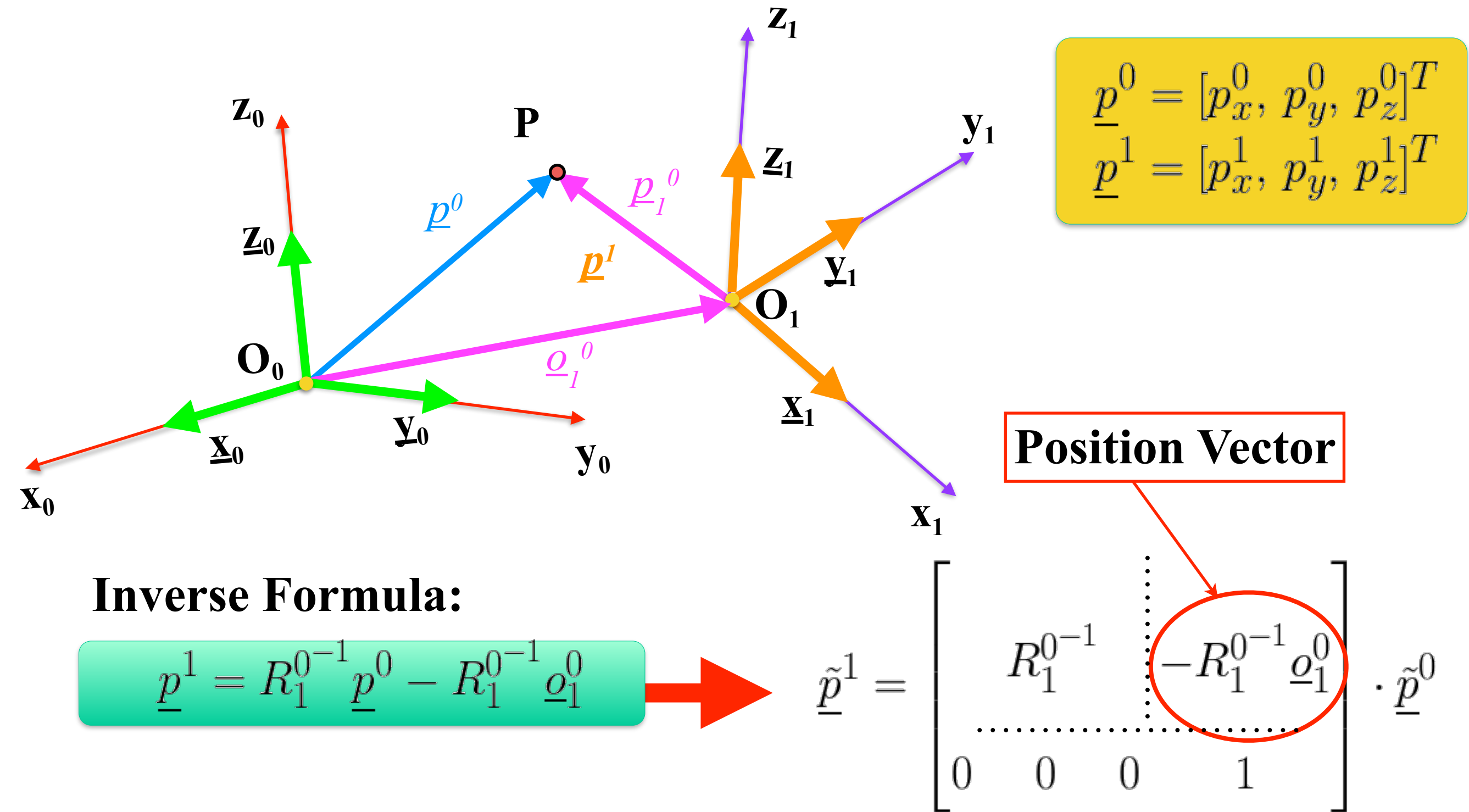
$$\underline{p}^1 = R_1^{0^{-1}} \underline{p}^0 - R_1^{0^{-1}} \underline{o}_1^0$$

$$\underline{\tilde{p}}^1 = \begin{bmatrix} R_1^{0^{-1}} & -R_1^{0^{-1}} \underline{o}_1^0 \\ \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \underline{\tilde{p}}^0$$

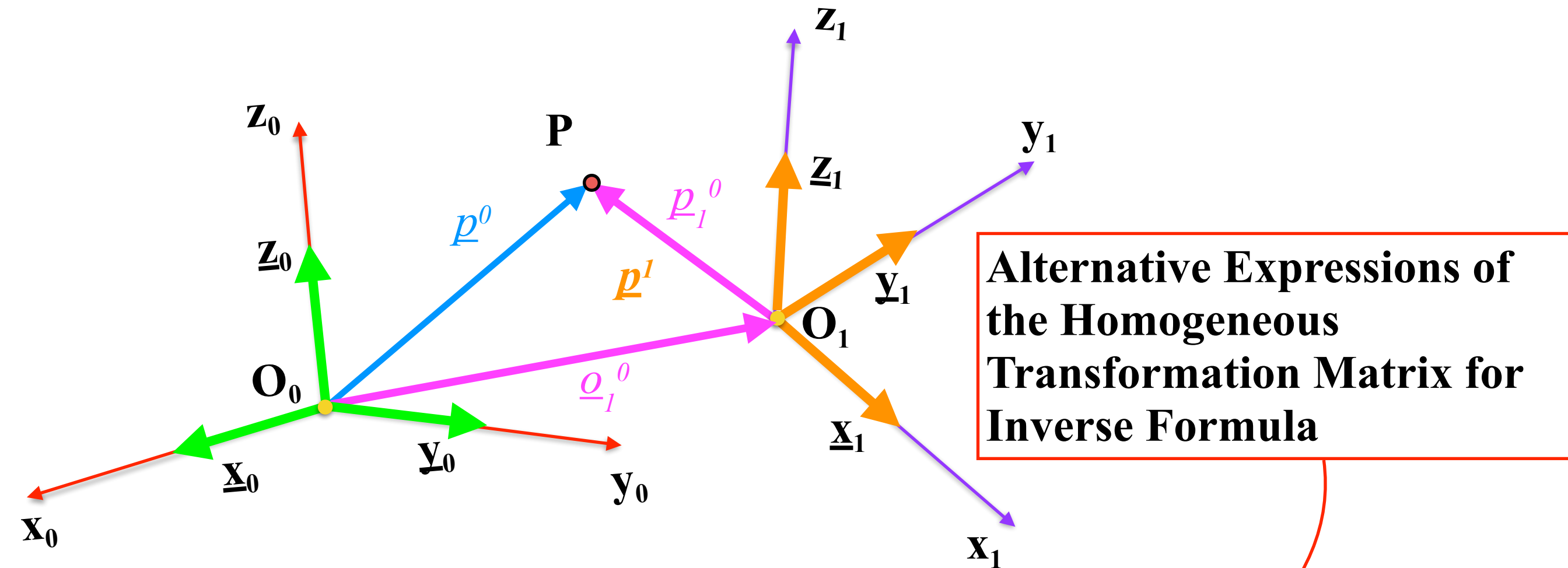
How to Describe Roto-Translations of Frames via Homogeneous Coordinates



How to Describe Roto-Translations of Frames via Homogeneous Coordinates



How to Describe Roto-Translations of Frames via Homogeneous Coordinates



Alternative Expressions of the Homogeneous Transformation Matrix for Inverse Formula

$$A_0^1 = \begin{bmatrix} R_1^{0^T} & \vdots & -R_1^{0^T} \underline{q}_1^0 \\ \dots & \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} R_0^1 & \vdots & -R_0^1 \underline{q}_1^0 \\ \dots & \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

How to Describe Roto-Translations of Frames via Homogeneous Coordinates

Given

$$A_1^0 = \begin{bmatrix} R_1^0 & \vdots & \underline{o}_1^0 \\ \hdashline & & \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_0^1 = \begin{bmatrix} R_1^{0T} & \vdots & -R_1^{0T} \underline{o}_1^0 \\ \hdashline & & \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note that:

$$A_0^1 = (A_1^0)^{-1} \quad \rightarrow \quad (A_1^0)^{-1} \neq (A_1^0)^T$$



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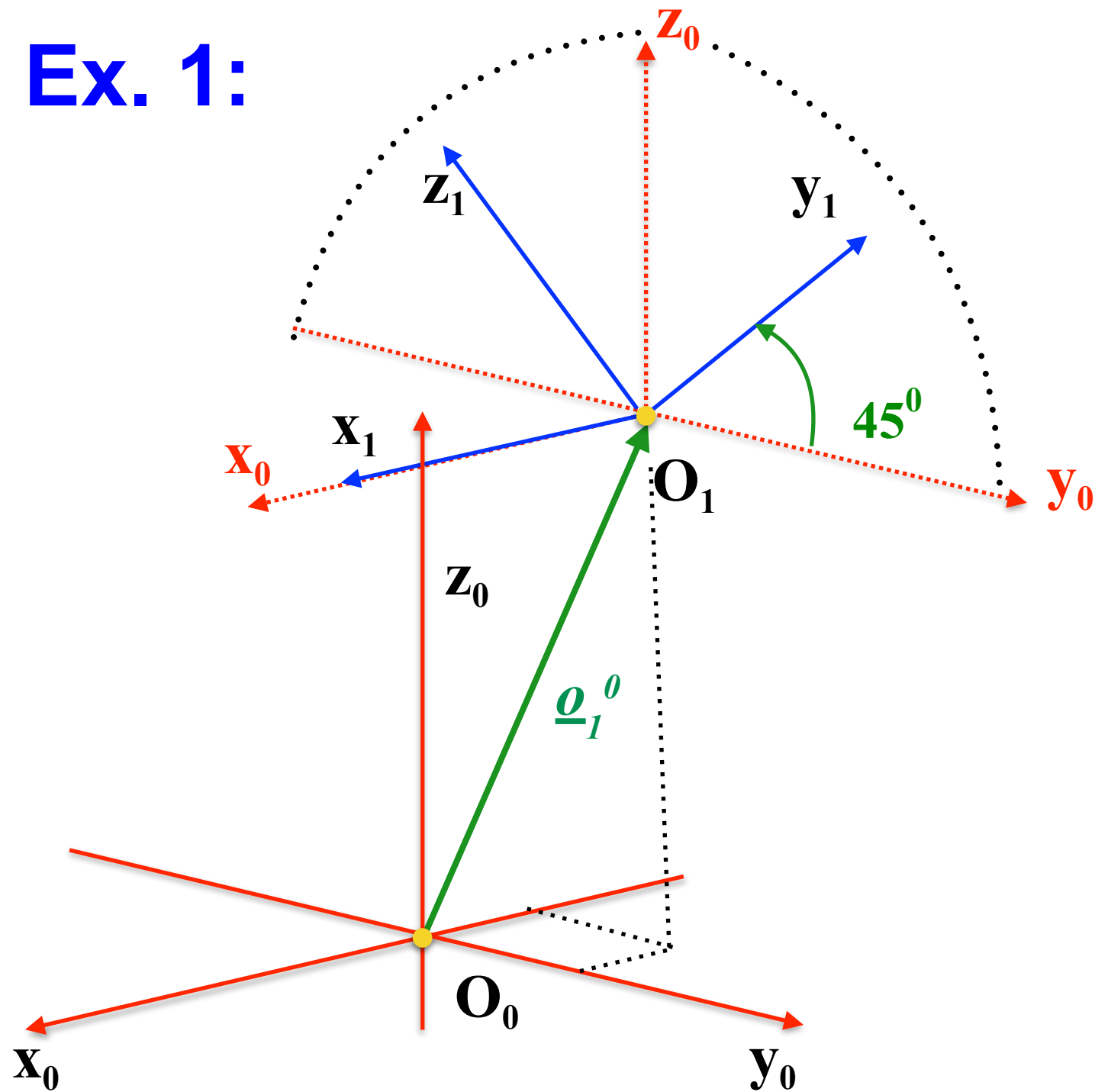
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Some Examples on Homogeneous Transformations

Examples on Homogeneous Matrices

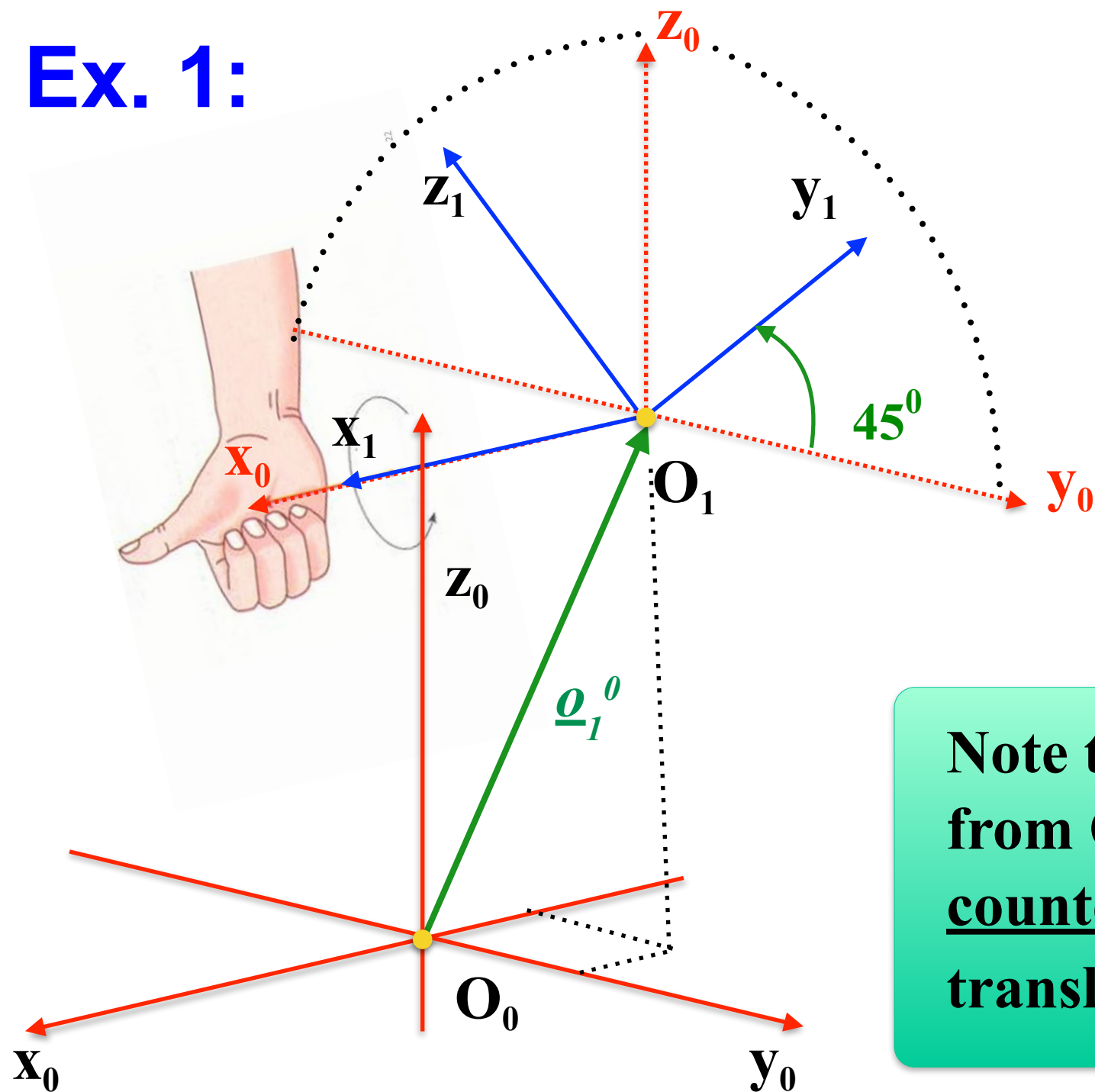
Ex. 1:



Given $\underline{O_1^0} = [-1, 2, 8]^T$
Find the Homogeneous Transformation Matrix
describing O_1 - $x_1y_1z_1$ w.r.t.
 O_0 - $x_0y_0z_0$

Examples on Homogeneous Matrices

Ex. 1:

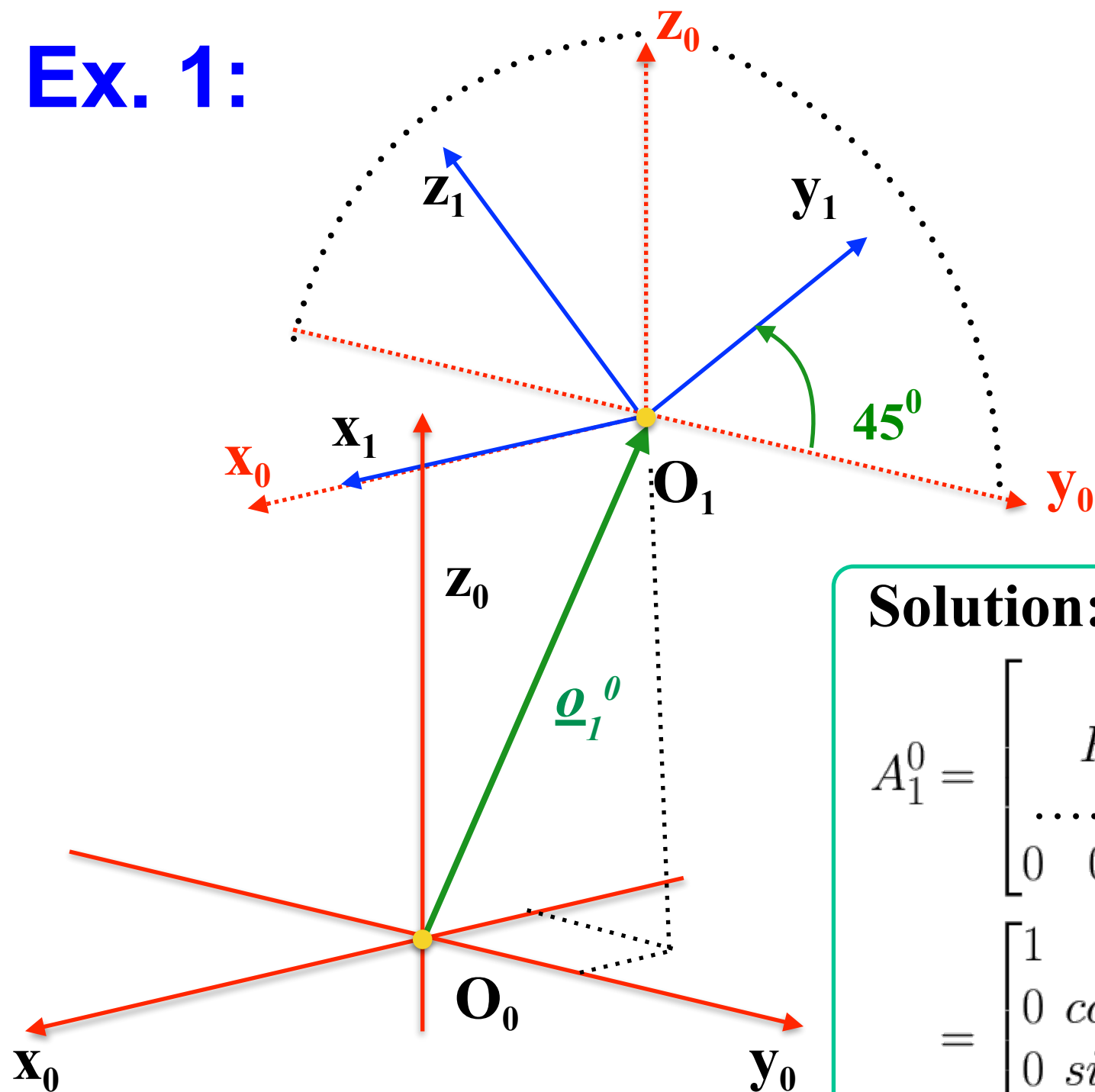


Given $q_1^0 = [-1, 2, 8]^T$
Find the Homogeneous Transformation Matrix describing O_1 - $x_1y_1z_1$ w.r.t. O_0 - $x_0y_0z_0$

Note that: O_1 - $x_1y_1z_1$ is obtained from O_0 - $x_0y_0z_0$ via an elementary counterclockwise rotation and a translation defined by q_1^0

Examples on Homogeneous Matrices

Ex. 1:



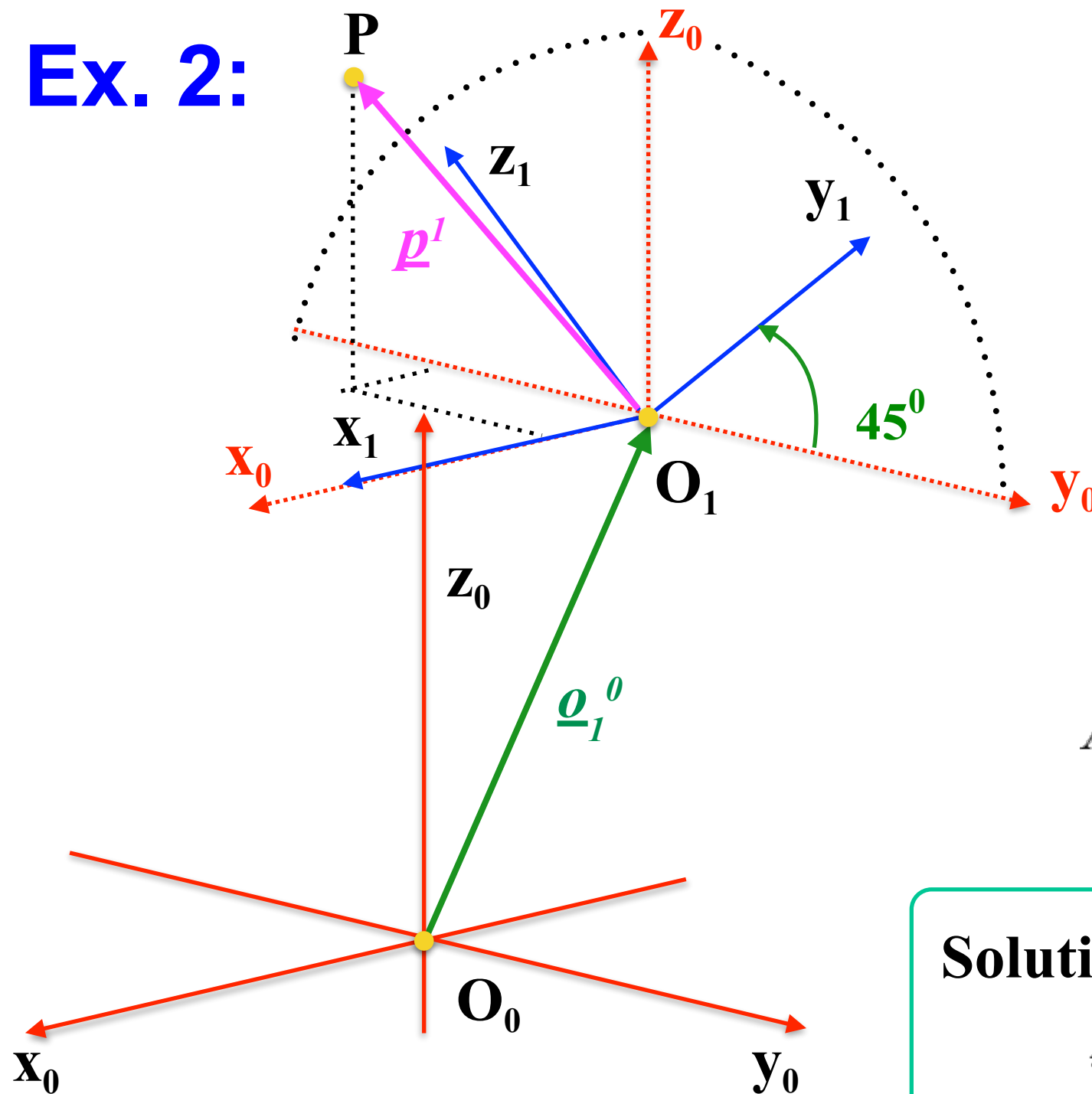
Given $\underline{q}_1^0 = [-1, 2, 8]^T$
Find the Homogeneous Transformation Matrix
describing O_1 - $x_1y_1z_1$ w.r.t.
 O_0 - $x_0y_0z_0$

Solution:

$$\begin{aligned}
 A_1^0 &= \begin{bmatrix} R_1^0 & \vdots & \underline{q}_1^0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} R_x(\gamma) & \vdots & \underline{q}_1^0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & \cos(45^\circ) & -\sin(45^\circ) & 2 \\ 0 & \sin(45^\circ) & \cos(45^\circ) & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

Examples on Homogeneous Matrices

Ex. 2:



Given $\underline{q}_1^0 = [-1, 2, 8]^T$
 $\underline{p}^1 = [1, -2, 5]^T$

Find the coordinates of point P w.r.t. O_0 - $x_0y_0z_0$

$$A_1^0 = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \leftarrow \text{From Ex.1}$$

Solution:

$$\underline{\tilde{p}}^0 = A_1^0 \cdot \underline{\tilde{p}}^1 = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 2 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 5 \\ 1 \end{bmatrix}$$



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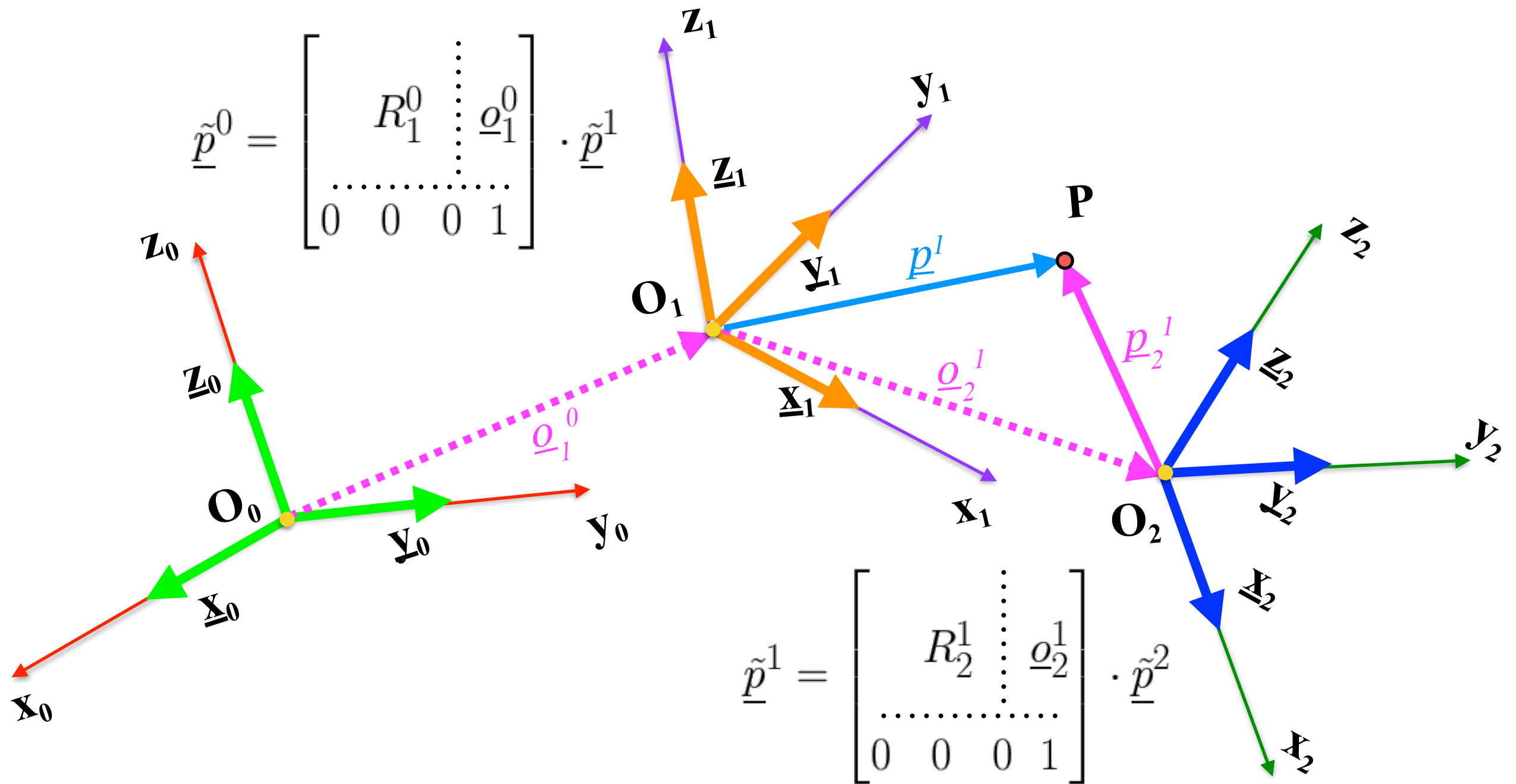
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Prof. Antonella Ferrara

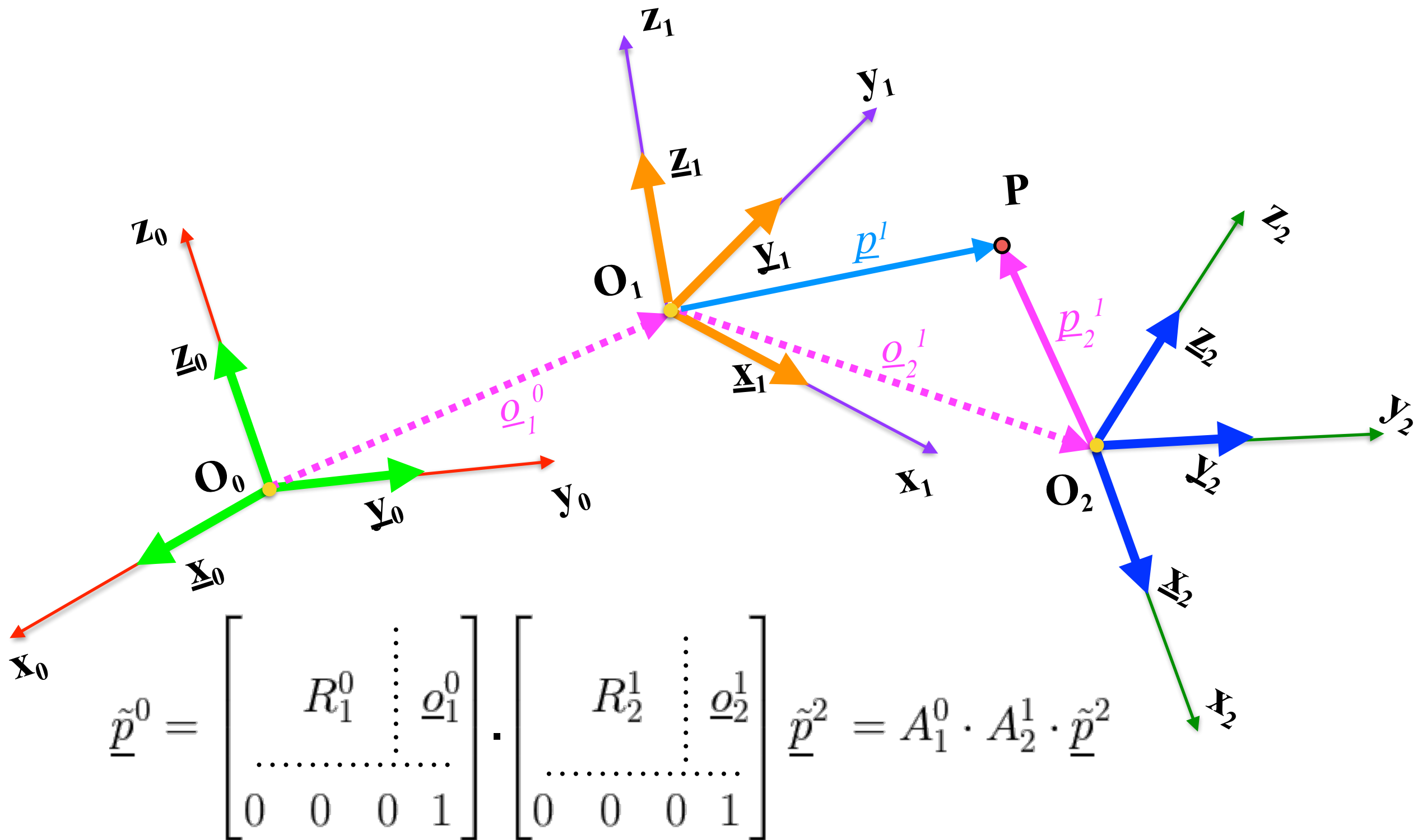


Composition of Roto-Translations

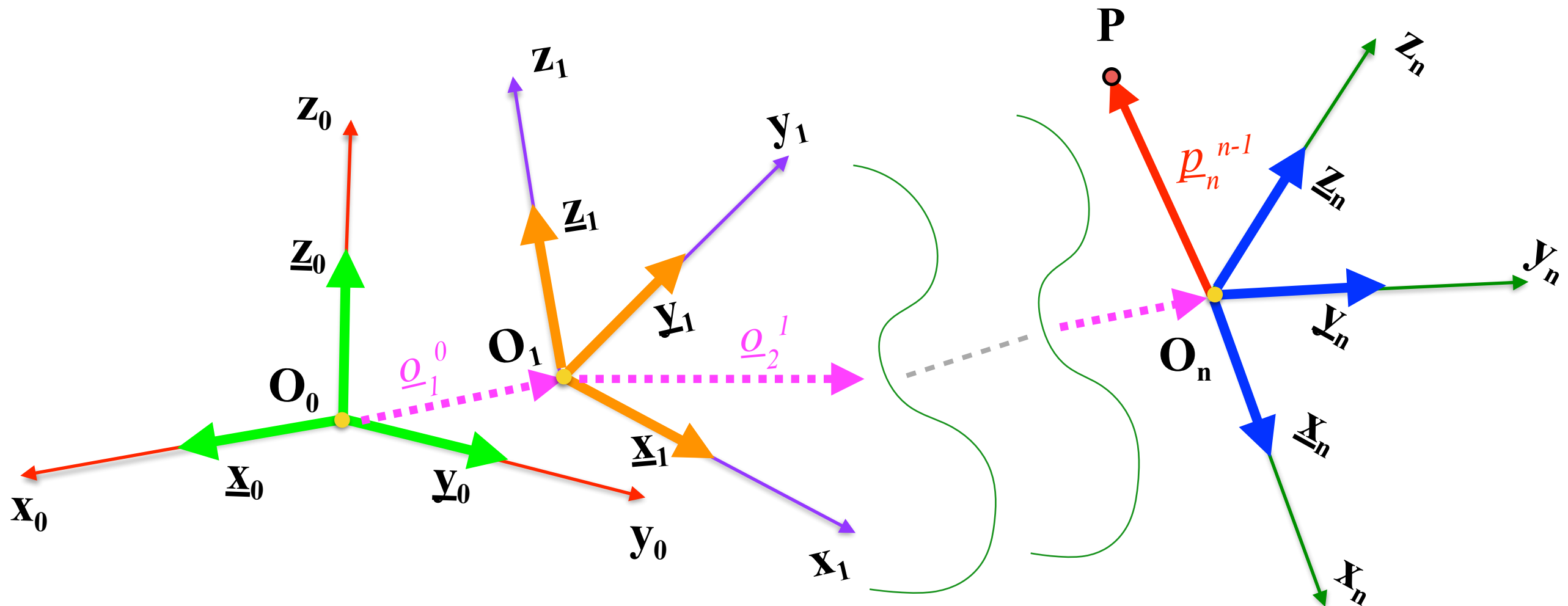
Composition of Roto-Translations



Composition of Roto-Translations

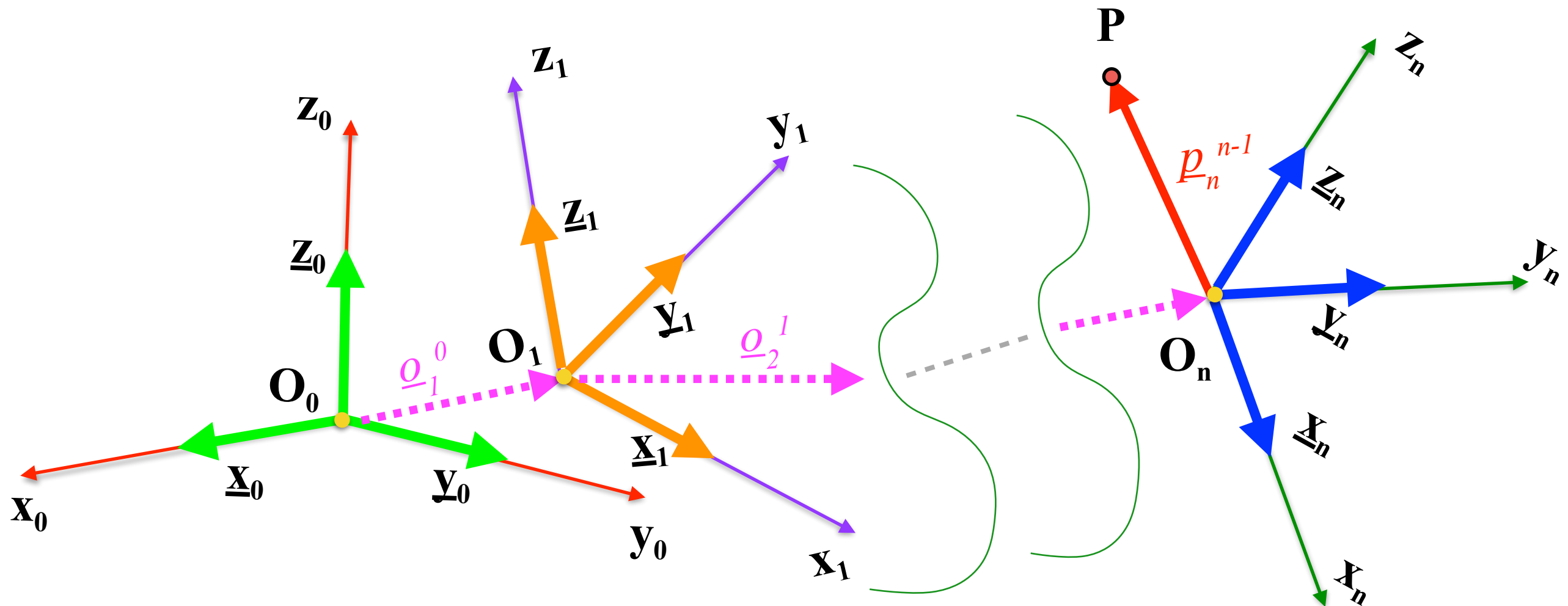


Composition of n Roto-Translations



$$\underline{\tilde{p}}^0 = A_1^0 \cdot A_2^1 \cdots A_{n-1}^{n-2} \cdot A_n^{n-1} \cdot \underline{\tilde{p}}^n$$

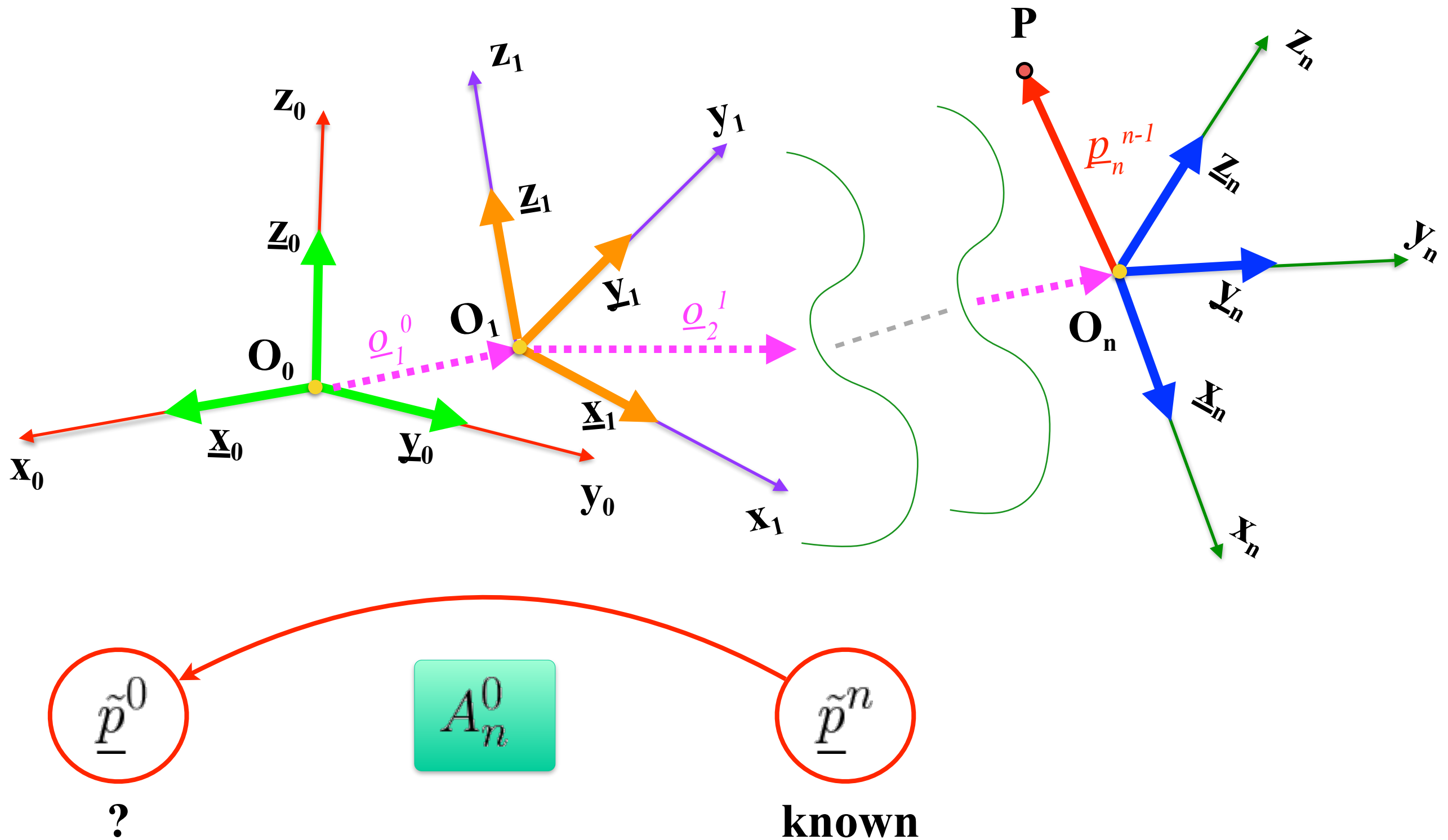
Composition of n Roto-Translations



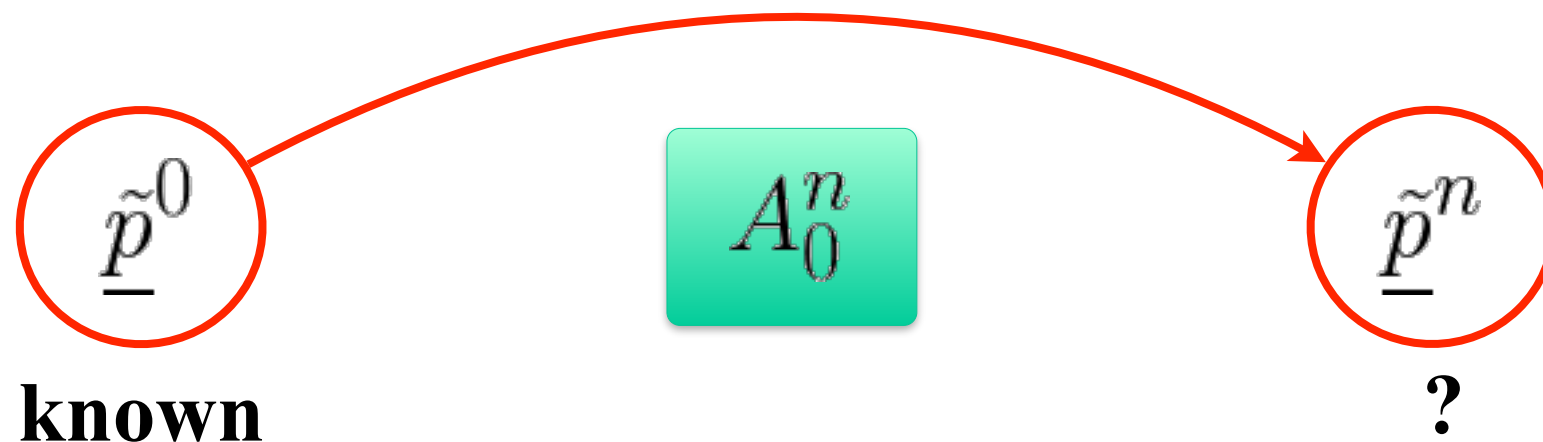
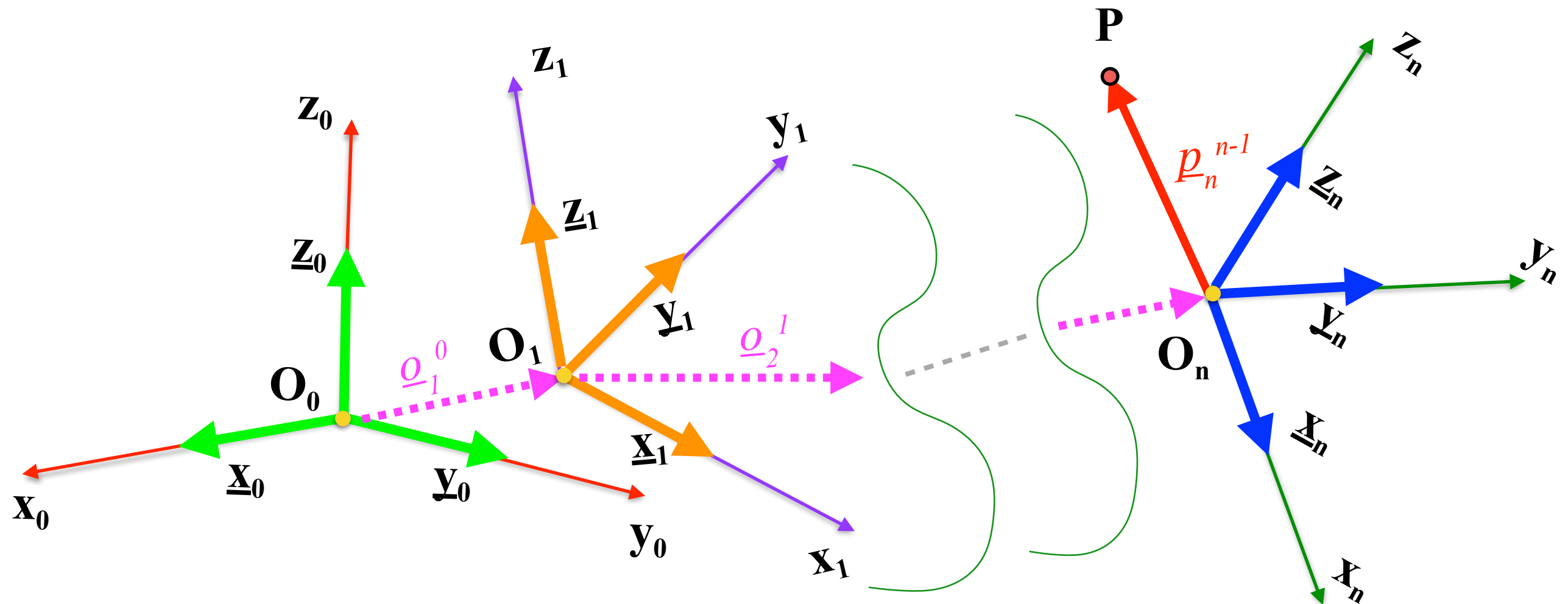
$$\underline{\tilde{p}}^0 = A_1^0 \cdot A_2^1 \cdots A_{n-1}^{n-2} \cdot A_n^{n-1} \cdot \underline{\tilde{p}}^n$$

$$A_n^0$$

Composition of n Roto-Translations



Composition of n Roto-Translations



known

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Dipartimento di
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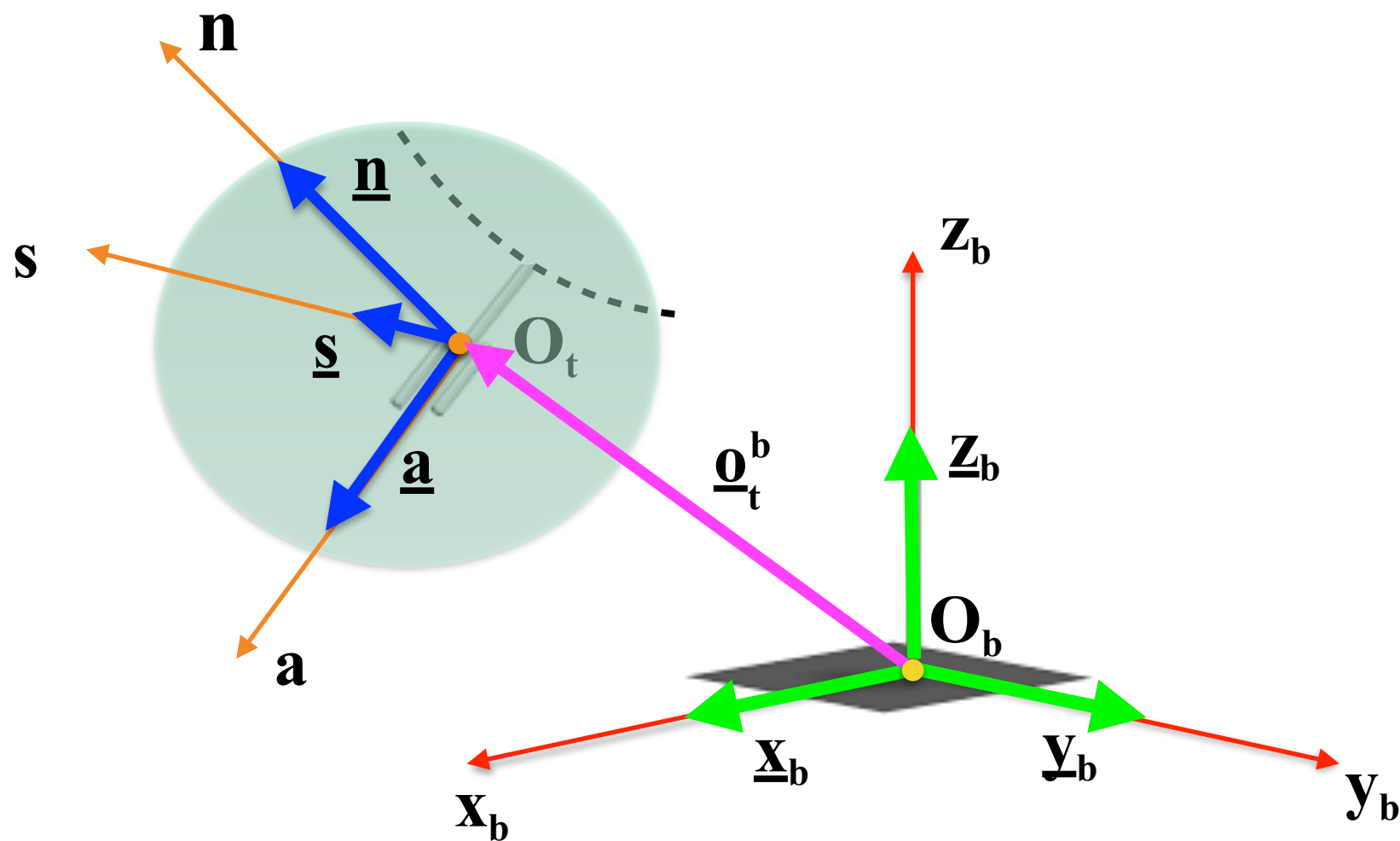
Prof. Antonella Ferrara



“Pose” of Rigid Objects in 3D Space

Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object
with respect to the Base Reference Frame:

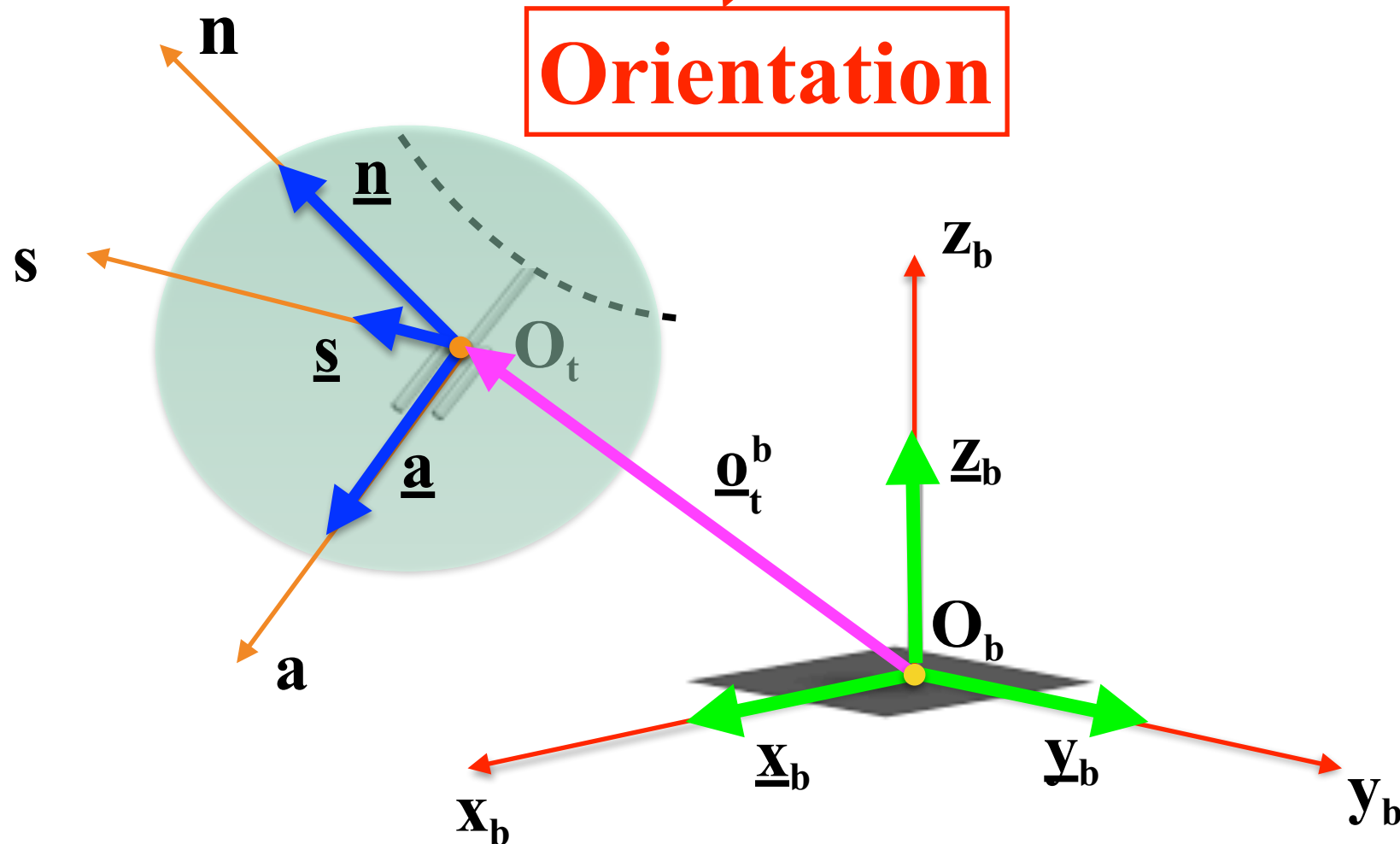


Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object
with respect to the Base Reference Frame:

Position

Orientation



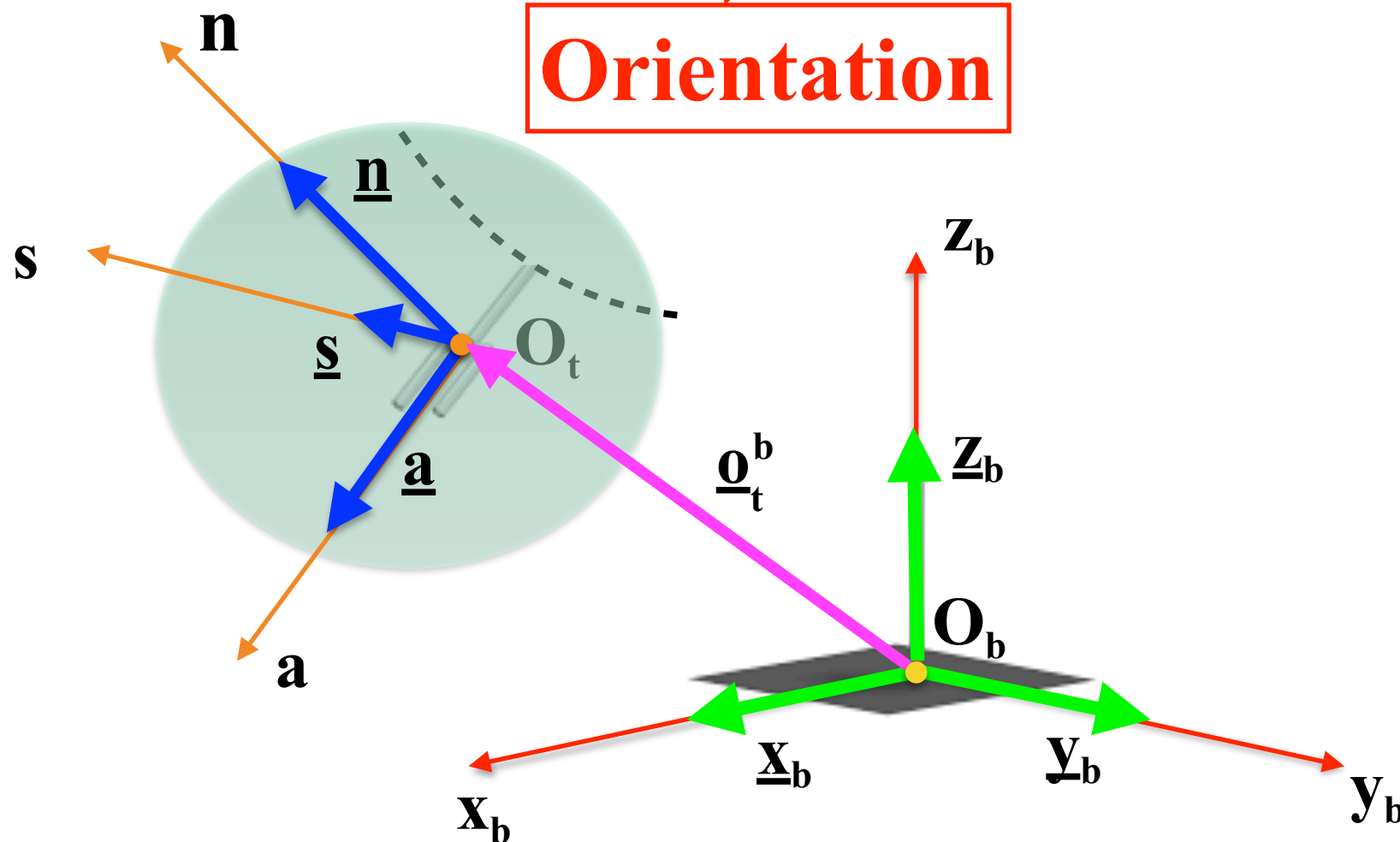
Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object
with respect to the Base Reference Frame:

Homogeneous
Transformation
Matrix

Position

Orientation



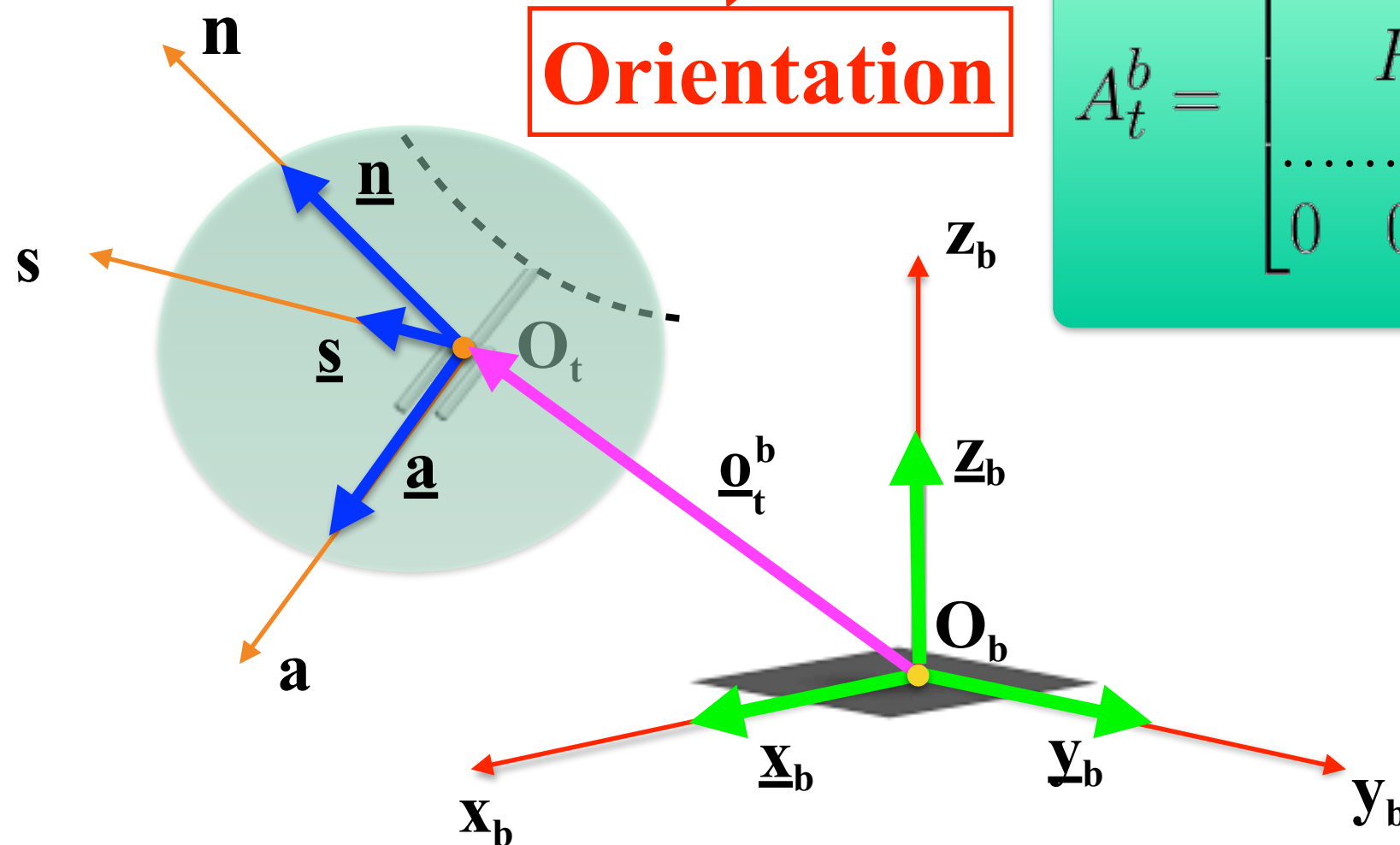
Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object with respect to the Base Reference Frame:

Homogeneous Transformation Matrix

Position

Orientation



$$A_t^b = \begin{bmatrix} R_t^b & \underline{q}_t^b \\ \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \underline{n}^b & \underline{s}^b & \underline{a}^b & \underline{q}_t^b \\ \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

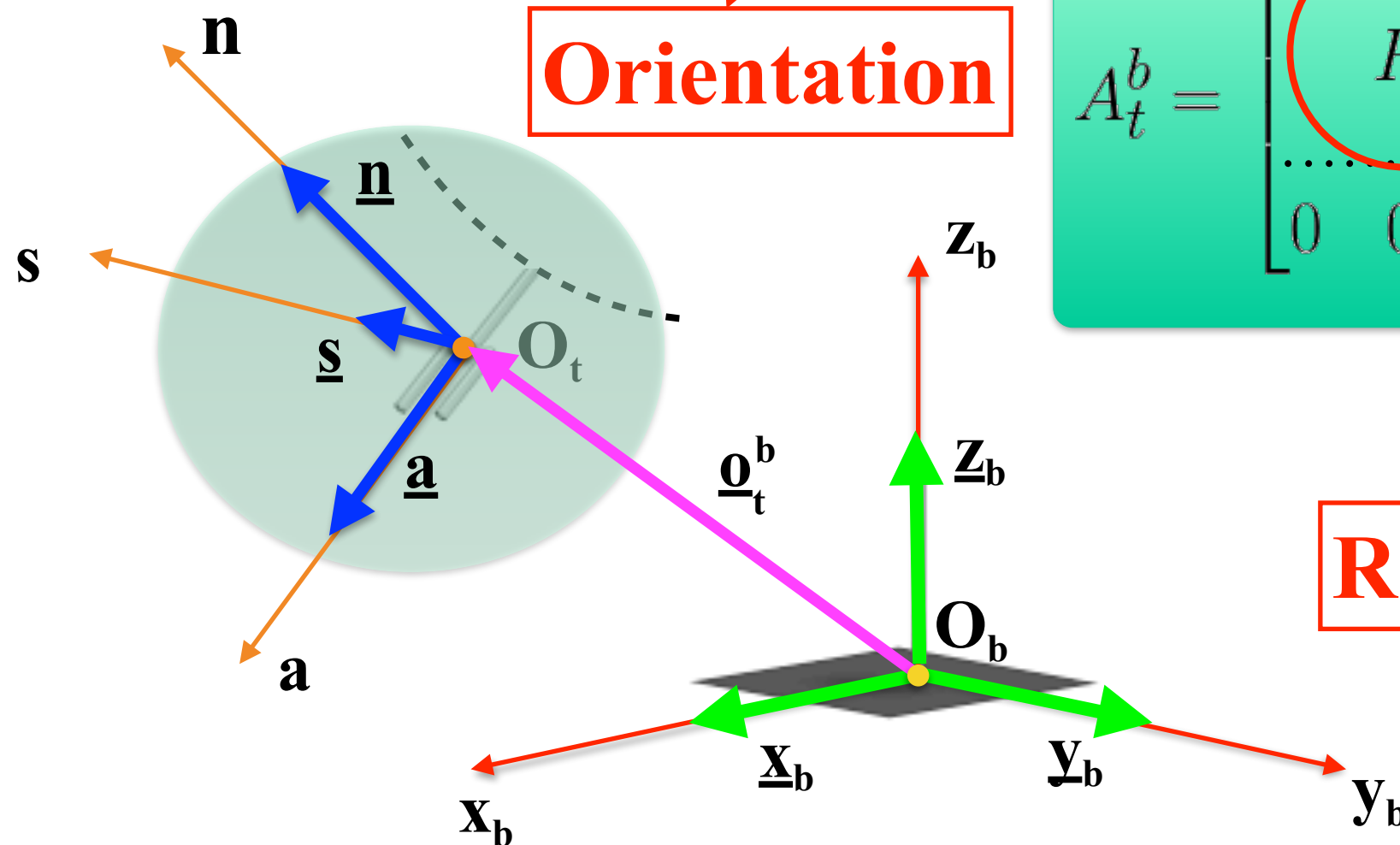
Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object with respect to the Base Reference Frame:

Homogeneous Transformation Matrix

Position

Orientation



$$A_t^b = \begin{bmatrix} R_t^b & \underline{q}_t^b \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \underline{n}^b & \underline{s}^b & \underline{a}^b & \underline{q}_t^b \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation Matrix

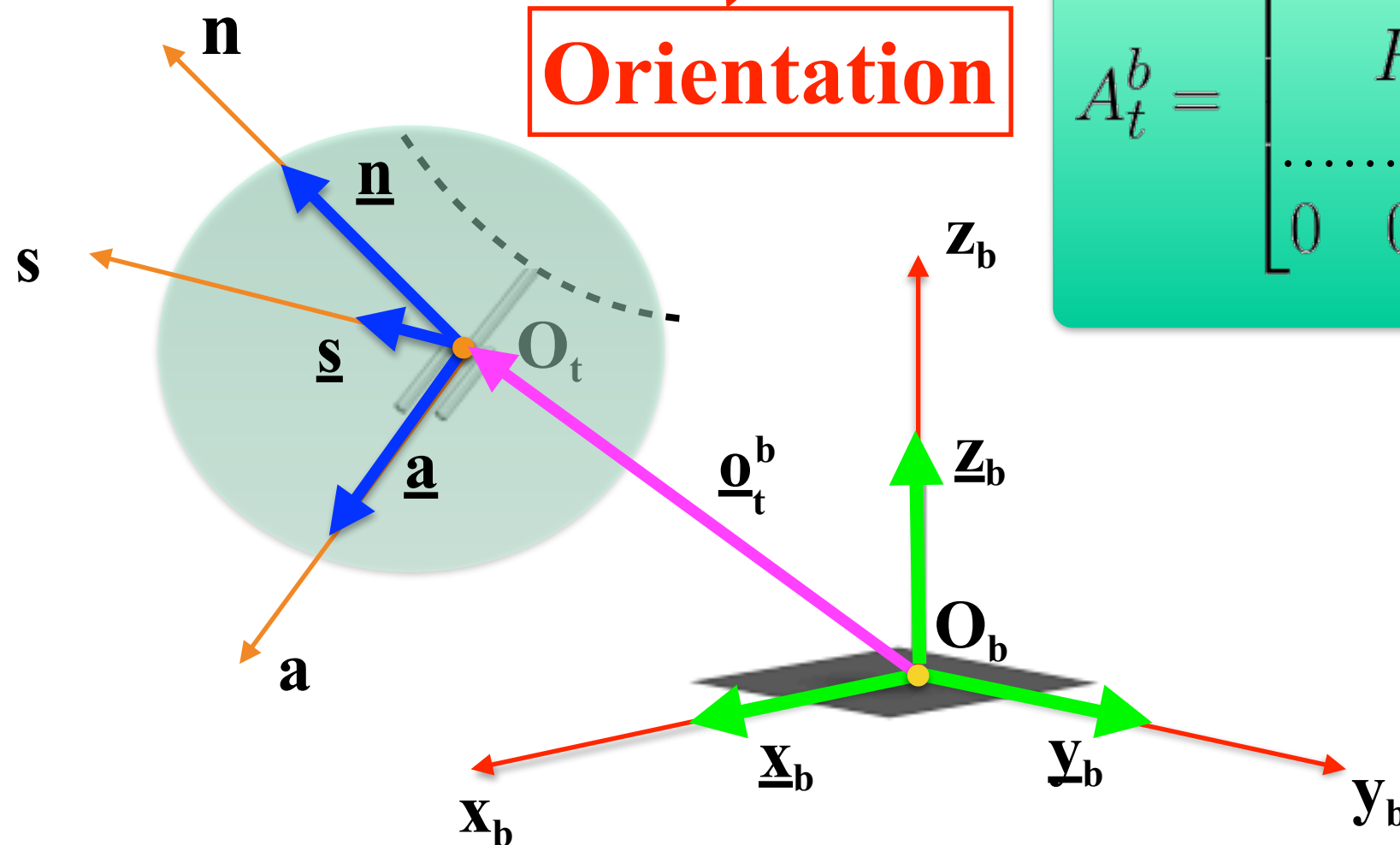
Going Back to Rigid Objects in 3D Space

Description of the Pose of a Rigid Object
with respect to the Base Reference Frame:

Homogeneous
Transformation
Matrix

Position

Orientation



$$A_t^b = \begin{bmatrix} R_t^b & \underline{q}_t^b \\ \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \underline{n}^b & \underline{s}^b & \underline{a}^b & \underline{q}_t^b \\ \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Position Vector